

# Where to Stop: Tile-Adaptive Early Exit in a Learned Image Decoder

## Supplementary Material

Anonymous CVPR submission · Paper ID \*\*\*\*

### A. Test settings, implementation and reproducibility

#### A.1. What this section covers

This section is the recipe and the protocol: the released codec the work starts from and the exact file it is, the test set, the conventions every decibel is measured under, every module with its shape and its parameter count, the training configuration with a column naming where each choice was measured, the hardware, the seeds and what varies between two runs of one command. Numbers are not repeated across sections; where a quantity belongs to a later one, this section names it and stops. One fact governs the rest and is stated here rather than buried: the checkpoint every headline number comes from has seen the training set exactly once, and A.10 reports how much the numbers move with a second pass.

#### A.2. Test settings: the released codec, and what is frozen

There is no traditional-codec invocation in this work and none is needed. The axis measured here is decoder-side compute against a fixed learned decoder, on a bitstream that decoder produced, so the anchor is a checkpoint rather than a command line, and the released codec's own standing against VTM is published with it. That checkpoint is identified by content and not by path, because the released file is a bare state dict carrying no epoch, no optimiser and no step, so a path plus a hash is the whole of its provenance.

The anchor is `~/DCVC/checkpoints/cvpr2026_image.pth.tar`, sha256 `b3b900de23f30e4fc437010ddffe6a5413a56d6cfd97fb6235adabcd2b6973302`, recorded with that reading in `results/dmc_ld_recon_audit.json`.

The codec is imported rather than reimplemented: `flexuf/model.py` sets `DCVC_ROOT` to the DCVC checkout and imports `src.models.image_model.DMCI` from it, so the entropy model, the hyperprior, the quantisation-step tables and the 64 quality indices are the released code running unmodified. Our decoder is those weights rearranged: an opening upsampling block, 12 `DepthConvBlocks` at `C = 384` channels and a head that pixel-shuffles back to RGB, with the 12 blocks grouped into `K = 6` exits of `b = 2` blocks each and the first `j = 2` groups running once over the whole frame before the decode splits into tiles. Exits 0 and 1 therefore lie below the split and cannot be selected; the four reachable exits are 2 to 5.

The encoder is frozen for the whole of training, and that is asserted rather than assumed: `scripts/signalled_curve.py` compares our encoder state dict against the released one tensor by tensor and requires the largest absolute difference to be exactly zero before a single frame is measured. A trained decoder therefore consumes byte by byte the stream the released encoder produces, the bit rate is identical by construction, and the whole of the measured difference is decoder-side.

#### A.3. The test set, and what tiling does to it

| class   | sequences | resolution | padded to | tiles | added px |
|---------|-----------|------------|-----------|-------|----------|
| UVG     | 7         | 1920x1080  | 2048x1280 | 40    | 26.4%    |
| MCL-JCV | 30        | 1920x1080  | 2048x1280 | 40    | 26.4%    |
| HEVC B  | 5         | 1920x1080  | 2048x1280 | 40    | 26.4%    |
| HEVC E  | 3         | 1280x720   | 1280x768  | 15    | 6.7%     |
| HEVC C  | 4         | 832x480    | 1024x512  | 8     | 31.3%    |
| HEVC D  | 4         | 416x240    | 512x256   | 2     | 31.3%    |
| all     | 53        |            |           |       |          |

**Table 28.** The test set, and what tiling does to each resolution. Take from it that granularity is not constant across the set: a 1080p frame carries 40 tiles for the allocation to work with and a 416x240 frame carries two, and the small classes also pay the most padding. That single fact explains most of the per-class spread reported in section D.

Sources: `results/per_class_RECIPES12.json` for the class membership and tile counts, `results/tile_definition.json` for the padded sizes; both on `runs/RECIPES12/ckpt_PAPER.pth.tar`. The sequences are UVG [23], MCL-JCV [24] and HEVC classes B, C, D and E [7]; the full list of 53 names is recorded in the measured field of `results/signalled_RECIPES12_ctc53.json`, with `not_measured` empty.

Frames are the leading frames of each sequence, taken consecutively. The headline file uses two frames on each of the 53 sequences of Table 28, so 106 frames; the routed, rate-rank and partial signalling tables use one. Every results file records `frames_per_seq` and every table in this supplement states it. No frame is dropped and no sequence excluded: the `not_measured` field of every curve file is empty.

Border padding is an *estimator* of the unseen neighbour, and the seam is its error, so the four rules below are four estimators compared on the same frames with early exit switched off, leaving tiling as the only difference from a full-frame decode.

| Border estimator      | q0           | q32          | q63          |
|-----------------------|--------------|--------------|--------------|
| Zeros (stock)         | 0.126        | 0.287        | 0.677        |
| Replicate             | <b>0.071</b> | <b>0.123</b> | <b>0.218</b> |
| Linear extrap.        | 0.198        | 0.286        | 0.384        |
| AR(1) per channel [9] | 0.061        | 0.107        | 0.197        |

**Table 29. Border estimators**, dB below the released decoder on the same latent, 256 px tiles, full CTC, with every tile at full depth. Lower is better. The ordering is not the obvious one. Replication, a zero-order hold, beats linear extrapolation by a wide margin, because extrapolating the local gradient past a boundary amplifies whatever noise sits on that boundary and assuming local constancy does not: 0.384 dB against 0.218 dB at q63. The fitted per-channel AR(1) rule wins on quality, reaching 0.197 dB, and loses on cost at +10.7% of decode wall-clock, which against a 0.1 dB budget does not close; section H carries that verdict in full.

`results/ctc_seam_p256.json`, generated into `paper/tables/padding.tex` by `scripts/make_paper_tables.py`. The shipped configuration is replicate padding.

Tile size is the other half of what tiling costs, and it is free in arithmetic: a tiled decode's multiply-accumulate count does not depend on the tile side at all, because every tile runs the same graph and the tiles partition the same feature map. What the side buys is granularity for the allocation, and what it costs is seam. The two sides we have measured are below.

| Estimator, q63 | 128 px | 256 px | ratio         |
|----------------|--------|--------|---------------|
| zeros          | 1.249  | 0.677  | $\times 0.54$ |
| replicate      | 0.372  | 0.218  | $\times 0.58$ |
| arls           | 0.332  | 0.197  | $\times 0.59$ |

**Table 30. Tile size, at q63.** dB below the released decoder with every tile at full depth, so this is tiling penalty and nothing else. Doubling the side roughly halves the penalty, which is what a perimeter-driven error predicts, and it costs no computation. Against that, 256 px gives a 1080p frame 40 tiles to allocate across and 128 px gives it 160; section H reports what happened when we tried to spend that granularity.

results/ctc\_seam\_p256.json and results/ctc\_seam\_p128.json, generated into paper/tables/tilesizes.tex by scripts/make\_paper\_tables.py. The two tile sizes come from separate training runs, which is the confound section H states.

#### A.4. The measurement conventions

- **Colour.** Sources are YUV 4:2:0, 8 bit, converted to 4:4:4, divided by 255 and shifted by  $-0.5$  for the network. PSNR is the released codec's own metric,  $(6 \cdot Y + U + V)/8$ , computed after converting the reconstruction back to 4:2:0 on the 0 to 255 scale. Measuring in 4:4:4 would compare against a chroma upsample of the source rather than the source, and would report a chroma PSNR no published DCVC-UF number could be compared with.
- **Padding.** Frames are padded by replication, bottom and right only, to a whole number of tiles; results/tile\_definition.json records the site as the RGB frame before the encoder and the mode as "replicate, bottom and right only: F.pad(x, (0, pw, 0, ph))". The stock codec aligns to 16, which is enough for the transform and not for a tile grid. Rate is charged on the true pixel count and PSNR computed after cropping back, so padding can cost and cannot flatter, and it does cost: a tile containing replicated rows gives up about twice the quality of one that does not.
- **Decibels.** The main paper's protocol box fixes the convention; what it does not say is that pooling all tiles into one mean squared error first, which is the other live convention, moves the answer by about a third of a 0.1 dB budget. Section E measures that gap. No table here mixes the two and every table says which it is on.
- **A second metric.** MS-SSIM is reported beside PSNR at the operating point, computed as Wang et al. 2003, 5 scales, 11-tap Gaussian sigma 1.5, valid region, luma plane in 4:2:0 on 0..255. The implementation is self-checked at start-up: identical inputs score 1.000000 and a 3×3 box blur scores 0.7804, both recorded in results/supp\_opquality\_PAPER.json.
- **The map is billed.** The exit map is entropy coded from a per-frame histogram and its cost added to the rate before any saving is computed: 64 to 91 bits per frame at the 0.1 dB budget, at most 8.92e-05 bpp. Every row records map\_bits and the bpp it added.
- **The reference.** The released decoder re-expressed in the same ladder and selected by K, so a K = 12 run cannot be quoted against the K = 6 remap.

The padding figure and the second metric are both from results/supp\_opquality\_PAPER.json on the pinned checkpoint, 53 sequences at the 0.1 dB operating point: mean per-tile penalty 0.218 dB over the 547 tiles containing replicated rows against 0.113 dB over the 1,218 that do not, at q0. The map-bit range is from results/signalised\_RECIPES12\_ctc53.json, field map\_bits.

Two limits belong beside that list. Sweeping a multiplier reaches only the lower convex hull of the achievable set [26, 27], so a budget falling in a gap of the hull is met at the nearest reachable point; every row carries a reachability flag and a floor, and rows whose floor already exceeds the budget are reported as unreachable rather than dropped. And 0.1 dB is a reporting convention, not a perceptual threshold. Subjective work measures the smallest noticeable change in quantisation parameter or in a video quality metric, not in tenths of a decibel of PSNR, so nothing published licenses a claim that 0.1 dB is invisible; 0.3 dB and 0.5 dB are reported beside it so that the choice carries no weight.

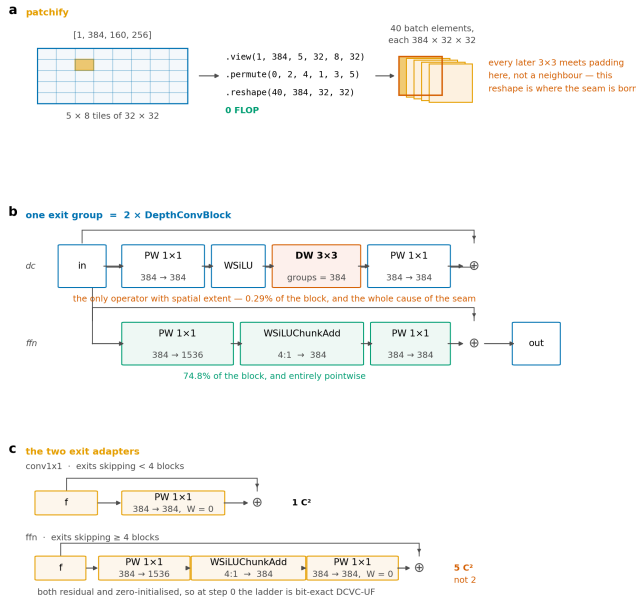
#### A.5. The ladder, module by module

| module                  | runs              | in → out              | params     | share |
|-------------------------|-------------------|-----------------------|------------|-------|
| upsample                | frame             | 256 → 384, $\times 2$ | 1,431,936  | 10.1  |
| groups 0 to 1, stem     | frame             | 384 → 384             | 4,154,880  | 29.2  |
| groups 2 to 5, routable | tile              | 384 → 384             | 8,309,760  | 58.4  |
| one DepthConvBlock      | tile              | 384 → 384             | 1,038,720  | 7.3   |
| adapters 0 to 3, FFN    | tile, at its exit | 384 → 1536 → 384      | 739,200    | 5.2   |
| adapter 4, 1×1          | tile, at its exit | 384 → 384             | 147,840    | 1.0   |
| seam repair, grid       | after stitching   | 384 → 384             | 152,704    | 1.1   |
| head, PixelShuffle 8    | frame             | 384 → 192 → 3         | 335,232    | 2.4   |
| router head (B)         | tile              | 161 → 6               | 144,024    | 1.0   |
| decoder, released       |                   |                       | 14,231,808 | 100.0 |
| decoder, ours           |                   |                       | 3,257,344  | +22.9 |
| encoder, frozen         | encode            |                       | 27,956,246 | 196   |

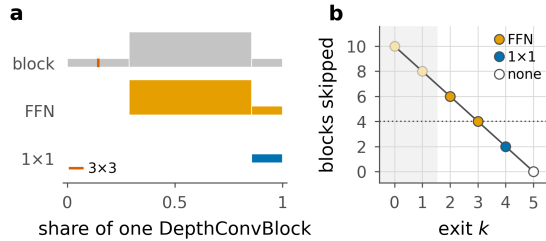
**Table 31.** Where every module sits, what shape it is and what it weighs, with the last column as a percentage of the released decoder's own parameters; adapter rows are per adapter and group rows cover the whole run of groups. Take from it three things. The ladder adds 22.9% to the decoder's parameters, almost all of it in the four feed-forward adapters. Of that, 1,478,400 parameters can never run, because at  $j = 2$  the decoder clamps every exit to at least 2 and adapters 0 and 1 are unreachable; they still take gradient from the distillation term, which taps every exit, and dropping them would forfeit the ability to re-run the same checkpoint at another split depth. And the frozen encoder is larger than the whole decoder, which is why an encoder-side search is the expensive half of this system and a decoder-side one is not.

Source: results/supp\_module\_shapes.json, derived by scripts/module\_shapes.py from the state dict of runs/RECIPES12/ckpt\_PAPER.pth.tar (md5 daa3760937ea0bd8) on the CPU. Parameter counts are tensor sizes read straight from the state dict. The same file also carries each module's multiply-accumulates per feature pixel, and the three aggregate shares that fall out of them, 8.16% for the upsample, 89.44% for the twelve-block trunk and 2.40% for the head, reproduce the forward-hook audit of results/mac\_audit.json to four decimals. Section B prices the decode module by module and derives the per-exit cost; none of that is repeated here.

One initialisation detail behind the seam-repair row of Table 31 is load-bearing and is not visible in the arithmetic. The pass multiplies its correction by a learned gate indexed by position within a tile, a 32×32 array of scalars shared across all 384 channels, and that gate starts as a decaying function of the distance to the nearest tile edge. At step zero the module therefore already knows where the seams are and training refines rather than discovers, while its output convolution is still zero, so the whole module is exactly the identity and the warm-start controls of A.9 still hold.



**Figure 32. The three pieces a reader has to re-implement.** **a**, patchify is a pure reshape and costs nothing, and it is where both the seam and every operation of saving come from. **b**, one exit group is two DepthConvBlocks, whose only operator with spatial extent is a single 3×3 depthwise worth 0.33% of the block (9C of the block's 7C<sup>2</sup> + 9C, results/mac\_audit.json), so an early exit gives up almost purely pointwise capacity. **c**, the two adapters that replace it, both residual and both zero-initialised. Shapes are those of a 1080p frame padded to 2048×1280 and split into 40 tiles.



**Figure 33. Inside an exit adapter.** **a**, one DepthConvBlock and the two adapters, drawn to scale: bar length is a share of the block, bar height the channel width written. Neither adapter contains the block's only 3×3 (the hairline rule), so neither adds receptive field or seam penalty, which is the design constraint the seam measurement imposes. **b**, blocks skipped per exit, coloured by adapter; the rule switches to the FFN at four skipped blocks. Lengths are counted with hooks off the modules themselves.

What the adapters are worth is measured rather than argued, and the measurement is available because the identity is exactly what they were initialised to. Setting each adapter back to that identity leaves the ladder otherwise untouched, so the difference is what training put into them.

|      | q0    | q32     | q63   |         |       |         |
|------|-------|---------|-------|---------|-------|---------|
| Exit | with  | without | with  | without | with  | without |
| 2    | 0.137 | 2.022   | 0.179 | 2.949   | 0.214 | 4.543   |
| 3    | 0.061 | 1.026   | 0.086 | 1.827   | 0.099 | 3.163   |
| 4    | 0.043 | 0.400   | 0.055 | 0.563   | 0.058 | 0.874   |
| 5 +  | 0.028 | 0.028   | 0.036 | 0.036   | 0.042 | 0.042   |

**Table 32. What the adapters are worth.** dB below the released decoder with every tile at that exit, with the trained adapters and with each set back to the identity it was initialised to. The deepest exit has no adapter by construction and is the control, and it moves by exactly zero. Without the adapters the shallowest exit costs 4.54 dB at q63, forty-four times the working budget, and the adapter buys 4.33 dB of that back.

results/adapter\_ablation.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar, generated into paper/tables/adapters\_ablation.tex by scripts/make\_paper\_tables.py.

## A.6. Tiles, and the exit clamp

patchify is a view, a permute and a reshape with no arithmetic at all: a feature map of shape [1, 384, 160, 256] becomes 40 batch elements of [384, 32, 32], as panel a of Figure 32 shows. Tile order is row major, which is what the exit map is indexed by and what flexuf/eval.py reproduces with the same permutation when it measures per-tile error. One tile is 256 px of RGB, 32 px of feature map and 16 px of latent, at 16 pixels per latent position. patchify asserts divisibility rather than padding: an unpadded 1080p frame gives a 40×40 feature map that is not divisible by 32, and the run stops instead of producing a ragged last row.

The clamp is the most consequential line in the decoder. forward() applies exit\_map.clamp(min = j, max = K - 1) before the per-tile loop, so a map naming exit 0 at j = 2 runs group 2 and wears exit 2's adapter. Four separate places have to apply that same clamp, and getting any of them wrong is silent rather than loud, which is why they are listed rather than described.

| file and function                    | what it does                                                                                             |
|--------------------------------------|----------------------------------------------------------------------------------------------------------|
| decoder.py, forward                  | clamps the map to [j, K-1] before the per-tile loop                                                      |
| cost.py, exit_costs                  | charges max(k, j) blocks, and bills the adapter the clamped exit wears rather than the one the map names |
| eval.py, tiled_exit_mses             | fills the columns below j with the decode at exit j, so an argmin over all K lands on a reachable choice |
| head2.py, forward and oracle_ce_loss | masks the logits below the split depth with -inf, and clamps the oracle label to the same bound          |

**Table 33.** The exit clamp, and the four places that have to repeat it. A clamp in the decoder without the matching clamp in the cost model bills a shallow map for compute it never ran, which is a saving no meter agrees with. The mask in the router is -inf and not a large negative constant, because nothing in a cross-entropy penalises a common offset added to every logit, so a finite mask can stop being the smallest entry in its row.

Source: the four files named in the first column, under flexuf/.

## A.7. The deployed decode, and how training differs from it

```

input latent y, quantisation step q, exit map m
1 f = upsample(y)
2 for g = 0 ... j-1: f = groups[g](f) frame, no borders
3 tiles, nh, nw = patchify(f, 32) reshape, 0 MAC
4 m = clamp(m, j, K-1) below j cannot run
5 active = all tiles; work = tiles
6 for g = j ... K-1:
7   work = groups[g](work) borders meet padding
8   leaving = active tiles whose m equals g
9   out = adapters[g](work[leaving]) raw if g = K-1
10  if training: out = out · gate[leaving]
11  canvas[leaving] = out
12  drop leaving from work and from active the saving
13  canvas = unpatchify(canvas, nh, nw)
14  canvas = canvas + G[r mod P, c mod P] · seam(canvas)
15 return pixel_shuffle(head(canvas · q), 8)

```

**Table 34.** The deployed routed decode, line for line. Take from it that line 12 is the entire saving: every group runs on fewer tiles than the last, and nothing else in the procedure is cheaper than a full decode. Line 10 is the only difference between training and inference, and it is numerically the identity.

Line 10 of Table 34 deserves the space. A router trained jointly with the decoder needs gradient to reach its logits and an argmax blocks it, so the gate is p / p.detach() with p the selected exit's probability under a Gumbel-softmax relaxation: its value is exactly 1 and the reconstruction bit-identical to the deployed path, while its gradient is the straight-through estimator. At inference the gate is absent and the exit is a hard argmax or a transmitted symbol. That is the one place where the trained object and the shipped object are not the same computation.

A second execution path exists and is bit-identical. With sorted execution the tiles are ordered by depth with a stable argsort before line 6, so "still active at group g" becomes a contiguous prefix, every group is

a slice rather than a masked gather, and the loop costs one host read instead of one synchronisation per group. Only the wall clock moves, by 1.2 points, and section B reports it.

### A.8. The training data

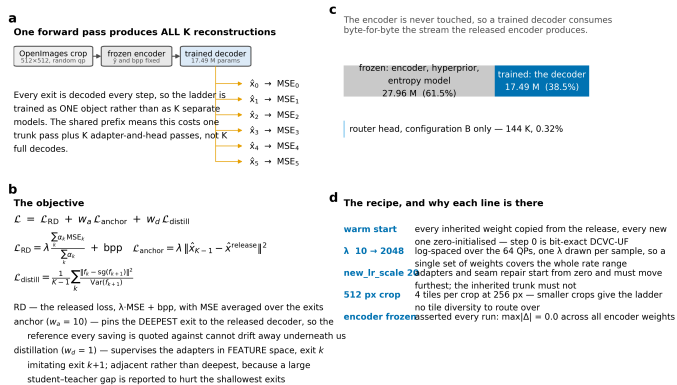
Training uses Open Images [22], fetched from the CVDF public mirror by `scripts/fetch_openimages.sh` and prepared by `scripts/prepare_openimages.py` into the flat layout DCVC-UF's own ImageFolder expects. Of 384,795 files scanned, 380,126 survive a minimum-side filter of 512 px, 4,669 are dropped as too small and 0 as unreadable. The filter is not cosmetic: the loader zero-pads any image smaller than the current crop, and an image that is half black border teaches the decoder to reconstruct black borders. Every 400th entry of the sorted survivor list is then held out, giving 512 validation images and 379,614 training images with no overlap, deterministically, so every run and every control sees the same held-out set.

Each sample is a random crop at the schedule's patch size with a random horizontal flip, converted to YCbCr, scaled to [0,1] and shifted by -0.5. One quality index is drawn uniformly over the 64 levels per sample and the matching multiplier comes from the log-spaced table running 10 to 2048, so one set of weights covers the whole rate range rather than a point on it. The crop is 512 px, which is what the released recipe's final phase asks for and also the smallest crop carrying more than one tile: at 256 px tiles a 512 px crop holds 4 and a 256 px crop holds one. A single tile has no border against a neighbour, so patched training at the smaller crop would train a configuration that cannot occur at inference. Evaluation uses a deterministic centre crop with no flip, because the loader's random crop swung one model between 29.54 and 30.58 dB at q0 across two runs of one command.

### A.9. The warm start, and the controls it asserts

`scripts/warmstart_from_release.py` performs the rearrangement, and it is a key remap and nothing else: the opening upsample takes `dec_1[0]`, group g block i takes `dec_1[1 + g-b + i]`, the head takes `dec_2`, and the adapters and the seam-repair gate are new and zero-initialised. The encoder, hyperprior, entropy model and quantisation-step tables are copied verbatim. The transfer moves 398 tensors and creates 28 adapters at initialisation. Two controls follow from zero initialisation and both are asserted at every build rather than assumed: the deepest exit reproduces the released decoder with a largest absolute difference of 0.0, and every untrained exit equals the released decoder truncated at that depth, also 0.0. A third lives in the checkpoint loader, which tolerates a missing or extra tensor only under the adapter, seam-repair, pad-coefficient and router prefixes and raises otherwise; both sides of a tolerated mismatch are zero-initialised identities, so dropping one is exact rather than approximate.

### A.10. The training run



**Figure 34. The training scheme.** a, one forward pass produces all K reconstructions, so the ladder is trained as one object. b, the three loss terms. c, the encoder is never touched. d, why each line of the recipe is there. The figure is a schematic drawn from the model definition; the parameter counts it carries are the ones tabulated in A.5.

The recipe, laid out in Figure 34, is Microsoft's own, applied at the point in their schedule that matches what we inherit. Their schedule is a 105-entry table of learning rate and crop size indexed by epoch, and the released weights are the output of its last entry, so applying entry zero would kick a converged codec with the from-scratch learning rate. The run enters at offset 99, inside their final 512 px phase at a learning rate of  $1 \times 10^{-5}$ ; all 852 records in the run's log carry that rate and that crop, which is the check that the offset did what it was meant to. Because the offset lands in the phase the experiment needs, no crop-size override is passed at all, where every other warm-started run here raises the crop floor by hand.

One deviation from a plain fine-tune is deliberate. The adapters and the seam-repair gate start at zero and have to move furthest, while the inherited trunk must not move at all if the deepest exit is to stay the released decoder, and the cost of choosing wrongly between them is measured: at the from-scratch rate the trunk drifts 0.20 dB of anchor at q32 in a single epoch. The new modules therefore get their own parameter group at 20 times the schedule's rate, selected by name prefix, with the multiplier re-applied whenever the schedule sets the base rate.

The loss has three terms. The rate-distortion term is the released loss, a multiplier times mean squared error plus bits per pixel, with the error averaged over the exits under fixed weights. The anchor term pins the deepest exit to a frozen copy of the released decoder, because every saving in the paper is quoted against that decoder and a reference drifting underneath the measurement makes the measurement meaningless. The distillation term supervises each adapter in feature space against the exit one step deeper, normalised by the teacher's own variance, following the adjacent-teacher form rather than the deepest-teacher form [15], since a large gap between student and teacher is reported to hurt the shallowest exits most. Gradients are clipped at a norm of 0.1 and a batch whose norm is not finite is dropped; across all 852 records the count of steps skipped for that reason is 0 and the median gradient norm is 0.273.

An epoch is 47,451 optimiser steps, which is 379,614 images at batch 8. The run logs every 200 steps and the median interval is 299.8 s, so a step costs about 1.5 s and an epoch about 19.8 hours. Training is fp32 throughout on a single NVIDIA RTX A6000 with eight data workers.

The pinned checkpoint is the first epoch boundary of a run whose launcher was given 16 epochs. Section H tabulates what the later checkpoints of this run and of its sibling measure at the same budget on the same frames; the one figure that belongs here is the floor, the quality given up before any tile exits early, which falls at every rate between the two epoch pins: it spans 0.0222 to 0.0485 dB at epoch 0 and 0.0295 to 0.0595 at epoch 1. No comparison in this paper between two runs at different epochs can be read as a comparison between two configurations, and where such a comparison appears it is labelled with both epochs. Nothing here extrapolates to a converged run.

Floors from results/signalled\_RECIPES12\_ctc53.json (runs/RECIPES12/ckpt\_PAPER.pth.tar, epoch 0) and results/signalled\_RECIPES12\_e1.json (runs/RECIPES12/ckpt\_PIN\_e1.pth.tar, epoch 1), both 53 sequences at the 0.1 dB budget.

$$\mathcal{L} = \mathcal{L}_{RD} + w_a \mathcal{L}_{anchor} + w_d \mathcal{L}_{distill}. \quad (8)$$

The distillation term is written in feature space and not in pixels because the head is a fixed map from feature to RGB, so matching the deeper feature is the stronger constraint, with 384 dense channels of target instead of three. Each exit imitates its neighbour, exit k following exit k+1, rather than the deepest exit.

One choice in the recipe matters more than the weights on those three terms. The adapters are trained *through the tiled decode path they are deployed in*, which is what the `train_patched` flag of the launcher above selects. Training them full frame and tiling only at inference loses 0.14 to 0.24 dB; training through the deployed path gains 0.51 to 0.90 dB, so the sign of the effect flips. A ladder that has never seen a seam does not know it has to compensate for one.

Those two ranges are read from the run log recorded in DECISIONS.md for the warm-start and train\_patched comparisons; no file in results/ holds them.

### A.11. Every hyperparameter, and where each was measured

| knob                                | value                                         | searched, and where                                    |
|-------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| <b>ladder geometry</b>              |                                               |                                                        |
| exits K, blocks per exit b          | 6, 2                                          | signalled_BEST128 and signalled_FINE12, at K = 12      |
| split depth j                       | 2                                             | supp_seam_vs_split, j from 0 to 6                      |
| RGB tile                            | 256 px                                        | ctc_seam_p128 against ctc_seam_p256; tilesize_adaptive |
| halo, latent and trunk              | 2, 0                                          | no; the trunk halo was priced at 2.25× and rejected    |
| adapter kind                        | scaled                                        | adapter_ablation                                       |
| adapter expansion                   | 4                                             | no; the block's own 4:1                                |
| seam repair                         | grid                                          | supp_seam_vs_split                                     |
| tile padding                        | replicate                                     | padding_ablation: zeros, replicate, linear, arls       |
| tile coupling                       | off                                           | coupling_ablation                                      |
| full-frame head                     | on                                            | the per-tile arm exists to measure it                  |
| <b>decoder training</b>             |                                               |                                                        |
| optimiser                           | AdamW                                         | no; the released recipe's                              |
| schedule                            | 105 entries, at offset 99                     | no; verified by scripts/verify_recipe.py               |
| learning rate                       | $1 \times 10^{-5}$                            | no; set by the schedule                                |
| multiplier, new modules             | ×20                                           | no; set from one anchor-drift measurement              |
| crop                                | 512 × 512                                     | no; set by the schedule                                |
| batch, workers, precision           | 8, 8, fp32                                    | no                                                     |
| gradient clip                       | 0.1, non-finite dropped                       | no; released recipe's                                  |
| multiplier range                    | 10 to 2048, 64 indices                        | no; released                                           |
| quality index                       | uniform, per sample                           | no                                                     |
| anchor weight                       | 10                                            | two settings run; 10 kept                              |
| distillation                        | 1.0, adjacent                                 | adjacent against deepest                               |
| auxiliary weight                    | 1.0, constant                                 | no                                                     |
| training path                       | deployed tiled decode                         | against the full-frame arm                             |
| encoder                             | frozen                                        | no; asserted at each evaluation                        |
| seed                                | none set                                      | no; see A.15                                           |
| <b>router head, configuration B</b> |                                               |                                                        |
| steps, batch, crop                  | 3,000, 6, 512                                 | no                                                     |
| learning rate, decay                | $1 \times 10^{-3}$ cosine, $1 \times 10^{-4}$ | no                                                     |
| regret weight                       | 1.0                                           | no                                                     |
| seed                                | 0                                             | fixed, so the input ablation is comparable             |
| live inputs                         | stem, latent, scales, qp                      | router_ablation, six variants                          |
| <b>evaluation</b>                   |                                               |                                                        |
| multiplier bisection                | 60 halvings on [0, 1]                         | no                                                     |
| outer correction                    | 6 passes, tolerance 5e-4 dB                   | no                                                     |
| rate-rank bisection                 | 40 halvings                                   | no                                                     |
| router bisection                    | on beta, bracket [−2e4, 2e4]                  | no                                                     |
| timing                              | 40 iterations after 20 of warm-up             | protocol in section B                                  |
| batch, when timing                  | 1                                             | supp_latency_batch, at 1, 2 and 4                      |

**Table 35.** Every knob this work sets, its value, and whether it was searched. Take from it that almost nothing on the training side is tuned: the optimiser, the schedule, the clip, the crop and the multiplier range are the released recipe's, and the four settings that are ours exist to protect a property the measurement depends on rather than to improve a number. What was searched is the geometry, and the third column names the file in results/ that holds each comparison, so a reader can price the choice

instead of taking it on trust.

Values come from scripts/launch\_recipe512.sh, train\_flexuf\_image.py, scripts/train\_router2.py and the four evaluation scripts; the third column names files in results/. The launcher's flags, without the two leading hyphens the shell needs, are: freeze\_encoder train\_patched epoch\_offset 99 new\_lr\_scale 20 anchor\_weight 10 seam\_repair grid adapter\_kind scaled distill\_weight 1.0 distill\_teacher adjacent lambdas 10 2048 batch\_size 8 n 8 e 16 num\_exits 6 split\_depth 2 latent\_patch 16 latent\_halo 2 aux\_weight 1.0 tile\_pad replicate.

### A.12. Finding the operating point

A budget is a quality target rather than a multiplier, so the multiplier has to be found, and it is found twice over: an inner bisection on a cheap table of per-tile distortions, and an outer correction against what a real decode delivers. The two levels exist because the table measures each tile with every other tile at the same exit, while a routed frame is mixed and a tile's border sees whatever depth its neighbour chose. The residual between them is small and nearly constant, so shifting the inner target by it converges in two or three passes, and each pass costs one decode per frame rather than one per bisection step.

Section G writes that search out line by line, with the shape each step returns. Three details of it change what the numbers mean. A rate whose floor exceeds its budget is written out with a reachability flag rather than dropped, so a saturated or unreachable row is visible instead of absent. The saving is measured and not modelled: a forward hook on every convolution and linear layer counts the multiply-accumulates the routed decode performed and divides by the same count for the released decode on the same latent, so the ratio needs no calibration constant and a module skipped by an early exit costs nothing because its hook never fires. The gap between model and meter is reported in every row: at the 0.1 dB budget it runs from 0.58 to 0.77 points, always in the direction of the model flattering us, and it is the measured column the paper quotes.

Two saving columns exist and only one is quoted. One divides by our own deepest exit, the other by the released decoder, and the released decoder is the denominator every claim is stated against; it is exactly 1 in the units of the cost table, so the seam-repair tax stays inside the figure it is supposed to be net of. The same two-level scheme with different constants runs in scripts/raterank\_curve.py, scripts/router\_curve.py and scripts/hybrid\_curve.py, whose constants Table 35 lists.

### A.13. What the work cost to produce

| item                      | cost                      | source                     |
|---------------------------|---------------------------|----------------------------|
| one decoder epoch         | 19.8 h on one A6000       | train_log.jsonl            |
| the pinned checkpoint     | 1 epoch                   | ckpt_PAPER records epoch 0 |
| the run so far            | 3 epochs and 27,400 steps | train_log.jsonl            |
| one router head           | 3,000 steps, 0.91 h       | router_ablation_stem       |
| encoder search, per frame | 4.52 decodes              | supp_encoder_cost          |
| decoder search, per frame | none                      | the map is signalled       |

**Table 36.** What producing these numbers costs, priced rather than asserted. Take from it that the decoder is cheap to build and cheap to run, and that the whole of the allocation cost falls on the encoder, which already has the source and already runs an analysis pass. Sections B and F price the encoder-side search and the router heads against what they buy; this table only says what the work took.

Sources: results/router\_ablation\_stem.json for one head's wall\_s field, results/supp\_encoder\_cost\_PAPER.json for the search cost (pinned checkpoint, q63, 40 tiles, 15 iterations, one tiled decode at 114.9 ms). Decoder wall clock is from runs/RECIPE512/train\_log.jsonl; no file in results/ records it.

### A.14. The pinned checkpoint, and what every table rests on

Results are measured on runs/RECIPE512/ckpt\_PAPER.pth.tar, md5 daa3760937ea0bd89b89c555c2eefa88, which records epoch 0 inside itself and is byte identical to that run's ckpt\_epo0.pth.tar. The name exists because a filename is not a checkpoint here: a per-epoch watcher overwrites runs/\*/ckpt\_eval.pth.tar as each epoch lands, so a results file

naming that path names a file that no longer exists. scripts/make\_paper\_tables.py and scripts/check\_paper.py both hold the pinned name and, among candidates for one quantity, prefer a file whose own ckpt field records it, falling back to the most recently written; hand-written order breaks ties and nothing else.

Every run named in this document warm-starts from the released decoder through one of two key remaps, one per ladder size, and freezes the encoder, so all of them consume the identical bitstream and differ only in the ladder flags of A.11. They are compared only at labelled epochs, for the reason section H measures.

| results file                | backs                    | checkpoint       | seq×fr |
|-----------------------------|--------------------------|------------------|--------|
| signalled_RECIPES12_ctc53   | main results, complexity | pinned, e4       | 53×1   |
| saturation_RECIPES12_ctc53  | operating range          | pinned, e4       | 53 fr  |
| static_RECIPES12_b01        | static controls          | pinned, e4       | 53     |
| per_class_RECIPES12         | per class                | pinned, e4       | 53 fr  |
| router_RECIPES12_b01_e4head | A against B              | pinned, e4       | 53×1   |
| router_RECIPES12_b03_e4head | A against B              | pinned, e4       | 53×1   |
| raterank_RECIPES12_b01      | rate rank                | pinned, e4       | 53×1   |
| raterank_RECIPES12_b03      | rate rank                | pinned, e4       | 53×1   |
| raterank_RECIPES12_b05      | rate rank                | pinned, e4       | 53×1   |
| hybrid_RECIPES12_b01_fixe d | partial signalling       | pinned, e4       | 53×1   |
| hybrid_raterank_b01         | rate rank, second run    | pinned, e4       | 53×1   |
| hybrid_v3_b01               | rate rank, second run    | R512/eval        | 53×1   |
| latency_RECIPES12_sorted    | latency                  | R512/eval        | -      |
| router_latency              | operating range          | R512/eval        | -      |
| adapter_ablation            | adapter ablation         | pinned, e4       | 16 fr  |
| combined_RECIPES12_b01      | blend                    | pinned, e4       | 53×1   |
| hybrid_RECIPES12_b03_fixe d | macros only              | pinned, e4       | 53×1   |
| hybrid_lorenz_b01           | macros only              | R512/eval        | 53×1   |
| coupling_ablation           | macros only              | pinned, e4       | 16 fr  |
| map_transfer                | macros only              | pinned, e4       | -      |
| band_collapse               | macros only              | none             | -      |
| band_collapse_BEST          | macros only              | none             | -      |
| router_retrain_compare      | macros only              | none             | -      |
| signalled_BEST_ctc53        | ladder configurations    | BEST/eval, e1    | 53×1   |
| signalled_BEST128_ctc53     | ladder configurations    | BEST128/step, e1 | 53×1   |
| signalled_FINE12_ctc53      | ladder configurations    | FINE12/step, e1  | 53×1   |
| ctc_seam_p256               | padding, tile size       | none             | -      |
| ctc_seam_p128               | tile size                | none             | -      |
| raterank_BEST_compare       | rate rank, second run    | none             | -      |
| bdrate                      | BD-rate                  | none             | -      |
| check_paper                 | claim check              | none             | -      |

**Table 37.** Provenance of every generated table in the main paper. These are the 31 results files scripts/make\_paper\_tables.py reads, and the checkpoint column is read from each file at build time rather than typed. Take from it that 16 of the 31 are on the pinned checkpoint and the rest are not, so a reader comparing two rows of two different tables should check this column first. R512 abbreviates the run RECIPES12, "eval" is a per-epoch file the watcher overwrites in place, and "none" means the file records no checkpoint field.

The pinned group of Table 37 carries the headline saving, the operating range, the per-class breakdown, the static controls and both routed configurations, so the paper's central claim rests on one epoch of one run measured consistently. Section H states what the rest of the column costs in reproducibility.

## A.15. Seeds, variation, and the interval that does exist

No seed is set anywhere in the decoder's trainer. Four things vary between two runs of one command: the sampler is reshuffled every epoch from the global generator, the quality index is drawn per sample, the tiled training path draws a fresh random exit for every tile at every step, and cuDNN selects kernels by autotuning. The honest statement is that the recipe reproduces and the run does not, and this work has no repeated run of one configuration from which a training-side spread could be quoted. The router experiments are the exception and were built to be: every variant of the input ablation fixes seed 0 and shares architecture, parameter count, optimiser, image order and quality-index draws, so the only difference between them is which inputs the head may see, and their agreement figures carry a standard error across 1,200 frames.

| quality index | mean  | sd    | min   | median | max   | worst sequence |
|---------------|-------|-------|-------|--------|-------|----------------|
| q0            | 36.83 | 3.71  | 24.22 | 38.75  | 39.12 | BQSquare       |
| q16           | 33.14 | 8.15  | -0.95 | 36.14  | 39.12 | BQSquare       |
| q32           | 29.52 | 10.20 | -0.95 | 32.63  | 39.12 | BQSquare       |
| q48           | 27.85 | 10.70 | -0.95 | 31.09  | 39.12 | BlowingBubbles |
| q63           | 26.69 | 10.77 | -0.95 | 29.87  | 39.12 | BQSquare       |

**Table 38.** The interval this work can honestly report: saving in percent against the released decoder at the 0.1 dB budget, across the 53 test sequences, on the pinned checkpoint. Take from it that the spread is content and not noise. It widens with rate, from a standard deviation of 3.7 points at q0 to 10.8 at q63, and the minimum turns negative at the high rates because a sequence whose tiles all stay at the deepest exit costs about 1% more than the released decoder rather than less. This measures the test set, not the training draw, and is not a substitute for the run-to-run interval that does not exist.

Source: results/supp\_per\_sequence\_PAPER\_b010.json, the rows\_vs\_release block, computed from results/supp\_paper\_curve\_PAPER.json on runs/RECIPES12/ckpt\_PAPER.pth.tar. The file carries the same spread in our own deepest-exit denominator as well; the two differ by more than rounding and are never mixed.

| quality index | path 1 | path 2 | gap  |
|---------------|--------|--------|------|
| q0            | 36.08  | 33.68  | 2.41 |
| q16           | 31.19  | 28.40  | 2.79 |
| q32           | 24.80  | 22.28  | 2.52 |
| q48           | 19.38  | 17.44  | 1.94 |
| q63           | 11.66  | 11.48  | 0.18 |

**Table 39.** Two implementations of the same quantity do not agree. Saving in percent at the 0.1 dB budget, computed by the curve-sweeping path and by the signalling path, which differ in how they pool distortion across tiles and frames. Take from it a floor on protocol sensitivity: the gap reaches 2.79 points, larger than several of the differences the paper draws conclusions from, so numbers from the two paths are never mixed inside one table.

Source: results/crosscheck\_paths.json. The file records no checkpoint field.

Evaluation itself is deterministic once a checkpoint is fixed: the sequence list, the leading frames and the tile grid are all fixed and no crop is random. scripts/check\_paper.py re-reads 88 numerical claims out of the paper's prose and checks each against the file it is supposed to come from. As recorded in results/check\_paper.json, 82 of them pass. One does not: "band: raw spread at 0.1 dB", which is listed among the limitations of section H rather than presented as passing.

## A.16. Data, code and provenance

- **Training images.** Open Images [22], from the CVDF public mirror named in scripts/fetch\_openimages.sh. The terms are the ones that mirror publishes and nothing here redistributes them.
- **Test sequences.** UVG [23] from ultravideo.fi, MCL-JCV [24] and HEVC classes B, C and D from the public mirrors named in scripts/fetch\_mcljcv.sh and scripts/fetch\_hevc\_bcd.sh, and HEVC class E from media.xiph.org. Each fetch script records its source URL, and no sequence is re-encoded before it is measured.
- **The released codec.** The DCVC-UF intra checkpoint [12], identified by sha256 in results/dmc\_ld\_recon\_audit.json and used under the licence its repository ships.

- **Our code and checkpoints.** The trainer, the ladder, the cost model, the meter, the router and every evaluation script named here are in the repository this supplement is built from; the pinned checkpoint is identified by md5 in A.14.
- **Human subjects.** None, which is why A.4 declines to claim that 0.1 dB is invisible.

## B. Architecture and complexity accounting

This section states what the decoder is, module by module and convolution by convolution, derives what a decode costs from those shapes, and then checks the derivation against counts taken off decodes that were run. It closes with what the saving is worth in the units a deployment measures, which are milliseconds, joules, megabytes and frames per second, and none of which is the unit the allocation is optimised in.

Two units first. A *multiply-accumulate* (MAC) is one multiply and one add, and MAC/px is per pixel of the grid the operator runs on, which is not the same grid for every operator. The *released decoder* is DCVC-UF's intra decoder [12] with none of this work added, and it is the unit every cost below is quoted in. The ladder's geometry is section A's:  $N = 12$  trunk blocks at width  $C = 384$ ,  $K = 6$  exits one every  $b = 2$  blocks, a split depth of  $j = 2$  that leaves the first four blocks full-frame, and a tile of 256 RGB px.

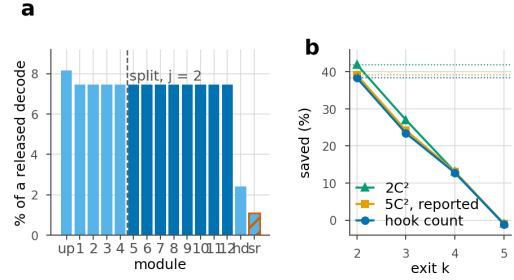
### B.1. The decoder, module by module

The released intra decoder is 14,231,808 parameters in 72 convolutions and nothing else: no attention, no normalisation with parameters of its own on the critical path, and no operator that changes the spatial grid except two pixel shuffles. It has three parts. An *upsample* takes the decoded latent, widens it with a  $1 \times 1$  and doubles its grid with a pixel shuffle, then runs one DepthConvBlock at  $C = 384$ . A *trunk* of 12 further DepthConvBlocks runs at that width and grid throughout. A *head* narrows to 192 channels with a  $1 \times 1$ , runs one more DepthConvBlock at that narrower width, and pixel-shuffles by 8 straight to RGB, with no convolution after the shuffle. Table 40 is that pipeline with its shapes.

| Module                  | Where        | Grid      | Out ch. | MAC/px    | % dec. |
|-------------------------|--------------|-----------|---------|-----------|--------|
| Latent y, decoder input | entropy dec. | 68×120    | 256     | 0         | 0.00   |
| 1×1, widen              | upsample     | 68×120    | 1536    | 393,216   | 0.71   |
| PixelShuffle 2          | upsample     | 136×240   | 384     | 0         | 0.00   |
| DepthConvBlock          | upsample     | 136×240   | 384     | 1,035,648 | 7.45   |
| DepthConvBlock × 12     | trunk        | 136×240   | 384     | 1,035,648 | 89.44  |
| 1×1, narrow             | head         | 136×240   | 192     | 73,728    | 0.53   |
| DepthConvBlock          | head         | 136×240   | 192     | 259,776   | 1.87   |
| PixelShuffle 8          | head         | 1088×1920 | 3       | 0         | 0.00   |
| <b>Ours, added</b>      |              |           |         |           |        |
| Conv1x1Adapter          | exit 4       | 32×32     | 384     | 147,456   | 1.06   |
| FFNAdapter              | exits 0 to 3 | 32×32     | 384     | 737,280   | 5.31   |
| GridSeamRepair          | post-stitch  | 136×240   | 384     | 150,912   | 1.09   |
| Router head             | stem         | per tile  | $K = 6$ | n/a       | 0.163  |

**Table 40.** The decode, module by module, at a 1920×1080 frame padded to 1920×1088. Grids are height × width, and “Where” names the released decoder's three parts: the upsample dec<sub>1</sub>[0], the trunk dec<sub>1</sub>[1..12] and the head dec<sub>2</sub>. The last column is the module's share of one released decode, and for the two adapters it is the share if every tile wears that adapter. Take from the table that the grid changes exactly twice and never inside the trunk, so an exit ladder cut into the trunk needs no resampling; that the head is cheap because it runs at 192 channels rather than 384, which is a quarter of the arithmetic per pixel; and that the FFN adapter is the most expensive thing this work adds, at 5.31% of a decode, more than half a trunk block.

Released modules and their shapes: results/mac\_audit.json, forward hooks on every convolution of an IntraDecoder at the real tensor shapes. A MAC count depends on shapes and not on values, so no weights are loaded for it. Adapters and seam repair: results/adapter\_cost.json, hooks on the modules of runs/RECIPES12/ckpt\_PAPER.pth.tar. Router: results/router\_latency.json, measured on runs/RECIPES12/ckpt\_eval.pth.tar; the head is the same size in both checkpoints.



**Figure 35.** Where the arithmetic is, and what an exit saves. **a**, one released decode by module: the opening upsample, the twelve trunk blocks, the head, and the seam-repair pass this work adds (hatched). Light bars always run; dark bars are the eight blocks after the split that a tile can skip. **b**, the saving at each reachable exit under three prices for the same architecture, with each price's ceiling dotted. Section B.5 is the table behind it.

### B.2. Every convolution in the decode

Table 41 opens up the modules of Table 40, giving every convolution in the decode in the  $(K, C_{in}, C_{out}, S)$  convention that the architecture figures of this literature use, with the group count added because two of the shapes are depthwise. The 72 convolutions of the released decoder have only 8 distinct shapes between them, because the DepthConvBlock repeats.

| Convolution          | K | $C_{in}$ | $C_{out}$ | S | Grp | MAC/px  | Used |
|----------------------|---|----------|-----------|---|-----|---------|------|
| up.conv.0            | 1 | 256      | 1536      | 1 | 1   | 393,216 | 1    |
| dc.0, 1×1 in         | 1 | 384      | 384       | 1 | 1   | 147,456 | ×13  |
| dc.2, 3×3 depthwise  | 3 | 384      | 384       | 1 | 384 | 3,456   | ×13  |
| dc.3, 1×1 out        | 1 | 384      | 384       | 1 | 1   | 147,456 | ×13  |
| ffn.0, 1×1 C→4C      | 1 | 384      | 1536      | 1 | 1   | 589,824 | ×13  |
| ffn.2, 1×1 4C→C      | 1 | 384      | 384       | 1 | 1   | 147,456 | ×13  |
| dec_2.adapter        | 1 | 384      | 192       | 1 | 1   | 73,728  | 1    |
| head dc.0            | 1 | 192      | 192       | 1 | 1   | 36,864  | 1    |
| head dc.2, depthwise | 3 | 192      | 192       | 1 | 192 | 1,728   | 1    |
| head dc.3            | 1 | 192      | 192       | 1 | 1   | 36,864  | 1    |
| head ffn.0           | 1 | 192      | 768       | 1 | 1   | 147,456 | 1    |
| head ffn.2           | 1 | 192      | 192       | 1 | 1   | 36,864  | 1    |
| Conv1x1Adapter       | 1 | 384      | 384       | 1 | 1   | 147,456 | 1    |
| FFNAdapter pw_in     | 1 | 384      | 1536      | 1 | 1   | 589,824 | ×4   |
| FFNAdapter pw_out    | 1 | 384      | 384       | 1 | 1   | 147,456 | ×4   |
| Seam repair, 3×3 dw  | 3 | 384      | 384       | 1 | 384 | 3,456   | 1    |
| Seam repair, 1×1     | 1 | 384      | 384       | 1 | 1   | 147,456 | 1    |

**Table 41.** Every distinct convolution in the decode. K is the kernel side, S the stride, Grp the group count; MAC/px is per pixel of that operator's own grid, which is the latent grid for the first row and the feature grid for the rest. “Used” counts the instances in one decode: the DepthConvBlock appears 13 times, once in the upsample and 12 times in the trunk, and there are  $K - 1 = 5$  adapters of which four are of the FFN kind. Take from it that only two of the shapes have a kernel wider than  $1 \times 1$  and both are depthwise, so the 14 instances of them in one decode carry 0.34% of the arithmetic between them.

results/mac\_audit.json for the released rows, results/adapter\_cost.json for ours. The two depthwise convolutions are the whole of the decoder's spatial footprint, which is why a tile boundary is cheap to repair.

That last figure is the premise of the whole method, so it is worth stating separately: 99.66% of the released decoder's arithmetic is  $1 \times 1$  convolution, which has no neighbours to miss. The receptive field of the trunk plus head grows by exactly one feature pixel per block, from 1 feature pixel with no trunk blocks to 13 with all 12, which is 104 RGB px per tile edge (results/dmc\_ld\_rf\_probe.json, measured on the released DCVC checkpoints rather than ours). A tile decoded alone is therefore wrong only in a band, and the band is narrower for a tile that exits early.

The two DepthConvBlock widths obey the same closed form. Writing the block as a  $1 \times 1$  projection, a  $3 \times 3$  depthwise, a second  $1 \times 1$  and a feed-forward pair whose first  $1 \times 1$  expands to 4C and whose gated activation folds 4C back to C for free, the closing  $1 \times 1$  is  $C \rightarrow C$  rather than  $4C \rightarrow C$  and the block is

$$M_{\text{block}}(C) = C^2 + 9C + C^2 + 4C^2 + C^2 = 7C^2 + 9C, \quad (9)$$

which is 1,035,648 MAC/px at  $C = 384$  and 259,776 at  $C = 192$ , both matching Table 41 row by row. The arithmetic model in `flexuf/cost.py`, which runs inside the allocation's `argmin` and nowhere else, writes the block as  $8C^2 + 9C = 1,183,104$  instead, so it prices a block 14.24% high. Section B.6 measures what that is worth against a hook count.

### B.3. What the ladder adds

Three modules are new. An exit that stops early hands the head a feature map the remaining blocks would have refined, and an *adapter* stands in for them. The  $1 \times 1$  adapter is  $f + Wf$  with  $W$  a  $C \times C$  matrix. The FFN adapter repeats the block's feed-forward pair, a  $1 \times 1$  expanding  $C \rightarrow 4C$ , the same gated activation, and a  $1 \times 1$   $C \rightarrow C$ . Both are zero-initialised, so the ladder is the released decoder exactly at step zero and any quality the adapters add is attributable to them. Neither has a spatial footprint, so neither widens the seam. The *seam repair* is a gated depthwise  $3 \times 3$  followed by a zero-initialised  $1 \times 1$ , run once on the stitched canvas after unpatchify and before the head, which is the only place a  $3 \times 3$  can see across a tile boundary. The *router head* reads the shared stem and predicts a per-tile exit; Section F is its architecture and this section only prices it.

| Module               | Operators                                                        | MAC/px  | In $C^2$ | Blocks | Of decode |
|----------------------|------------------------------------------------------------------|---------|----------|--------|-----------|
| $1 \times 1$ adapter | $1 \times 1$ $C \rightarrow C$                                   | 147,456 | 1.00     | 0.1424 | 1.061%    |
| FFN adapter          | $1 \times 1$ $C \rightarrow 4C$ , $1 \times 1$ $C \rightarrow C$ | 737,280 | 5.00     | 0.7119 | 5.306%    |
| Seam repair          | $3 \times 3$ dw, $1 \times 1$ $C \rightarrow C$                  | 150,912 | 1.02     | 0.1457 | 1.086%    |
| Router head          | two $1 \times 1$ , then an MLP                                   | n/a     | n/a      | n/a    | 0.163%    |

**Table 42.** What the ladder adds, priced against one released decode. “Blocks” is the module in units of a trunk block. Take from it that the FFN adapter costs  $5C^2$  per pixel: the gated activation is free, so the pair is  $4C^2 + C^2$  rather than  $4C^2 + 4C^2$ , and the module is 0.71 of a whole block. The routed allocation lives at exits 2 and 3, which both wear it, so it is priced where the reported saving leans hardest.

`results/adapter_cost.json` for the two adapters, on the pinned checkpoint. The seam-repair row is  $9C + C^2$  per pixel; Section B.6 confirms that figure against a hook count rather than taking it from the source. Router: `results/router_latency.json`.

The ladder assigns the FFN adapter to any exit skipping four or more blocks and the  $1 \times 1$  adapter otherwise, which at  $K = 6$  and  $b = 2$  means the FFN at exits 0 to 3, the  $1 \times 1$  at exit 4, and no adapter at exit 5, whose feature is the trunk's own output with nothing standing in for a skipped block. The arithmetic at exit 5 is therefore the released decoder's, plus the repair pass, which is why the deepest exit costs slightly more than the release rather than the same.

### B.4. From the block to the decoder

Write  $s_{\text{up}}$ ,  $s_{\text{trunk}}$  and  $s_{\text{head}}$  for the shares of a released decode taken by the upsample, the trunk and the head;  $a_k$  for the adapter at exit  $k$ ; and  $r$  for the seam-repair pass. A tile assigned an exit shallower than  $j$  still runs group  $j$ , because the decoder clamps the map, so the cost of exit  $k$  in units of one released decode is

$$C_k = s_{\text{up}} + s_{\text{head}} + r + s_{\text{trunk}} \frac{(\max(k, j) + 1)b}{N} + a_{\max(k, j)}, \quad (10)$$

and the frame-level cost of an exit map  $a$  over  $T$  tiles amortises the shared part over the frame rather than over the tile,

$$C(a) = s_{\text{up}} + s_{\text{trunk}} \frac{jb}{N} + s_{\text{head}} + r + \frac{1}{T} \sum_{i=1}^T u_{a(i)}, \quad (11)$$

with  $u_k$  the per-tile suffix, meaning the blocks after the split plus the adapter. Only the last term depends on the map. That is the structural reason a ladder over a decoder of this shape has a ceiling well below 100%: at the split depth the upsample, the first  $jb = 4$  blocks, the head and the repair pass are already 41.5% of a released decode, and they run whatever the map says.

None of the four shares has to be taken on trust. `results/mac_audit.json` gives  $s_{\text{up}} = 0.0816$ ,  $s_{\text{trunk}} = 0.8944$  and  $s_{\text{head}} = 0.0240$  by hook count, so

one trunk block is 0.074533 of a decode. That figure has an independent check. `results/static_RECIPES12_b01.json` records the frame-level saving of a map that sends every tile to exit  $e$ , for  $e = 2$  to 5, on the pinned checkpoint; differencing consecutive entries gives one trunk block as 0.074533, which agrees with the audit to  $7 \times 10^{-7}$  of a decode. One is a shape audit of the released decoder with no weights loaded and the other is a set of measured savings on ours, which is the reason to report both.

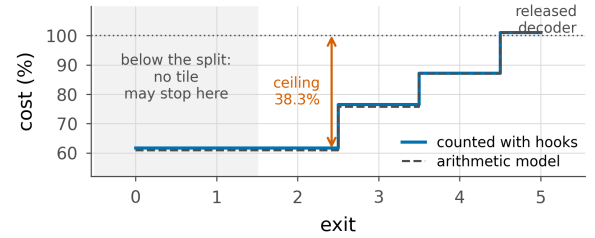
### B.5. The per-exit cost vector and the ceiling

| Exit $k$    | Blocks | Adapter      | c model | c hook | Saved % |
|-------------|--------|--------------|---------|--------|---------|
| 0 (clamped) | 6      | FFN          | 0.6088  | 0.6167 | 38.33   |
| 1 (clamped) | 6      | FFN          | 0.6088  | 0.6167 | 38.33   |
| 2           | 6      | FFN          | 0.6088  | 0.6167 | 38.33   |
| 3           | 8      | FFN          | 0.7578  | 0.7658 | 23.42   |
| 4           | 10     | $1 \times 1$ | 0.8697  | 0.8724 | 12.76   |
| 5           | 12     | none         | 1.0095  | 1.0109 | -1.09   |

**Table 43.** The per-exit cost vector, in units of one released decode, priced two ways: the arithmetic model the allocation's `argmin` runs on, and the hook count assembled in Section B.4, which is what every saving in the paper is quoted from. The two differ by most at the shallowest reachable exit, which wears the FFN adapter, and by least at exits 4 and 5, which wear the cheap adapter or none. The deepest exit costs 1.09% more than the release, which is the seam-repair pass.

Exits 0 and 1 are unreachable: the decoder clamps the map at  $j = 2$ , so a tile nominally assigned them leaves through exit 2 and wears exit 2's cost. The model vector is recorded in `results/why_qp_PAPER.json`; a cost vector is a property of the model rather than of the weights.

The ceiling is  $100(1 - c)$ , the saving when every tile takes the shallowest reachable exit. It is a property of the architecture and no allocation can pass it. The model reads 39.1% and the meter 38.33%, which is the value the paper's tables use, and the next subsection is the demonstration.



**Figure 36. The ladder, priced.** What a tile costs at each exit, as a percentage of one released decode, counted with hooks on the executed pass and beside the arithmetic model the `argmin` uses. Exits below the split depth are not distinct, because the first  $j$  groups run for every tile whatever it does. The deepest exit costs slightly more than the release, which is the full-frame deblocking pass. Reported savings in the paper are the hook count.

### B.6. Predicted against hook-measured

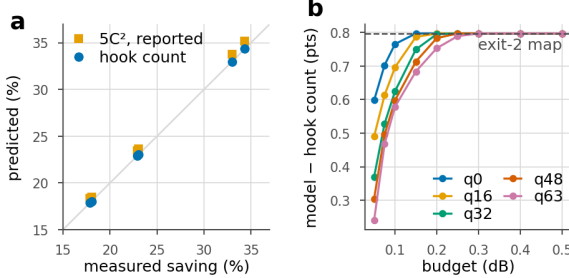
A cost model and the decoder it models drift apart. The check that catches it is a hook count: `flexuf/measure.py` attaches a forward hook to every convolution and linear layer and accumulates MACs from the output shape of each call. A group skipped by an early exit never fires its hook and so costs nothing; a group run on 28 of 40 tiles costs 28 tiles' worth, because tiles are the batch dimension and the output shape carries them. There is no bookkeeping to get wrong, and the count is the ground truth the model is judged against.

| Condition | Map 2/3/4/5 | Measured | 5C <sup>2</sup> | diff   | Hook    | diff    |
|-----------|-------------|----------|-----------------|--------|---------|---------|
| 720p q0   | 11/4/0/0    | 34.3524  | 35.1494         | +0.797 | 34.3524 | +0.0000 |
| 720p q32  | 6/4/2/3     | 23.0606  | 23.6547         | +0.594 | 23.0606 | +0.0000 |
| 720p q63  | 4/3/4/4     | 18.0177  | 18.4971         | +0.479 | 18.0177 | +0.0000 |
| 1080p q0  | 28/10/1/1   | 32.9763  | 33.7436         | +0.767 | 32.9763 | +0.0000 |
| 1080p q32 | 16/10/6/8   | 22.8829  | 23.4682         | +0.585 | 22.8829 | +0.0000 |
| 1080p q63 | 11/7/11/11  | 17.8489  | 18.3184         | +0.470 | 17.8489 | +0.0000 |

**Table 44.** Predicted against measured, on six decodes with six different exit maps at two resolutions. “Measured” is the hook count off the decode that was executed and timed. Take from it that the assembly of Section B.4 reproduces the count exactly in every condition, to four decimal places and in fact to floating point, while the model the paper reports is optimistic by 0.47 to 0.80 points and always in the same direction.

results/supp\_power.json, pinned checkpoint, 0.1 dB budget, NVIDIA RTX A6000. The map is the file’s pooled exit histogram rounded to the frame’s tile count, which is what scripts/power\_profile.py decodes; its four columns are the tiles leaving at exits 2, 3, 4 and 5.

Because a hook count is exactly linear in the exit map, six mixtures spanning four reachable exits over-determine the per-exit vector and it can simply be solved for. The six equations are consistent to  $3 \times 10^{-16}$ , and the solution differs from the assembled vector by at most  $2 \times 10^{-15}$  at any exit. The identity is worth stating plainly: the per-module counts of results/mac\_audit.json and results/adapter\_cost.json are not a model of what the meter reports, they are what the meter reports, rearranged. The residual against the model is therefore not noise and not a tolerance but a closed form: three modules are priced as a fraction of a block, the model’s block is  $8C^2 + 9C$  where the meter’s is  $7C^2 + 9C$ , and the mispricing is largest at the exits wearing the FFN adapter, which are exactly the exits the saving leans on. It comes to 0.7970 points of saving at exits 2 and 3 and 0.1354 at the deepest.



**Figure 37. The model against the meter.** **a**, six decodes at two resolutions: the hook-count assembly lies on the identity line and the model the paper reports lies above it. **b**, the same residual over the whole budget grid, 9 budgets by 5 quality indices on 53 sequences. It is positive everywhere, grows with the budget, and saturates at the value an all-exit-2 map implies.

Figure 37b runs the same comparison over the whole operating range rather than six points of it. The residual is positive in all 45 cells, from 0.241 to 0.797 points, and it saturates at exactly that exits-2 value. The paper’s headline figures are the hook counts, so no subtraction is left for a reader to do; had the model been quoted instead, every figure would have been between 0.58 and 0.77 points higher. At the 0.1 dB budget the model reads 34.46% at the lowest rate where the meter reads 33.7%, and 24.10% at the highest where the meter reads 23.5%.

results/signalled\_RECIPES12\_grid.json and results/signalled\_RECIPES12\_ctc53.json, both on the pinned checkpoint. Both columns are in the files: saving\_pct\_vs\_release is the model, saving\_pct\_measured is the hook count, and model\_minus\_measured is their difference.

| Decoder                        | GMAC | % of release | ms  |
|--------------------------------|------|--------------|-----|
| Released DCVC-UF               | 454  | 100.0        | 111 |
| FLEX-UF, all tiles deepest     | 458  | 101.0        | 114 |
| FLEX-UF, 0.1 dB budget         | 328  | 72.4         | 78  |
| FLEX-UF, architectural ceiling | 263  | 58.1         | –   |

**Table 45. Decoder complexity at 1080p**, in the three units a reader is likely to want: arithmetic, arithmetic as a fraction of the release, and milliseconds. Our deepest exit costs slightly more than the released decoder because it still pays the deblocking filter, and that 1.0095, not 1.0, is what every saving in the paper is *not* divided by. Wall-clock is the sorted per-tile loop.

results/signalled\_RECIPES12\_ctc53.json and results/latency\_RECIPES12\_sorted.json, generated into paper/tables/complexity.tex by scripts/make\_paper\_tables.py. The architectural-ceiling row has no wall clock because no allocation on this test set reaches it at the budgets measured here.

## B.7. How the timings were taken

The rest of this section leaves arithmetic for the clock, so the protocol comes first. Everything below follows the same rules and they are stated rather than implied. Every timing, power and memory figure in this section is at the 0.1 dB budget, and every GPU figure is an NVIDIA RTX A6000, all eight cards in this machine being that model; the CPU rows are the only second device class this work can report.

- **Batch 1, and why.** Every wall clock here is one frame at a time, which is what a decoder does. A batch is genuinely different work for this decoder and not a repetition, because exit\_map is indexed over the whole batch’s tiles, so B frames of 40 tiles form one shrinking active set of 40B and the groups run wider. Section B.8 measures what the choice costs.
- **Warm-up, then interleaving, then the median.** 20 untimed iterations, then 40 rounds of one call of each variant in turn. This is a shared card, and timing all of the released decoder and then all of ours puts any drift in a neighbouring job’s load straight into the ratio, which is the only quantity these runs exist to produce.
- **What the timer wraps.** time.perf\_counter around a synchronised call, not CUDA events, because events do not exist on the CPU and the two device classes have to be timed alike. The timer starts at the decoded latent and stops at the reconstruction, so it includes patchify, every group, unpatchify, the seam-repair pass and the head.
- **What is deliberately outside it.** Entropy decoding, because the bitstream and therefore the rate do not depend on the exit map: including it would divide both sides of every ratio by the same constant and flatter the method. The map itself is charged, in bits rather than in time, at 80 bits per frame.
- **The router is charged in MACs and not in seconds.** Its arithmetic is 0.163% of a decode but 0.50% of decode time, 3.0× its arithmetic share, an extra 0.33 points bought by launching a small kernel over a large map. That is the general reason a MAC count cannot see a kernel launch, in miniature (results/router\_latency.json, 40 iterations at 2048×1280 on an NVIDIA RTX A6000, measured on runs/RECIPES12/ckpt\_eval.pth.tar). The router does not run at all in the signalled configuration, where the encoder chooses the map.
- **The masked path, not the sorted one.** The shipped configuration leaves sorted\_tiles off, so every group is a masked gather over the full tile batch. Sorting the tiles by depth turns each group into a contiguous slice and recovers 1.2 points at 1080p for bit-identical output (results/latency\_RECIPES12\_sorted.json, on runs/RECIPES12/ckpt\_eval.pth.tar). None of the timings below use it, so they are the slower of the two implementations.

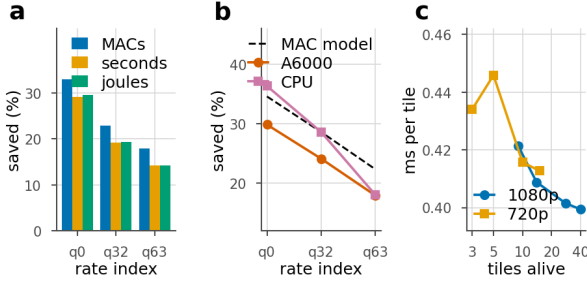
## B.8. Where the seconds go, and on what device

| Stage                        | MAC % | 1080p tiles | 1080p ms | 720p tiles | 720p ms |
|------------------------------|-------|-------------|----------|------------|---------|
| Stem: upsample + groups 0, 1 | 37.97 | frame       | 41.39    | frame      | 16.30   |
| Patchify                     | 0.00  |             | 0.21     |            | 0.09    |
| Group 2 (2 blocks, per tile) | 14.91 | 40          | 15.98    | 15         | 6.19    |
| Group 3 (2 blocks, per tile) | 14.91 | 28          | 11.24    | 10         | 4.16    |
| Group 4 (2 blocks, per tile) | 14.91 | 14          | 5.72     | 5          | 2.23    |
| Group 5 (2 blocks, per tile) | 14.91 | 9           | 3.79     | 3          | 1.30    |
| Unpatchify                   | 0.00  |             | 0.21     |            | 0.09    |
| Seam repair (full-frame)     | 1.09  | frame       | 2.07     | frame      | 0.83    |
| Head                         | 2.40  | frame       | 4.36     | frame      | 1.72    |

**Table 46.** Where a routed decode goes, in arithmetic and in milliseconds. The MAC column is each stage’s share of a released decode at full occupancy and sums to 101.09%, which is the deepest exit’s cost. Take from the table that patchify and unpatchify are free arithmetically and nearly free on the clock, that the repair pass is 1.09% of the arithmetic and 2.4% of this decode’s clock, and that the head takes 5.1% of the clock for 2.40% of the arithmetic, being bandwidth-bound rather than arithmetic-bound.

results/supp\_latency\_1920x1080.json and results/supp\_latency\_1280x720.json, pinned checkpoint, q32, 0.1 dB budget, NVIDIA RTX A6000 taken under the evaluation lock so the card was not shared. Tile counts are that decode’s own exit map, which leaves 28, 14 and 9 of 40 tiles alive in groups 3, 4 and 5.

Patchify and unpatchify are reshapes and cost no arithmetic at all. They require the feature map to divide by the 32 px feature tile, which the encoder-side replicate padding guarantees: 1920×1080 becomes 2048×1280 and 40 tiles, 1280×720 becomes 1280×768 and 15 (results/tile\_definition.json).



**Figure 38. What the arithmetic is worth.** **a**, one exit map measured three ways at 1080p. **b**, the same map on two device classes against what the MAC model predicts: the A6000 falls short of the prediction at every rate, the CPU does not. **c**, why a GPU falls short: a tile costs more to decode in a group that has fewer tiles left in it.

Two effects separate seconds from multiply-accumulates and the figure isolates both. The first is the tiled path itself, which the released decoder does not pay: patchify, unpatchify and the repair pass are 2.9% of the routed clock at 1080p. Timing our own decoder with every tile at the deepest exit isolates it: that variant does the same arithmetic as the release plus the tiling machinery, and it is 2.6 to 3.8% slower across nine measurements on the pinned checkpoint, against 3.6% for the figure the main paper quotes. The second is occupancy. A group late in the ladder launches kernels over a handful of 32×32 feature maps, and a GPU sized for the first group is partly idle inside them. Dividing each group of Table 46 by the tiles alive in it, at 1080p a tile costs 5.5% more in group 5, on 9 tiles, than in group 2 on 40. At 720p the same curve rises by 8.0% from 15 tiles to 5 and then falls back at 3, so the trend is clear at 1080p and not monotone at 720p.

Device class and batch size are the two knobs a reader would ask about next, and Table 47 sweeps both. All eight cards in this machine are RTX A6000s, so the only second device class reachable here is the CPU. It is also the unflattering comparison to make, in the sense that it is the one our own explanation predicts we will look better on: a CPU decode is closer to arithmetic-bound, so if the shortfall on the GPU is scheduling rather than arithmetic it should shrink.

| Device      | Rate | Batch | ms/frame rel. | ms/frame ours | fps ours | Saved % | MAC model % |
|-------------|------|-------|---------------|---------------|----------|---------|-------------|
| A6000       | q0   | 1     | 111.3         | 78.1          | 12.8     | 29.84   | 34.58       |
| A6000       | q0   | 2     | 111.1         | 76.7          | 13.0     | 30.97   | 34.58       |
| A6000       | q0   | 4     | 110.7         | 76.5          | 13.1     | 30.96   | 34.58       |
| A6000       | q32  | 1     | 111.5         | 84.6          | 11.8     | 24.11   | 28.59       |
| A6000       | q32  | 2     | 111.3         | 83.3          | 12.0     | 25.16   | 28.59       |
| A6000       | q32  | 4     | 110.8         | 82.9          | 12.1     | 25.17   | 28.59       |
| A6000       | q63  | 1     | 111.5         | 91.5          | 10.9     | 17.92   | 22.33       |
| A6000       | q63  | 2     | 111.2         | 89.9          | 11.1     | 19.14   | 22.33       |
| A6000       | q63  | 4     | 110.7         | 89.6          | 11.2     | 19.06   | 22.33       |
| CPU, 8 thr. | q0   | 1     | 8836          | 5625          | 0.178    | 36.34   | 34.58       |
| CPU, 8 thr. | q32  | 1     | 8153          | 5822          | 0.172    | 28.59   | 28.59       |
| CPU, 8 thr. | q63  | 1     | 8280          | 6788          | 0.147    | 18.02   | 22.33       |

**Table 47.** Wall clock at 1080p by device class and batch size, on one exit map per rate. “rel.” is the released decoder timed in the same loop; fps is one over our per-frame time. Take from it that batch 1 costs about a point of saving against batch 4 and nothing beyond it, so the batch-1 convention this work uses is close to free; and that the A6000 falls short of the MAC model by 3.2 to 4.7 points at every rate and every batch size, while the CPU meets or beats it at the two lower rates.

results/supp\_latency\_batch\_1920x1080.json and results/supp\_latency\_cpu\_1920x1080.json, pinned checkpoint, 0.1 dB budget, exit maps read from results/supp\_paper\_curve\_PAPER.json. GPU rows: 20 warm-up and 40 interleaved iterations. CPU rows: 8 threads, 1 warm-up and 5 iterations only, on a shared machine.

The CPU rows carry a noise floor and it should be read before the conclusion. The released decode does not depend on the rate, so its three timings should be equal; they range from 8.15 to 8.84 s, a spread of 8.4%, which is what 5 iterations on a shared machine buys. That is enough to support the gap at q0 and q32, where the CPU is 6.5 and 4.5 points above the A6000 on the same map, and not enough to support anything at q63, where the two agree to a tenth of a point. The direction is the one the occupancy explanation predicts, and a firmer statement needs a machine that is not shared.

## B.9. What the saving is worth in joules

Board power was sampled from the driver at 100 ms, which is about the rate the sensor updates. Each condition therefore runs for a fixed wall time of 20 s rather than a fixed iteration count, so that a 30 ms decode yields two hundred power samples rather than three. Idle is measured for 4 s immediately before and after every timed run on the same card, so a neighbouring process waking up mid-measurement appears as a drift between the two rather than as a saving. Energy per frame is the mean board power over the run times the median decode time.

| Condition | MACs % | Time % | Gap  | Energy % | Above idle % |
|-----------|--------|--------|------|----------|--------------|
| 720p q0   | 34.35  | 30.81  | 3.54 | 30.10    | 29.70        |
| 720p q32  | 23.06  | 18.13  | 4.93 | 18.45    | 18.63        |
| 720p q63  | 18.02  | 12.99  | 5.03 | 12.96    | 12.94        |
| 1080p q0  | 32.98  | 29.07  | 3.91 | 29.56    | 29.88        |
| 1080p q32 | 22.88  | 19.14  | 3.75 | 19.34    | 19.48        |
| 1080p q63 | 17.85  | 14.14  | 3.71 | 14.17    | 14.18        |

**Table 48.** The three units side by side, all as savings of the routed decode against the released decode on the same latent. Take from it the result of this subsection: energy saved tracks time saved to within 0.5 points in every condition, while arithmetic saved runs 3.5 to 5.0 points ahead of both. Subtracting idle power changes nothing, which is the check that the agreement is not an artefact of the static draw.

results/supp\_power.json, pinned checkpoint, 0.1 dB budget, NVIDIA RTX A6000 with a 300 W limit, taken under the evaluation lock. 720p is padded to 1280×768 and 15 tiles, 1080p to 2048×1280 and 40 tiles.

The engineering result should be stated plainly: the energy saving of this method tracks its time saving, not its arithmetic saving. A partly idle GPU still draws most of its static power, so the joules follow the seconds almost exactly, and the seconds fall short of the multiply-accumulates for the occupancy reason of Figure 38c. The routed decode draws the same board power as the released one, within 3 W in the worst of the six conditions and within 1 W in four of them, so routing does not lower the instantaneous draw of the card; it shortens the job. At 1080p and the

lowest rate the three read 32.98%, 29.07% and 29.56%; at the highest rate they read 17.85%, 14.14% and 14.17%. Quoting the arithmetic figure as an energy figure overstates it by three and a half to five points.

Two limits. The idle baseline drifts upward as the card warms through the six conditions, so the above-idle column of Table 48 is the weaker of the two energy figures, and it is only because it agrees with the unadjusted column to within 0.4 points that the conclusion survives. Power also comes from the driver counter rather than an external meter, so the absolute joules inherit whatever bias that sensor has; the savings are ratios of two measurements taken the same way on the same card minutes apart, which is what the bias mostly cancels in.

## B.10. Memory, frame rate, and what the encoder pays

| Frame (padded) | Tiles | MB rel.       | MB ours | Change % | fps rel. | fps ours |
|----------------|-------|---------------|---------|----------|----------|----------|
| 1024x512       | 8     | 180.0         | 204.0   | +13.33   | 43.89    | 49.88    |
| 1280x768       | 15    | 337.5         | 382.5   | +13.33   | 22.94    | 28.03    |
| 2048x1280      | 40    | 900.0         | 1020.0  | +13.33   | 9.03     | 11.18    |
| 3840x2304      | 135   | out of memory |         |          |          |          |

**Table 49.** Peak decoder memory and frame rate, released against ours, at every resolution that fitted. Take from it that routing costs memory rather than saving it, by a constant 13.33% at every resolution, because the tiled path holds the whole tile batch and the stitched canvas alive together; and that the frame-rate gain grows with the frame, from 43.9 to 49.9 fps at 832×480 up to 9.0 to 11.2 fps at 1080p, because a larger frame has more tiles to allocate across. The 4K row is a genuine failure and is reported as one: the job ran out of memory on a 48 GB card that four other processes were sharing, so no measurement above 1080p exists anywhere in this work.

results/supp\_footprint.json, pinned checkpoint, q32, 0.1 dB budget, NVIDIA RTX A6000. Frame rates here are lower than in Table 47 because that table's maps are at different rates and this one is q32 throughout. Nothing above 1080p is measured anywhere in this work.

The signalled configuration moves the choice of exit map to the encoder, which has to price the tiles before it can allocate them, and that cost is 4.52 of one tiled decode per frame per rate; section G prices the two ways of building the table and says why only one of them may be used to report quality. The asymmetry is the design rather than an accident: a decoder-side saving is bought with encoder-side work, which suits a compress-once decode-many deployment and does not suit live encoding.

results/supp\_encoder\_cost\_PAPER.json: pinned checkpoint, one sequence (Bosphorus), q63, 40 tiles, 15 iterations on an NVIDIA RTX A6000.

## C. Derivations

This section states the allocation problem from its objects upward and proves what the paper asserts about it. Nothing here needs a result from elsewhere in the supplement. Each statement carries the assumptions it uses, a proof, and a remark on what it leaves open.

| no. | claim                    | in  | checked in                  | ckpt  |
|-----|--------------------------|-----|-----------------------------|-------|
| P1  | separates over tiles     | C.2 | verify_theory               | eval  |
| P2  | monotone in $\lambda$    | C.2 | verify_theory               | eval  |
| P3  | blind sets are a hull    | C.3 | static_RECIPES12_b01        | PAPER |
| T4  | a Minkowski average      | C.3 | not measured                | -     |
| P5  | reaches only the hull    | C.3 | hull_gap                    | eval  |
| P6  | savings on a 1/N lattice | C.3 | theory_checks               | eval  |
| P7  | the floor                | C.4 | saturation_RECIPES12_ctc53  | PAPER |
| P8  | the saturation price     | C.4 | verify_theory               | eval  |
| P9  | the ceiling              | C.4 | per_class_RECIPES12         | PAPER |
| T10 | adaptivity gain          | C.5 | theory_check                | BEST  |
| P11 | plotted log-convexity    | C.6 | logconvexity                | none  |
| P12 | Lorenz bound             | C.7 | hybrid_RECIPES12_b01_e4head | PAPER |
| P13 | rank-1 threshold rule    | C.8 | raterank_RECIPES12_b01      | PAPER |
| P14 | rank-1 uses hull exits   | C.8 | raterank_RECIPES12_b01      | PAPER |

**Table 50.** The numbered results of this section, and the evidence for each. P is a proposition and T a theorem; the numbering is stable and is what the main paper cites. PAPER is runs/RECIPES12/ckpt\_PAPER.pth.tar, eval is runs/RECIPES12/ckpt\_eval.pth.tar and BEST is runs/BEST/ckpt\_eval.pth.tar. verify\_theory is scripts/verify\_theory.py, which reports 7 of 7 propositions passing on one sequence at one rate.

Files named without a directory are under results/ with a .json extension, and the table names them in full. T4 is proved and not measured: it describes a set of 120 vertices that no experiment enumerates.

## C.1. The setting, and four assumptions

**Tiles.** The decoder is cut at a fixed depth: everything above the cut runs once over the whole frame, everything below on square patches of the trunk's feature grid. Replicate padding makes the grid divide exactly, so the tiling is a partition. Tiles are indexed by t and N is their number in a frame, 40 at 1080p and 2 at 416×240; section A gives their geometry.

**Exits.** An exit is a point at which a tile may stop and be handed to the reconstruction head. There are  $K = 6$  over 12 residual blocks, evenly spaced, so exit k runs  $(k+1)b$  blocks with  $b = 2$ . At a split depth of  $j = 2$  the first  $jb$  blocks run full-frame whatever exit a tile takes, so the usable ladder has  $K - j = 4$  members and k ranges over j, ..., K-1 throughout.

**Distortion.**  $D_{t,k}$  is the mean squared error tile t incurs when it leaves at exit k, measured on the deployed tiled path so that what tiling costs sits inside it, and measured against the released decoder's decode of the same latent so that synthesis is the only difference between the two sides. Nothing forces  $D_{t,K-1}$  to zero, and C.4 returns to that.

**Decibels.** Distortion is reported as  $10 \log_{10}$  of the ratio to the reference, and the ratio can be formed at two levels. Pooling every tile of every frame into one mean squared error is the form the Lagrangian below is exact for; averaging a per-frame decibel is the codec convention, and it is what the budget is bisected against throughout. Section E measures how far apart the two read on one allocation, which is a substantial fraction of the working budget.

**Compute.** Compute is counted in multiply-accumulate operations and divided by those of one released full-frame decode, 453.5 GMAC at 1080p, so  $c = 1$  is one released decode and a saving of S% means the frame was decoded for  $(1 - S/100)$  of one. Every operator runs at a fixed multiple of the latent grid, which makes the cost of an exit resolution-independent. One tile at exit k costs

$$c_k = s_{\text{up}} + s_{\text{trunk}} \frac{(k+1)b}{N_{\text{blk}}} + s_{\text{head}} + a_k + r, \quad (12)$$

with  $s_{\text{up}}$  the opening upsample,  $s_{\text{trunk}}$  the trunk spread over  $N_{\text{blk}} = 12$  blocks of which exit k runs  $(k+1)b$ ,  $s_{\text{head}}$  the head,  $a_k$  the adapter exit k wears and r the full-frame deblocking filter; section B measures all five. Only the trunk term depends on k, which is why a ceiling exists: however shallow the ladder gets, the stem, the head and the filter are still paid.

| exit k | blocks | adapter | share | c_k    | 100(1-c_k) | c_k, stored |
|--------|--------|---------|-------|--------|------------|-------------|
| 2      | 6      | ffn     | 0.712 | 0.6088 | 39.12      | 0.5809      |
| 3      | 8      | ffn     | 0.712 | 0.7578 | 24.22      | 0.7300      |
| 4      | 10     | 1×1     | 0.142 | 0.8697 | 13.03      | 0.8697      |
| 5      | 12     | none    | 0.000 | 1.0095 | -0.95      | 1.0095      |

**Table 51. What each exit costs**, in units of one released full-frame decode, with the adapter share in units of one residual block. The deepest exit costs more than 1 because a tiled decode still pays the deblocking filter. The last column is the cost vector stored in results/tile\_table.json, which is the one the checks of Propositions 5, 6, 8 and 10 are computed on; each says so where it appears.

c\_k from the uniform-depth rows of results/static\_RECIPES12\_b01.json and adapter shares from results/adapter\_cost.json, both on runs/RECIPES12/ckpt\_PAPER.pth.tar. Section B audits the block count itself and prices the gap between the model the argmin runs on and the hook count every saving is quoted from.

The gaps between those costs are what the allocation trades in, and they are unequal: 0.1491, 0.1119 and 0.1398. The first is largest because exit 3 gives up the FFN adapter’s saving as well as two blocks.

**Assignments.** An assignment is a map  $a$  from tiles to exits. There are  $(K-j)^N$  of them,  $4^{40}$  at 1080p, so they will not be enumerated. The frame’s compute and distortion are

$$C(a) = \frac{1}{N} \sum_{t=1}^N c_{a(t)}, \quad D(a) = \frac{1}{N} \sum_{t=1}^N D_{t,a(t)}. \quad (13)$$

Everything after this rests on four assumptions, none of them a mathematical necessity. Each result names the ones it uses.

- **(A1) Compute is additive over tiles.** Exact when tiles decode independently, which is what a tiled decoder does.
- **(A2) Distortion is a tile mean**, and a tile’s error does not depend on which exits its neighbours took. The first half is exact for any per-pixel loss averaged over a frame; the second is not free, the deblocking filter running on the stitched canvas and seeing the whole exit map at once.
- **(A3) The reference is fixed.**  $D_{t,k}$  is measured on the deployed tiled path against the released decoder’s own decode of the same latent, so rate is identical on both sides and cancels.
- **(A4) The cost vector is strictly increasing** in  $k$  over  $k \geq j$ , which Table 51 shows it is.

The second half of (A2) is checked rather than asserted, over 60 allocations of one frame spanning savings of 1.2% to 41.9%: the largest disagreement between the per-tile prediction and a decode of the mixed map is  $5.5 \times 10^{-5}$  dB, four orders of magnitude below the budget. Nothing in (A1) to (A4) constrains  $D_{t,k}$  itself, and nothing requires it to fall as  $k$  grows.

Additivity check computed from results/tile\_table.json (Bosphorus, q32, 40 tiles), checkpoint runs/RECIPES12/ckpt\_eval.pth.tar. What it checks is a property of the decode path rather than of the weights.

## C.2. A price on compute

The encoder’s problem is to spend as little compute as possible while staying inside a distortion budget:

$$\min_a C(a) \quad \text{subject to} \quad D(a) \leq D_{\text{budget}}. \quad (14)$$

Constraint (14) couples the tiles: whether a tile can afford to leave early depends on how much of the budget the other 39 have used. The standard way out is to buy the constraint off with a price  $\lambda \geq 0$  charged per unit of compute:

$$L(a, \lambda) = D(a) + \lambda C(a) = \frac{1}{N} \sum_{t=1}^N [D_{t,a(t)} + \lambda c_{a(t)}]. \quad (15)$$

$\lambda$  is an exchange rate, in squared error per unit of relative compute, so  $\lambda c_k$  converts the compute an exit uses into the squared error that compute is deemed to be worth. Below it runs from about  $2 \times 10^{-7}$  to about  $1 \times 10^{-4}$ , against squared errors of order  $1 \times 10^{-4}$  on a 0 to 1 scale, and what matters

is where it sits relative to the differences between exits: at zero every tile takes its own lowest-error exit, and past a large enough price every tile has dropped as far as the ladder allows.

### Proposition 1 (separation)

For every  $\lambda \geq 0$ , an assignment minimises  $L(a, \lambda)$  over all assignments if and only if it minimises the bracket tile by tile:

$$k_t^*(\lambda) \in \arg \min_{k \geq j} [D_{t,k} + \lambda c_k]. \quad (16)$$

Assumes (A1) and (A2).

*Proof.* Write  $f_t(k) = D_{t,k} + \lambda c_k$ . Then  $N L(a, \lambda) = \sum_t f_t(a(t))$ , a sum in which the  $t$ -th term depends on  $a$  only through  $a(t)$ . For any assignment  $a$ , term by term  $f_t(a(t)) \geq \min_k f_t(k)$ , so  $N L(a, \lambda) \geq \sum_t \min_k f_t(k)$ , and the right-hand side is attained by any  $a$  that picks a minimiser in every tile. Conversely if  $a$  is optimal and some tile had  $f_t(a(t)) > \min_k f_t(k)$ , replacing  $a(t)$  by that minimiser and leaving every other tile alone would strictly lower  $L$ .

*Remark.* The result is about the frame mean and bounds no single tile. At the deployed 0.1 dB operating point the worst tile in the test set gives up 1.42 dB, so a frame inside its budget can hold a tile an order of magnitude outside it; section D tabulates that tail. No result in this section repairs it.

Per-tile tail from results/supp\_opquality\_PAPER.json, checkpoint runs/RECIPES12/ckpt\_PAPER.pth.tar, at the operating point read from results/signalled\_RECIPES12\_ctc53.json.

The  $N$ -dimensional search has become  $N$  independent searches over 4 numbers each, with ties broken toward the shallower exit so that  $k_t^*$  is a function. We call  $k_t^*(\lambda)$  the oracle’s choice, because it uses the true  $D_{t,k}$ , which only the encoder has. The construction is Shoham and Gersho’s [26] with compute in place of rate, made standard in coding by [27].

### Proposition 2 (monotonicity)

Let  $\lambda_1 < \lambda_2$  and let  $a_1, a_2$  be the corresponding oracle assignments. Then  $C(a_2) \leq C(a_1)$  and  $D(a_2) \geq D(a_1)$ . Assumes (A1), (A2) and Proposition 1.

*Proof.* Fix a tile and drop the index. Optimality at each price gives  $D_1 + \lambda_1 c_1 \leq D_2 + \lambda_1 c_2$  and  $D_2 + \lambda_2 c_2 \leq D_1 + \lambda_2 c_1$ , where  $(D_i, c_i)$  is the pair chosen at  $\lambda_i$ . Adding the two and cancelling  $D_1 + D_2$  leaves  $(\lambda_2 - \lambda_1)(c_2 - c_1) \leq 0$ , hence  $c_2 \leq c_1$ . Substituting that back into the first inequality gives  $D_2 \geq D_1 + \lambda_1(c_1 - c_2) \geq D_1$ . Averaging over tiles preserves both.

*Remark.* Monotone is not continuous. Cost moves in the jumps Proposition 6 measures, so bisecting  $\lambda$  converges on a reachable level rather than on the budget. The result also assumes exact minimisation of the true table, whereas the deployed search bisects a predicted table and corrects with a real decode, at most six passes to  $5 \times 10^{-4}$  dB.

| tile | bits   | D(2)   | D(3)   | D(4)   | D(5)   | exits as $\lambda$ rises          |
|------|--------|--------|--------|--------|--------|-----------------------------------|
| 34   | 10,446 | 1.608  | 1.638  | 1.679  | 1.655  | 2 throughout                      |
| 0    | 12,996 | 9.346  | 9.249  | 9.207  | 9.180  | 5, 4 at 1.9, 3 at 3.0, 2 at 6.5   |
| 18   | 16,989 | 16.249 | 15.795 | 15.638 | 15.606 | 5, 4 at 2.3, 3 at 11.2, 2 at 30.6 |

**Table 52. Three tiles of one frame, and what the price does to them.**  $D$  is in units of  $10^{-5}$  squared error and switch points in units of  $10^{-6}$  of  $\lambda$ . The cheapest tile prefers the shallowest exit at every price including  $\lambda = 0$ , its error not falling with depth at all, so depth is not always worth buying. The most expensive tile holds out against the shallowest exit 4.7 times longer than the middle one, which is why one uniform depth cannot suit both.

results/tile\_table.json (Bosphorus, q32, 40 tiles), checkpoint runs/RECIPES12/ckpt\_eval.pth.tar. The 60 prices stored in that file produce 33 distinct savings, the granularity C.3 quantifies.

## C.3. The achievable set, and what one price reaches

The pairs  $(C(a), D(a))$  an allocation can produce form a set whose shape decides both what a sweep can reach and what adaptivity is worth. There are two versions of it. Write

$$\bar{D}_k = \frac{1}{N} \sum_{t=1}^N D_{t,k}, \quad (17)$$

for the exit mean at  $k$ , the one place tiles are averaged before anything is chosen.

### Proposition 3 (fixed proportions)

Suppose tiles are assigned to exits in proportions  $p$ , one probability per exit, independently of their content. Then the achievable set of expected pairs is exactly the convex hull of the  $K$ - $j$  points whose coordinates are the cost  $c_k$  and the exit mean at  $k$ . *Assumes (A1) and (A2).*

*Proof.* Under content-independent proportions the expected cost is  $\sum_k p_k c_k$  and, by the tile mean, the expected distortion is  $\sum_k p_k$  times the exit mean at  $k$ . The map from  $p$  to that pair is affine and its domain is the probability simplex, whose extreme points are the unit vectors. An affine image of a simplex is the convex hull of the images of its vertices, and those images are the cost and exit mean of each exit.

*Remark.* The hull is a hull of expectations. One frame with  $N$  tiles realises proportions in multiples of  $1/N$ , so interior points are attained only to that granularity, and at  $N = 2$  the smallest test class reaches three mixtures per pair of exits.

The four uniform-depth decodes are the vertices of that hull, and they are everything a content-blind allocation can reach; section D tabulates them in dB at every rate beside the two shuffled controls. The mean over rates of the best single depth is 17.5%, against 27.6% for the per-tile allocation at the same budget.

results/static\_RECIPES12\_b01.json, checkpoint runs/RECIPES12/ckpt\_PAPER.pth.tar, 53 frames.

### Theorem 4 (per-tile assignment)

The convex hull of the pairs achievable by per-tile assignment is the Minkowski average of the  $N$  per-tile hulls,

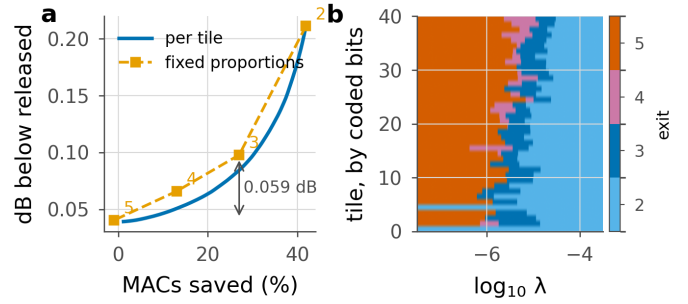
$$\mathcal{A}_{\text{tile}} = \frac{1}{N} \bigoplus_{t=1}^N \text{conv}\{(c_k, D_{t,k}) : k \geq j\}. \quad (18)$$

*Assumes (A1) and (A2).*

*Proof.* By (A1) and (A2) an achievable pair is the average of one point drawn from each per-tile set  $S_t$ , the set of pairs cost  $c_k$  against error  $D_{t,k}$ , so the achievable set is the Minkowski average of the  $S_t$ . Taking convex hulls commutes with Minkowski addition, the hull of a sum being the sum of the hulls, and with positive scaling; applying that  $N-1$  times moves the hull inside the sum.

*Remark.* The theorem bounds what any allocation reaches, a perfect router included, and says nothing about which of those points a price reaches, which is Proposition 5. It is the one result here with no measurement behind it, the set having up to  $N(K-j-1) = 120$  vertices at 1080p.

Proposition 3 is the special case in which every tile has the same row, and otherwise the per-tile set is strictly larger: the fixed-proportion hull has at most  $K-j = 4$  vertices and so 3 straight segments, while the Minkowski average is smooth at any plotted scale. Figure 39 draws both on one frame.



**Figure 39. The two achievable sets, on one frame.** **a**, the per-tile frontier of Theorem 4 traced by sweeping  $\lambda$ , against the fixed-proportion chain of Proposition 3 through the four uniform-depth points, labelled by exit. The per-tile curve lies below the chain everywhere between the vertices, by 0.059 dB at the exit-3 vertex. **b**, the exit each of the 40 tiles takes as  $\lambda$  rises, tiles ordered by the bits the entropy coder spent on them. Every row is non-increasing, which is Proposition 2 per tile, and the rows differ, which is what Theorem 4 buys over Proposition 3. The bottom row is a tile that never leaves the shallowest exit.

Both panels from results/tile\_table.json (Bosphorus, q32, 40 tiles), checkpoint runs/RECIPES12/ckpt\_eval.pth.tar, drawn by scripts/supp\_derivations\_figure.py on that file's own cost vector, the last column of Table 51. A grid of 4001 prices finds 93 distinct reachable savings on this frame, against the 95 the dynamic programme of Table 53 enumerates.

### Proposition 5 (a price reaches only the lower hull)

For every  $\lambda \geq 0$  the point  $(C, D)$  produced by the oracle assignment lies on the lower convex hull of the achievable set, and no  $\lambda$  produces a point strictly inside it. *Assumes (A1), (A2) and Proposition 1.*

*Proof.* By Proposition 1 the oracle assignment minimises  $D(a) + \lambda C(a)$  over all assignments, so its pair minimises the linear functional  $(C, D) \rightarrow D + \lambda C$  over the achievable set and hence over that set's convex hull. A minimiser of a linear functional over a compact convex set lies on its boundary, and because  $\lambda \geq 0$  the supporting line has slope  $-\lambda \leq 0$ , which places the point on the lower boundary. Conversely a point strictly inside the hull is a strict convex combination of achievable points and is therefore strictly beaten, at that same  $C$ , by some point of the hull, so it cannot minimise any such functional.

*Remark.* A budget between two hull vertices is unreachable by any price, and the proposition does not say what that costs, which is Proposition 6. The measurement below enumerates the exact Pareto set on one frame at one rate, on that file's own cost vector.

Everett's generalised multiplier method (1963) supplies the reassuring half of the converse: whatever compute the returned allocation consumes, it is optimal for that level. With four distinct exit costs the whole Pareto set can be enumerated by dynamic programming over tiles, so the other half can be priced exactly.

| budget dB | by sweep | exact Pareto | gap     |
|-----------|----------|--------------|---------|
| 0.05      | 12.398   | 12.421       | 0.0230  |
| 0.08      | 25.443   | 25.489       | 0.0452  |
| 0.10      | 30.475   | 30.496       | 0.0209  |
| 0.12      | 33.782   | 33.803       | 0.0211  |
| 0.15      | 37.439   | 37.461       | 0.0211  |
| 0.18      | 40.048   | 40.046       | -0.0018 |
| 0.20      | 41.166   | 41.165       | -0.0016 |

**Table 53. What convexity costs.** Savings in per cent at seven budgets on one frame. The sweep reaches 95 allocations against the 635 on the exact Pareto set, and the largest loss over these budgets is 0.045 saving points. The two negative entries are the sweep landing on a budget the enumeration's grid straddles rather than the sweep beating the Pareto set, which Proposition 5 forbids.

results/hull\_gap.json, Bosphorus q32, 40 tiles. The enumeration and the sweep read one cost vector, so the gap does not depend on which one.

### Proposition 6 (granularity)

If at most  $m$  tiles switch exit at any single breakpoint of the sweep, and each switch moves a tile by one rung, then consecutive reachable savings

differ by at most

$$\text{spacing} \leq \frac{100 \max_k (c_{k+1} - c_k)}{N} \%, \quad (19)$$

and the loss from convexity cannot exceed the largest spacing. *Assumes* (A1), (A4) and Proposition 2.

*Proof.* Between breakpoints the assignment is constant, so the reachable costs are exactly the costs at the breakpoints. At one breakpoint the mean cost changes by  $1/N$  times the sum of the individual changes, each at most  $\max_k (c_{k+1} - c_k)$  in magnitude, and there are at most  $m$  of them; multiplying by 100 puts it in saving points. For the second part, let a budget fall strictly between two reachable levels. By Proposition 2 the bisection returns the lower level, so the loss is at most the distance between them.

*Remark.* Both hypotheses are read off a measured table rather than guaranteed, and the bound is vacuous if either fails on another frame. It falls as  $1/N$ , so it says nothing useful at the smallest test class, where  $N = 2$  and one switch moves the frame by half the largest rung.

On the measured table  $m = 2$ , tiles with identical rows switching together, the largest rung is 0.1491 and  $N = 40$ , so the bound is 0.7453 saving points. The measured spacing is 0.7453, sitting on the bound, and the convexity loss of Table 53 is 0.0452. Nothing in the expression depends on content or on rate, and halving the tile side would put four times as many tiles on the lattice, each carrying a quarter of the step.

$m$ , the largest rung, the bound and the measured spacing from results/theory\_checks.json (Bosphorus q32, 40 tiles, runs/RECIPE512/ckpt\_eval.pth.tar), written by scripts/verify\_theory.py.

#### C.4. Floor, saturation and ceiling

The sweep has two ends, and a budget outside them buys nothing. The budget section measures where they sit at each rate; this one derives them.

##### Proposition 7 (the floor)

At  $\lambda = 0$  the allocation attains  $D_{\min} = (1/N) \sum_t \min_k D_{t,k}$ , and no allocation attains less. *Assumes* (A2) and Proposition 1. *Proof.* Immediate from Proposition 1 at  $\lambda = 0$ , and from  $D(a)$  being a mean of terms each bounded below by its own tile minimum.

*Remark.* The floor is a statement about the table, and the table is a product of training, which (A3) does not constrain. If the deepest exit drifts from the reference by  $\delta$  then the achievable set shifts up by  $\delta$  before any compute is saved, and a budget below  $\delta$  is unreachable at every allocation. Section H measures that drift, with and without the training objective's anchor term.

Because  $D_{t,k}$  is measured on the deployed tiled path,  $D_{\min}$  is not zero: it is what tiling costs before any tile exits early, and section E measures where it sits at each rate and which budgets fall below it.

##### Proposition 8 (saturation)

Define

$$\lambda_{\text{sat}} = \max_t \max_{k>j} \frac{D_{t,j} - D_{t,k}}{c_k - c_j}, \quad (20)$$

Then for every  $\lambda > \lambda_{\text{sat}}$  the oracle assigns exit  $j$  to every tile, the frame's cost is exactly  $c_j$ , and no larger price changes anything. *Assumes* (A1), (A4) and Proposition 1.

*Proof.* Exit  $j$  beats exit  $k$  on tile  $t$  precisely when  $D_{t,j} + \lambda c_j \leq D_{t,k} + \lambda c_k$ , that is when  $\lambda \geq (D_{t,j} - D_{t,k}) / (c_k - c_j)$ , the denominator being positive for  $k > j$  by (A4). Taking the maximum over  $k$  and then over  $t$  gives a single price beyond which exit  $j$  wins everywhere. The cost of the constant map is  $c_j$  by (13).

*Remark.*  $\lambda_{\text{sat}}$  is a maximum over the tiles of one frame, so a price chosen for a sequence saturates some of its frames and not others, and it constrains the price rather than the budget. On the worked frame the closed form gives  $3.045 \times 10^{-5}$ , exactly the last switch of the most

expensive tile in Table 52.

##### Proposition 9 (the ceiling)

The largest saving any allocation can reach is

$$S_{\max} = 100(1 - c_j) \%. \quad (21)$$

*Assumes* (A1), (A4) and Proposition 8. *Proof.*  $C(a)$  is a mean of costs each at least  $c_j$ ,  $j$  being the shallowest usable exit, so  $C(a) \geq c_j$  with equality only for the constant map, which Proposition 8 shows is attained.

*Remark.*  $S_{\max}$  depends on the split depth alone, so no amount of training raises it and the only lever is  $j$ . It is a ceiling in multiply-accumulates and not in seconds, and section B reports the wall clock falling short of it. Over the 18 class-and-rate cells measured at a saturating budget it spreads by 6e-06 saving points, which is single-precision noise.

Per-class invariance from results/per\_class\_RECIPE512.json at budgets of 0.5 dB and above, checkpoint runs/RECIPE512/ckpt\_PAPER.pth.tar; the ceiling itself is 39.1%, from results/saturation\_RECIPE512\_ctc53.json.

#### C.5. What adaptivity is worth

What content awareness is worth can be answered without building a predictor. Compare the best per-tile allocation with the best content-blind one at the same price:

$$J_{\text{ad}}(\lambda) = \frac{1}{N} \sum_t \min_k [D_{t,k} + \lambda c_k], \quad J_{\text{fix}}(\lambda) = \min_k [\bar{D}_k + \lambda c_k]. \quad (22)$$

$J_{\text{ad}}$  takes the minimum inside the average and  $J_{\text{fix}}$  takes it outside. By Propositions 1 and 3 the first is the best the per-tile family attains at price  $\lambda$  and the second the best the fixed-proportion family attains.

##### Theorem 10 (adaptivity gain)

For every  $\lambda \geq 0$ ,  $\Delta(\lambda) = J_{\text{fix}}(\lambda) - J_{\text{ad}}(\lambda) \geq 0$ , with equality if and only if some single exit  $k^*$  attains the per-tile minimum for every tile. *Assumes* (A1) and (A2).

*Proof.* Fix  $\lambda$  and let  $k$  be any exit. Then the exit mean at  $k$  plus  $\lambda c_k = (1/N) \sum_t [D_{t,k} + \lambda c_k] \geq (1/N) \sum_t \min_{k'} [D_{t,k'} + \lambda c_{k'}] = J_{\text{ad}}(\lambda)$ , because each summand dominates the corresponding minimum. The right-hand side does not depend on  $k$ , so taking the minimum over  $k$  on the left preserves the inequality and gives  $J_{\text{fix}} \geq J_{\text{ad}}$ . For equality, suppose some  $k^*$  attains the per-tile minimum everywhere; then choosing it makes every summand equal to its minimum and the inequality is tight. Conversely suppose  $\Delta = 0$  and let  $k^*$  attain  $J_{\text{fix}}$ . Then  $(1/N) \sum_t ([D_{t,k^*} + \lambda c_{k^*}] - \min_{k'} [D_{t,k'} + \lambda c_{k'}]) = 0$ . Every summand is non-negative, so every summand is zero, which says exactly that  $k^*$  attains the minimum on every tile.

*Remark.*  $\Delta$  upper-bounds what any predictor gains over a content-blind allocation at that price and says nothing about whether one can get any of it, which is what C.7 measures. It is a property of the ladder as much as of the content: improving the shallow exits makes more tiles agree on the argmin, so  $\Delta$  falls even as the saving rises and is not a figure of merit.

Read as a ratio  $\Delta/J_{\text{ad}}$  rather than in saving points, the gain rises with rate at a low price and falls to 0.00% to 0.03% at the highest price measured, which is Theorem 10's equality condition arriving: the oracle has itself collapsed onto one exit and nothing is left for a router to recover.

Ratios from results/theory\_check.json, checkpoint runs/BEST/ckpt\_eval.pth.tar, 40 sequences (UVG, MCL-JCV and HEVC class E; classes B, C and D not measured), on the last column of Table 51. The table below is the same quantity in saving points on the pinned checkpoint.

| q   | blind % | oracle % | gap  | blind dB | shuffled dB |
|-----|---------|----------|------|----------|-------------|
| q0  | 31.66   | 34.46    | 2.79 | 0.1147   | 0.1143      |
| q16 | 27.38   | 30.82    | 3.45 | 0.1213   | 0.1227      |
| q32 | 22.72   | 26.67    | 3.95 | 0.1238   | 0.1337      |
| q48 | 19.00   | 25.33    | 6.32 | 0.1321   | 0.1423      |
| q63 | 16.76   | 24.10    | 7.34 | 0.1370   | 0.1471      |

**Table 54. The same gap in the units the paper reports.** Columns 2 to 4 are savings at a 0.1 dB budget: the best content-blind mixture of uniform depths, the per-tile oracle, and the difference. Columns 5 and 6 are decibels at the oracle's own compute, what the best blind mixture would read there and what a random shuffle of the oracle's exit map actually reads. Adaptivity is worth 2.8 to 7.3 saving points, rising with rate, against an oracle reading 0.100 dB and a shuffle reading up to 0.147 dB.

Computed from results/static\_RECIPES12\_b01.json, checkpoint runs/RECIPES12/ckpt\_PAPER.pth.tar, 53 frames. The blind columns are the lower hull of the four uniform-depth points under Proposition 3, evaluated at the budget and at the oracle's compute; the shuffle column is a measured decode. At q0 the shuffle reads slightly better than the hull predicts, which bounds the interaction (A2) ignores at under 0.002 dB.

### C.6. Convexity in the units the frontier is plotted in

Theorem 4 gives  $D$  convex in  $C$ . The frontier is almost never plotted in those units: it is plotted as decibels against percentage saving, where convexity is a different statement, the logarithm being concave and increasing and so preserving neither convexity nor concavity. With  $S = 1 - C/C_{K-1}$ ,

$$dB(S) = \frac{10}{\ln 10} \ln D(C), \quad C = (1 - S)C_{K-1}. \quad (23)$$

#### Proposition 11 (plotted convexity is log-convexity)

Let  $D$  be twice differentiable along the frontier. Then  $dB(S)$  is convex if and only if  $D$  is log-convex in  $C$ :

$$DD'' \geq (D')^2 \iff \frac{d}{dC} \left( \frac{1}{\lambda} \right) \geq \frac{1}{D}. \quad (24)$$

Assumes twice differentiability along the frontier, and Proposition 5 for the second form.

*Proof.*  $C$  is affine in  $S$  and convexity is preserved under affine reparametrisation, so by (23)  $dB$  is convex in  $S$  if and only if  $\ln D$  is convex in  $C$ . Differentiating twice,  $(\ln D)'' = D''/D - (D'/D)^2 = [D D'' - (D')^2]/D^2$ , and  $D > 0$ , so  $(\ln D)'' \geq 0$  if and only if  $D D'' \geq (D')^2$ . For the second form, the slope of the frontier is  $D' = -\lambda$  by the supporting-line argument of Proposition 5, so  $D'' = -\lambda'$ ; substituting gives  $-\lambda' D \geq \lambda^2$ , and dividing by  $\lambda^2 > 0$  gives  $-\lambda'/\lambda^2 \geq 1/D$ , whose left-hand side is  $d(1/\lambda)/dC$ . The second form reads as a rate condition: distortion has to fall at least exponentially in compute for the plotted curve to be convex.

*Remark.* This is a property of the decoder rather than a consequence of Theorem 4, and a convex  $D$  can fail it. Differentiability fails in one of the two cases: under fixed proportions  $D$  is piecewise linear with  $K$ -j vertices, so the shape one sees depends on where one samples, whereas the per-tile hull is smooth at any plotted scale.

| q   | points | condition holds | margin | quartic fit |
|-----|--------|-----------------|--------|-------------|
| q0  | 16     | 100.0%          | 0.0024 | 84.0%       |
| q16 | 14     | 100.0%          | 0.0050 | 76.3%       |
| q32 | 13     | 100.0%          | 0.0088 | 75.3%       |
| q48 | 13     | 100.0%          | 0.0145 | 73.7%       |
| q63 | 13     | 100.0%          | 0.0230 | 74.7%       |

**Table 55. Testing log-convexity on the measured frontier.** The direct test asks whether the secant slopes of  $\ln D$  against  $S$  are non-decreasing, and the margin is the smallest increase attained. The condition holds at every point of every frontier and the margin grows tenfold from the lowest rate to the highest. Fitting a quartic and differentiating it twice instead reports 73.7% to 84.0%, and the difference is the fit rather than the data: near the ceiling the frontier is close to vertical and a polynomial overshoots, whereas the secant test has no fitted degree to choose.

results/logconvexity.json. That file records no checkpoint, so it cannot be attributed from disk; its five frontiers have 13 to 16 points each. The quartic column is the file's own

quartic\_frac\_ok field.

### C.7. How much of the oracle a partial signal recovers

Letting a predictor decide most cases and handing the hardest ones to something exact is the shape of selective prediction [36] and learning to defer [37, 38], and of the budgeted variant of the latter [39]. Two things make the case easier here. The expert is the encoder's own search, exact and free at test time, so what is scarce is the bits needed to say what it decided rather than the expert's time; and the selection rule is not learned, since a separable objective makes the optimal set of size  $s$  exactly the  $s$  largest regrets. What is left to measure is how concentrated the regret is.

The decoder cannot run the argmin of Proposition 1,  $D_{t,k}$  being an error against a source it never receives. A predictor can guess it. Writing  $p_{t,k}$  for a head's probability that tile  $t$  belongs at exit  $k$ , its rule is

$$\hat{k}_t = \arg \max_{k \geq j} [\log p_{t,k} - \beta c_k] = \arg \min_{k \geq j} \left[ \frac{-\log p_{t,k}}{\kappa} + \lambda c_k \right]. \quad (25)$$

The second form of (25) is the same rule with  $\beta = \lambda \kappa$ ,  $\kappa$  being a fixed calibration turning nats into squared error. Written that way it says that a router is the oracle of (16) with a surrogate in place of  $D_{t,k}$ , and that one price governs both. Define the per-tile regret at a fixed price:

$$r_t(\lambda) = [D_{t,\hat{k}_t} + \lambda c_{\hat{k}_t}] - [D_{t,k^*} + \lambda c_{k^*}] \geq 0. \quad (26)$$

Non-negativity is immediate from (16). The regret is what the frame's Lagrangian loses on tile  $t$  by trusting the prediction, and by Proposition 1 those losses add, so signalling the exits of a subset  $S$  removes exactly  $\sum_{t \in S} r_t$  from the objective and leaves the rest to the head.

#### Proposition 12 (Lorenz bound)

Fix  $\lambda$  and let  $r_{(1)} \geq r_{(2)} \geq \dots \geq r_{(N)}$  be the regrets in decreasing order with total  $R$ . For any subset  $S$  of size  $s$ , the fraction of the total regret that signalling  $S$  removes is at most

$$L(\rho) = \frac{1}{R} \sum_{i=1}^{\lceil \rho N \rceil} r_{(i)}, \quad \rho = s/N, \quad (27)$$

with equality if and only if  $S$  is a set of  $s$  largest regrets.  $L$  is the Lorenz curve of the regret distribution: non-decreasing, concave, running from 0 to 1, and lying above the diagonal by an amount the Gini coefficient summarises. Assumes (A1), (A2) and a fixed  $\lambda$ .

*Proof.* The sum of any  $s$  of the  $r_{(i)}$  is at most the sum of the  $s$  largest, since replacing an element of  $S$  by a larger one outside  $S$  cannot decrease the sum, and repeating that exchange terminates at a set of  $s$  largest values. Dividing by  $R$  gives the bound. Concavity holds because the increments  $r_{(i)}$  are non-increasing by construction.

*Remark.* The bound holds at a fixed price, and the deployed system re-bisects  $\lambda$  at every  $\rho$  so the frame lands back on its budget. Re-bisecting enlarges the feasible set: the overridden tiles and the predicted ones then carry two multipliers, and pairs of multipliers reach allocations no single one does. The measured recovery is therefore not bounded by  $L$ , and Proposition 12 is the wrong thing to call a bound on the deployed system.

| q   | Gini | L(.1) | rec(.1) | L(.2) | rec(.2) | L(.5) | rec(.5) |
|-----|------|-------|---------|-------|---------|-------|---------|
| q0  | 0.95 | 100%  | 36%     | 100%  | 53%     | 100%  | 86%     |
| q16 | 0.92 | 89%   | 34%     | 100%  | 45%     | 100%  | 83%     |
| q32 | 0.90 | 82%   | 38%     | 100%  | 57%     | 100%  | 83%     |
| q48 | 0.80 | 58%   | 37%     | 86%   | 60%     | 100%  | 88%     |
| q63 | 0.75 | 50%   | 31%     | 75%   | 52%     | 100%  | 83%     |

**Table 56. The Lorenz curve of the regret, and what signalling recovers.**  $L(\rho)$  is the share of the total regret carried by the  $\rho$  worst tiles, which Proposition 12 says is the most any signal of that size can remove at a fixed price;  $\text{rec}(\rho)$  is the share of the predictor-to-oracle gap the measured sweep recovers. Regret is concentrated at every rate, Gini 0.75 to 0.95: a tenth of the tiles carries 50% to 100% of the total regret and recovers 31% to 38% of the gap. Over every reachable cell the ratio  $\text{rec}/L$  runs from 22% to 100%, for the reason the remark gives.

results/hybrid\_RECIPES12\_b01\_e4head.json, checkpoint runs/RECIPES12/ckpt\_PAPER.pth.tar, 53 sequences at a 0.1 dB budget, against the head fitted to that checkpoint.  $\text{rec}(p)$  is computed here from the same file as the saving at  $p$ , relative to the  $p = 0$  and  $p = 1$  endpoints. Signalling half the tiles beats signalling all of them at 0 of 5 rates here; on a head fitted to other weights it did so at two of them, which is the re-bisection effect the remark predicts rather than a property of the decoder.

The signal is an entropy-coded mask over tiles plus an index per override, so its bill grows with  $p$  and is pure overhead at  $p = 1$ , where the mask names every tile; section G prices both objects. A fifth of the tiles costs 44 bits per frame and recovers over half the gap.

### C.8. The rate-rank rule, and the model it comes from

The decoder holds, before the trunk runs, the bits the entropy coder spent on each tile's latents; write  $b_t$  for that count over the frame's mean. Deriving the rule from a model rather than presenting it as a heuristic says when it must work and how it fails. The model is separability in log space: one scalar difficulty per tile, and one ladder profile applying to every tile in proportion.

$$\log D_{t,k} \approx u_t + \log \phi_k, \quad u_t = \alpha \log b_t + c. \quad (28)$$

Equivalently  $D_{t,k} = g_t \phi_k$  with  $g_t = \exp(u_t) > 0$ . The profile  $\phi$  is K-j numbers fitted once per rate and  $\alpha$  and  $c$  are two more, all three fitted leave-one-sequence-out. Nothing is learned per frame and nothing is transmitted, which is why the rule costs the decoder zero.

#### Proposition 13 (a rank-1 table gives a threshold rule)

If  $D_{t,k} = g_t \phi_k$  with  $g_t > 0$  then

$$\arg \min_{k \geq j} [g_t \phi_k + \lambda c_k] = \arg \min_{k \geq j} \left[ \varphi_k + \frac{\lambda}{g_t} c_k \right], \quad (29)$$

so a tile's exit depends on the tile only through the scalar  $\lambda/g_t$ , and by Proposition 2 the chosen exit is non-increasing in it. Sorting tiles by  $g_t$  and cutting the sorted list at K-j-1 thresholds therefore reproduces the oracle exactly. Assumes exact rank-1 structure, and Proposition 2.

*Proof.* Divide the bracket by  $g_t > 0$ , which does not move the argmin. The reduced problem is the one-tile problem of (16) with distortion  $\phi$  and price  $\lambda/g_t$ , so Proposition 2 applies and its solution is non-increasing in that price. Since  $\lambda/g_t$  is decreasing in  $g_t$ , the exit is non-decreasing in  $g_t$ : harder tiles go deeper. The thresholds are the breakpoints of that one common problem and there are at most K-j-1 of them.

*Remark.* Exactness needs an exactly rank-1 table, which no real table is, and the shortfall of the deployed rule measures the departure. The proposition also says the exit is monotone in  $g_t$  and not that a bit count estimates  $g_t$ , which is a separate and weaker claim.

#### Proposition 14 (a rank-1 rule uses only the profile's hull)

Under the same model, an exit  $k$  is chosen for some tile at some price only if  $(c_k, \phi_k)$  lies on the lower convex hull of the profile. Assumes exact rank-1 structure, and Propositions 5 and 13. *Proof.* By Proposition 13 the reduced problem is a single Lagrangian sweep over the points  $(c_k, \phi_k)$ , and by Proposition 5 such a sweep reaches only hull points.

*Remark.* The statement is about the fitted profile rather than the true table, so an exit off the hull is dropped by the rule and may still be the oracle's choice. Hull membership is not automatic: on the single-sequence table of results/tile\_table.json the same construction drops exit 4.

Section F prints the fitted  $\alpha$  and  $\phi$  at every rate and reports what the rule saves. Two properties of that fit belong here, with the propositions they bear on.  $\alpha$  is positive at every rate, so a tile the entropy coder spent bits on is a tile every exit reconstructs worse, and it falls 7-fold from the lowest rate to the highest. And the hull is no longer full: at q0, q16 the fitted profile puts exit 4 above the line joining its neighbours, so the Lagrangian rule cannot select it there whatever the multiplier. On the checkpoint this paper first reported all four exits survived; on the one it reports now, the two lowest rates have a rung the rule will not use. That is Proposition 14 costing something rather than nothing. We have two checkpoints and so cannot say whether further training removes more

rungs or puts this one back; what the two do establish is that hull membership is a property of the weights and not of the design, and a claim that it is full has to be re-checked whenever they move.

Fitted values from results/raterank\_RECIPES12\_b01.json, checkpoint runs/RECIPES12/ckpt\_PAPER.pth.tar, 53 sequences at a 0.1 dB budget; hull membership computed in the build against the shipped cost vector of Table 51.

The rule gives up 3.9 saving points to the Lagrangian oracle at the rate where it does worst, and the loss is not in the separability. On the measured tile table the separable model absorbs 99.96% of the variance of  $\log D$ . The quantity it has to resolve is small: the exit profile spans 0.0449 in log units while the non-separable residual has a root mean square of 0.0144, 32% of it. A model can explain almost all of a table and still be unable to order the differences that decide the argmin, which reads only the differences. Predicting a tile's difficulty from its bit count alone then leaves  $R^2 = 0.84$ , and by Proposition 13 an error in  $\log g_t$  is a proportional error in the price that tile faces. Coded bits correlate with the mean level of a tile's row at Spearman 0.83 to 0.88 and with the oracle's exit index at 0.41 to 0.51, the ordering the model predicts.

Residual analysis computed from results/tile\_table.json (Bosphorus, q32, 40 tiles), checkpoint runs/RECIPES12/ckpt\_eval.pth.tar;  $\alpha$  fitted there is 4.69 against 2.75 for the 53-sequence fit at the same rate, the difference being one sequence against 53. Spearman correlations from results/raterank\_RECIPES12\_b01.json, pinned checkpoint; the file stores the last against the negated exit index and it is reported here with the sign flipped, so a positive value means more bits and a deeper exit.

### C.9. What is proved, what is checked, and what neither covers

Table 50 lists every result with its evidence. Four gaps in it are worth naming. Theorem 10 is measured as a ratio only on runs/BEST/ckpt\_eval.pth.tar over 40 sequences and on the stored cost vector; the pinned measurement of Table 54 gives it in saving points instead. The convexity check of Table 55 records no checkpoint. The granularity and hull results of C.3 rest on one sequence at one rate, which is enough to check arithmetic and not enough to characterise a codec. And every proposition here concerns a frame mean under (A1) and (A2), so none bounds what a single tile gives up.

## D. Complete results

This section prints in full what the main paper samples: every class at every quality index and at three budgets, the frontier drawn class by class, and the distribution behind each mean, over sequences and over tiles. It also carries three measurements the main paper has no room for: a perceptual metric beside the decibel, the composition with weight quantisation, and the raw numbers behind the curves. It ends with the cases where the method does badly.

### D.1. Every class at every rate

| Class                 | q0    | q16   | q32   | q48   | q63   |
|-----------------------|-------|-------|-------|-------|-------|
| MCL-JCV               | 37.1  | 34.4  | 31.2  | 30.2  | 29.2  |
| UVG                   | 36.0  | 33.8  | 31.2  | 28.3  | 26.7  |
| HEVC B                | 34.9  | 31.3  | 25.1  | 21.9  | 19.4  |
| HEVC E                | 36.6  | 32.4  | 29.3  | 29.3  | 29.3  |
| HEVC C                | 25.6  | 17.1  | 7.8   | 6.6   | 4.8   |
| HEVC D                | 18.3  | 10.9  | 2.5   | 2.5   | 2.5   |
| All 53, pooled        | 34.4  | 30.8  | 26.6  | 25.2  | 24.1  |
| Quality given up (dB) | 0.100 | 0.100 | 0.099 | 0.099 | 0.100 |

**Table 57.** Compute saved against the released decoder, by class and by quality index, at the 0.1 dB budget; the main paper prints the q0, q32 and q63 columns of the first six rows. Read it in two directions. Along a row the saving falls with rate, because the residual a shallow exit fails to reconstruct grows relative to the quality being protected. Down a column it falls with resolution, from 37.1% on MCL-JCV to 18.3% on HEVC D at q0, because a 256 px tile is 40 tiles on a 1080p frame and two at 416x240. The last row is the quality the pooled allocation gave up, which is the budget to within a thousandth of a decibel at every rate.

results/supp\_per\_class\_budgets.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar, 53 sequences at one frame each, per-frame convention. The pooled row reads 29.7% at q0

where the main paper's 33.7% comes from results/signalled\_RECIPES12\_ctc53.json; the two bisect the same budget with different code and settle on  $\lambda = 5.126 \times 10^{-5}$  and  $5.150 \times 10^{-5}$ .

**a**

compute saved (%), 0.1 dB budget

|         |      |      |      |      |      |
|---------|------|------|------|------|------|
| MCL-JCV | 37.1 | 34.4 | 31.2 | 30.2 | 29.2 |
| UVG     | 36.0 | 33.8 | 31.2 | 28.3 | 26.7 |
| HEVC B  | 34.9 | 31.3 | 25.1 | 21.9 | 19.4 |
| HEVC E  | 36.6 | 32.4 | 29.3 | 29.3 | 29.3 |
| HEVC C  | 25.6 | 17.1 | 7.8  | 6.6  | 4.8  |
| HEVC D  | 18.3 | 10.9 | 2.5  | 2.5  | 2.5  |

**b**

compute saved (%), 0.3 dB budget

|         |      |      |      |      |      |
|---------|------|------|------|------|------|
| MCL-JCV | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |
| UVG     | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |
| HEVC B  | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |
| HEVC E  | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |
| HEVC C  | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |
| HEVC D  | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |

**c**

quality given up (dB), 0.1 dB budget

|         |       |       |       |       |       |
|---------|-------|-------|-------|-------|-------|
| MCL-JCV | 0.109 | 0.111 | 0.110 | 0.107 | 0.109 |
| UVG     | 0.117 | 0.108 | 0.106 | 0.108 | 0.107 |
| HEVC B  | 0.076 | 0.078 | 0.078 | 0.081 | 0.076 |
| HEVC E  | 0.137 | 0.136 | 0.149 | 0.144 | 0.137 |
| HEVC C  | 0.066 | 0.062 | 0.062 | 0.059 | 0.061 |
| HEVC D  | 0.034 | 0.043 | 0.035 | 0.049 | 0.064 |

q0      q16      q32      q48      q63

**Figure 40.** The same measurement as a grid, with the looser budget under it. **a**, compute saved at the 0.1 dB budget. **b**, the same at 0.3 dB, where every class at the three lowest rates has reached the architectural ceiling of 39.1% and only q48 and q63 are still making a choice. **c**, the quality actually given up at the 0.1 dB budget, which the set meets and no class does: at q0 the allocation spends 0.116 dB on MCL-JCV and 0.028 dB on HEVC D. The small classes are cheap in quality for the same reason they are poor in saving, that with two or eight tiles there is no fine-grained way to spend a budget. The 0.5 dB grid is not drawn because every one of its thirty cells is at the ceiling.

Drawn by scripts/supp\_results\_figs.py from results/supp\_per\_class\_budgets.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar.

The resolution effect is a property of the tiling and not of the content, which is the argument against reading a single mean over a test set that mixes resolutions: MCL-JCV is 30 of the 53 sequences and all of them 1080p, so it carries most of the pooled row.

## D.2. Adaptivity against its controls

The main paper prints this comparison transposed and compressed to five columns. The full form is below, one block per quality index, because the uniform rows are the ones a reader is most likely to want to read off directly: they say what a decoder that made no per-tile choice at all would deliver at the same budget.

| q  | Allocation           | $\Delta$ PSNR (dB) | MACs saved (%) |
|----|----------------------|--------------------|----------------|
| 0  | uniform, exit 2†     | 0.139              | 39.1           |
|    | uniform, exit 3      | 0.061              | 24.2           |
|    | uniform, exit 4      | 0.041              | 13.0           |
|    | uniform, exit 5      | 0.026              | -1.0           |
|    | random (matched mix) | 0.114              | 34.5           |
|    | rate-ranked (free)   | 0.104              | 34.5           |
|    | <b>oracle</b>        | <b>0.100</b>       | <b>34.5</b>    |
| 16 | uniform, exit 2†     | 0.172              | 39.1           |
|    | uniform, exit 3      | 0.080              | 24.2           |
|    | uniform, exit 4      | 0.055              | 13.0           |
|    | uniform, exit 5      | 0.034              | -1.0           |
|    | random (matched mix) | 0.123              | 30.8           |
|    | rate-ranked (free)   | 0.104              | 30.8           |
|    | <b>oracle</b>        | <b>0.100</b>       | <b>30.8</b>    |
| 32 | uniform, exit 2†     | 0.218              | 39.1           |
|    | uniform, exit 3†     | 0.105              | 24.2           |
|    | uniform, exit 4      | 0.067              | 13.0           |

| q  | Allocation           | $\Delta$ PSNR (dB) | MACs saved (%) |
|----|----------------------|--------------------|----------------|
| 48 | uniform, exit 5      | 0.042              | -1.0           |
|    | random (matched mix) | 0.134              | 26.7           |
|    | rate-ranked (free)   | 0.105              | 26.7           |
|    | <b>oracle</b>        | <b>0.100</b>       | <b>26.7</b>    |
|    | uniform, exit 2†     | 0.253              | 39.1           |
|    | uniform, exit 3†     | 0.122              | 24.2           |
|    | uniform, exit 4      | 0.074              | 13.0           |
| 63 | uniform, exit 5      | 0.048              | -1.0           |
|    | random (matched mix) | 0.142              | 25.3           |
|    | rate-ranked (free)   | 0.107              | 25.3           |
|    | <b>oracle</b>        | <b>0.100</b>       | <b>25.3</b>    |
|    | uniform, exit 2†     | 0.288              | 39.1           |
|    | uniform, exit 3†     | 0.138              | 24.2           |
|    | uniform, exit 4      | 0.081              | 13.0           |
|    | uniform, exit 5      | 0.054              | -1.0           |
|    | random (matched mix) | 0.147              | 24.1           |
|    | rate-ranked (free)   | 0.111              | 24.1           |
|    | <b>oracle</b>        | <b>0.100</b>       | <b>24.1</b>    |

**Table 58. Adaptivity against two controls** at the 0.1 dB budget, in full. Uniform rows are every tile at one depth. The dagger marks a uniform depth whose distortion exceeds the budget, so it is not an admissible allocation at all: at q48 and q63 the only uniform depth inside the budget is the deepest one, which costs 1.0% rather than saving anything. *Random* draws each frame's map from the oracle's own exit histogram and shuffles it across tiles, so the mix of depths and the average cost are unchanged and only the content dependence is discarded. *Rate-ranked* keeps that histogram and orders it by the bits the entropy model spent per tile. Both shuffled rows are matched to the oracle on compute, so they are compared on the decibel column and not on the saving column.

results/static\_RECIPES12\_b01.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar, seed recorded in the file, generated into paper/tables/static.tex by scripts/make\_paper\_tables.py.

Two readings follow from the shuffled rows and they are easy to conflate. Quality falls sharply when the same histogram is assigned at random, so what the allocation buys is knowing *which* tiles can afford to run shallower and not the fact that some fraction of them does. And ordering that same histogram by a free decoder-side signal recovers 77% of the oracle's advantage over chance at q63, which is the control section F turns into a complete routing rule.

## D.3. Where the tiles exit

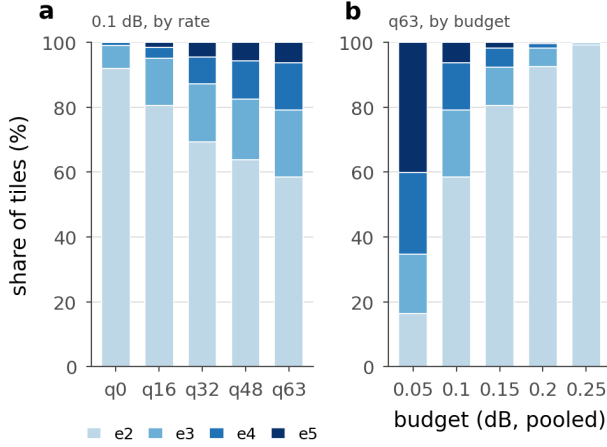
| Rate | e2   | e3   | e4   | e5  | Blocks skipped | Saved |
|------|------|------|------|-----|----------------|-------|
| q0   | 84.0 | 13.5 | 2.1  | 0.5 | 5.62           | 34.4  |
| q16  | 71.7 | 18.8 | 7.0  | 2.6 | 5.19           | 30.8  |
| q32  | 60.2 | 20.7 | 10.8 | 8.3 | 4.66           | 26.6  |
| q48  | 55.3 | 19.6 | 15.8 | 9.3 | 4.42           | 25.2  |
| q63  | 50.1 | 21.5 | 18.5 | 9.9 | 4.24           | 24.1  |

**Table 59.** The exit histogram at the 0.1 dB budget, as a percentage of the 1765 tiles in one pass over the test set, with the mean number of trunk blocks skipped and the saving that buys. At q0 two thirds of tiles take the shallowest selectable exit and the ladder above it is barely used. At q63 the mass is spread almost evenly over the four exits and the mean depth skipped has fallen from 5.04 blocks to 2.82. Exits e0 and e1 are omitted because the exit clamp cannot select them: the split depth is 2, so a map naming a shallower exit runs to e2 and wears e2's adapter.

Histograms summed over the six classes of results/supp\_per\_class\_budgets.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar. Blocks skipped per exit from results/adapt\_cost.json, same checkpoint.



**Figure 41. How deep each class has to go.** Mean exit taken by the tiles of each test class at the 0.1 dB budget, ordered by that mean. Resolution is the strongest predictor: the 416x240 class has two tiles a frame and almost no choice, and the 1080p classes have forty and use the whole ladder. This is the dependence the main paper’s tile-count section measures, seen per class rather than pooled.



**Figure 42. How the ladder fills up.** **a**, exit shares at the 0.1 dB budget as the rate rises. **b**, exit shares at q63 as the budget loosens, until at 0.3 dB 96% of tiles are already at the shallowest exit and there is almost nothing left to allocate. Both panels are pooled over the test set.

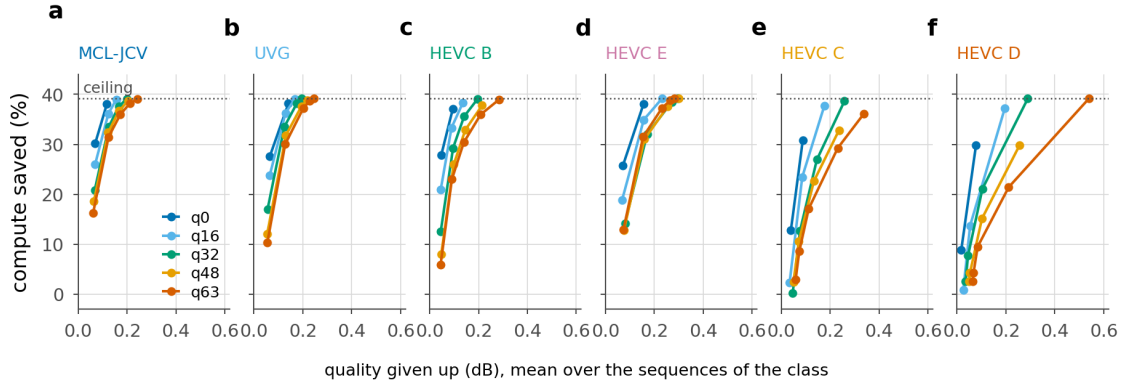
Drawn by scripts/supp\_results\_figs.py from results/supp\_paper\_curve\_PAPER.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar, pooled convention, so the budgets on the horizontal axis of **b** are looser than the identically labelled ones in the table above.

| Class   | e2   | e3   | e4   | e5   | e2   | e3   | e4   | e5   |
|---------|------|------|------|------|------|------|------|------|
| MCL-JCV | 88.5 | 9.3  | 1.8  | 0.3  | 57.8 | 19.7 | 14.4 | 8.1  |
| UVG     | 80.4 | 18.2 | 1.4  | 0.0  | 40.0 | 31.1 | 27.5 | 1.4  |
| HEVC B  | 75.5 | 21.0 | 2.0  | 1.5  | 26.5 | 20.5 | 32.5 | 20.5 |
| HEVC E  | 84.4 | 13.3 | 2.2  | 0.0  | 57.8 | 24.4 | 6.7  | 11.1 |
| HEVC C  | 18.8 | 68.8 | 12.5 | 0.0  | 0.0  | 12.5 | 18.8 | 68.8 |
| HEVC D  | 0.0  | 62.5 | 25.0 | 12.5 | 0.0  | 0.0  | 25.0 | 75.0 |

**Table 60.** The same histogram split by class at the two ends of the rate range: the first four columns are q0 and the second four are q63, each as a percentage of that class’s tiles. HEVC C and HEVC D reach neither of the two shallowest exits at q63, and three quarters of their tiles run to full depth, which is the 2.5% saving of the first table seen tile by tile. UVG behaves differently from MCL-JCV at q63 despite being the same resolution, which is content and not geometry.

results/supp\_per\_class\_budgets.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar.

#### D.4. The frontier, class by class



**Figure 43.** Compute saved against quality given up, one panel per class and one curve per quality index, from the seven budgets at which the frontier was measured. The dotted line is the architectural ceiling of 39.1%. Three things are visible that a table of operating points hides. The curves are steep and then flat, so most of the saving is bought in the first tenth of a decibel and the rest of the band buys little. The rate ordering is preserved at every budget, so a curve never crosses another within a panel. And the panels move to the right as resolution falls: MCL-JCV reaches the ceiling by 0.3 dB at every rate, while HEVC D needs 0.5 dB at q48 and does not reach it at q63 within the measured range.

Drawn by scripts/supp\_results\_figs.py from results/supp\_paper\_curve\_PAPER.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar. A point is the mean over the sequences of that class at a global operating point, so the multiplier is the one the whole set was bisected to and not a per-class retuning. Class membership per sequence from results/supp\_opquality\_PAPER.json, same checkpoint.

Two limits bound every panel. A budget below the floor cannot be met at all, because our deepest exit already differs from the released decoder by 0.026 dB at q0 and 0.054 dB at q63 in the per-frame convention; a budget above 0.139 dB at q0 or 0.288 dB at q63 changes nothing, because every tile is already at the shallowest selectable exit. Between those two the multiplier is doing work, and the 0.1 dB budget is inside the window at every rate.

### D.5. Per-sequence spread

A set mean is not what any single clip receives. One multiplier prices compute for the whole set, so each sequence lands wherever its own content puts it, and the spread of those landings is what a deployment would have to plan against.

| Rate | n  | Mean | SD   | Min  | p25  | Median | p75  | Max  |
|------|----|------|------|------|------|--------|------|------|
| q0   | 53 | 36.8 | 3.7  | 24.2 | 35.8 | 38.8   | 39.1 | 39.1 |
| q16  | 53 | 33.1 | 8.1  | -1.0 | 30.4 | 36.1   | 38.8 | 39.1 |
| q32  | 53 | 29.5 | 10.2 | -1.0 | 26.1 | 32.6   | 37.6 | 39.1 |
| q48  | 53 | 27.9 | 10.7 | -1.0 | 23.2 | 31.1   | 36.5 | 39.1 |
| q63  | 53 | 26.7 | 10.8 | -1.0 | 22.6 | 29.9   | 34.3 | 39.1 |

**Table 61.** Compute saved per sequence at the 0.1 dB budget: the distribution behind the mean, over all 53 sequences. The interquartile range widens from 9.9 points at q0 to 15.0 at q63, and the gap between best and worst sequence widens from 33.1 to 40.1. The minimum is negative at four of the five rates, because a sequence whose tiles all run to full depth costs 0.95% more than the released decoder: our deepest exit is the more expensive of the two. Counting over rates as well as sequences, 15 of the 265 pairs save less than a tenth of a decode at this budget.

results/supp\_per\_sequence\_PAPER\_b010.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar, read from its rows\_vs\_release block so that the denominator is the released decoder. Pooled convention, because the operating points come from results/supp\_paper\_curve\_PAPER.json.

| Sequence                | q0   | q63  | q63 dB |
|-------------------------|------|------|--------|
| BQSquare_416x240        | 24.2 | -1.0 | 0.146  |
| RaceHorses_416x240      | 24.2 | -1.0 | 0.064  |
| RaceHorses_832x480      | 31.7 | 3.9  | 0.079  |
| PartyScene_832x480      | 27.9 | 5.7  | 0.098  |
| BlowingBubbles_416x240  | 31.7 | 6.0  | 0.072  |
| BasketballDrill_832x480 | 30.3 | 7.4  | 0.001  |
| videoSRC27              | 39.1 | 39.1 | 0.127  |
| videoSRC29              | 39.1 | 39.1 | 0.126  |

**Table 62.** The six hardest sequences at q63 and the two easiest, with their q0 value beside them and the decibel each actually received. All six of the hard ones are 832x480 or 416x240. The delivered decibel is not the budget on any of them: across the 53 sequences at q63 it runs from 0.013 dB to 0.237 dB against a 0.1 dB target, so a single multiplier for the set means some clips overshoot the budget by more than a factor of two while others use an eighth of it.

results/supp\_per\_sequence\_PAPER\_b010.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar. Savings are against the released decoder; the decibel is the sequence's own, in the pooled convention.

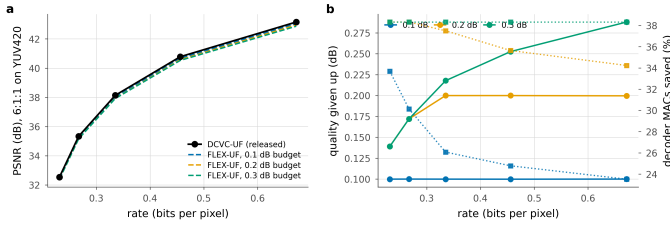
### D.6. A second metric, and the tiles that pay for the mean

The whole mechanism is denominated in PSNR, and PSNR is known not to track visual quality closely. Two things follow that are worth measuring rather than asserting: whether a perceptual metric agrees that the loss is small, and how the loss is distributed over tiles, since a mean of 0.1 dB is compatible with a few tiles losing a great deal.

| Rate | bpp    | PSNR   | ours   | lost   | MS-SSIM | ours    | lost   |
|------|--------|--------|--------|--------|---------|---------|--------|
| q0   | 0.2314 | 32.536 | 32.435 | 0.1012 | 0.89934 | 0.89818 | 0.0498 |
| q16  | 0.2671 | 35.335 | 35.235 | 0.1003 | 0.94904 | 0.94855 | 0.0417 |
| q32  | 0.3351 | 38.135 | 38.036 | 0.0990 | 0.97565 | 0.97544 | 0.0375 |
| q48  | 0.4560 | 40.777 | 40.675 | 0.1014 | 0.98790 | 0.98780 | 0.0334 |
| q63  | 0.6716 | 43.155 | 43.044 | 0.1117 | 0.99419 | 0.99414 | 0.0390 |

**Table 63.** PSNR and MS-SSIM at the 0.1 dB operating point, over the whole test set. MS-SSIM agrees with PSNR about the direction and is kinder about the size: expressed as  $-10\log_{10}(1 \text{ minus MS-SSIM})$ , the loss is 0.052 dB at q0 against 0.102 dB of PSNR, and 0.046 against 0.118 at q63, so on this metric the budget costs about half what the decibel it is written in suggests. Each metric is given as the released decoder, ours, and what was lost, with the MS-SSIM loss expressed as  $-10\log_{10}(1 \text{ minus MS-SSIM})$  so that the two loss columns are in the same unit. The bpp column is the released bitstream, which early exiting does not change beyond the exit map.

results/supp\_opquality\_PAPER.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar, 53 sequences at one frame each. MS-SSIM is Wang et al. 2003 at five scales with an 11-tap Gaussian of  $\sigma = 1.5$  on the valid region, on the luma plane in 4:2:0 on 0 to 255, the domain the PSNR column uses; the implementation self-checks at start-up and the file records that identical inputs score 1.000000 and a 3x3 box blur 0.7797.  $\lambda$  is read from results/signalised\_RECIPE512\_ctc53.json so the allocation is the deployed one, and the delivered decibel recomputed here matches the stored value to four decimals.



**Figure 44.** **a**, rate against quality over the five quality indices, the released DCVC-UF intra decoder and FLEX-UF at three budgets, on the 53 CTC intra frames. Both axes come from one pass over the same frames, and the encoders are asserted bit-identical on the weights before a shared rate is reported, because a shared x-axis is a claim rather than a convenience. **b**, the same difference magnified: the decibels each budget delivered, against the decoder arithmetic it bought on the right-hand axis.

Drawn by scripts/rd\_vs\_uf\_figure.py from results/rd\_absolute\_PAPER.json and results/signalled\_RECIPES12\_ctc53.json, both on runs/RECIPES12/ckpt\_PAPER.pth.tar.



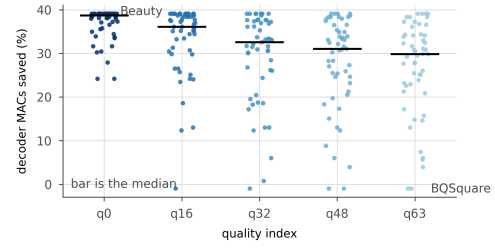
**Figure 45.** **a**, the rate-quality curve of the released decoder and of ours at the 0.1 dB budget, over the five quality indices. At plot scale they are one curve, which is what a 0.1 dB budget means. **b**, the gap that panel **a** cannot show, in PSNR and in MS-SSIM expressed as a decibel.

Drawn by scripts/supp\_results\_figs.py from results/supp\_opquality\_PAPER.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar.

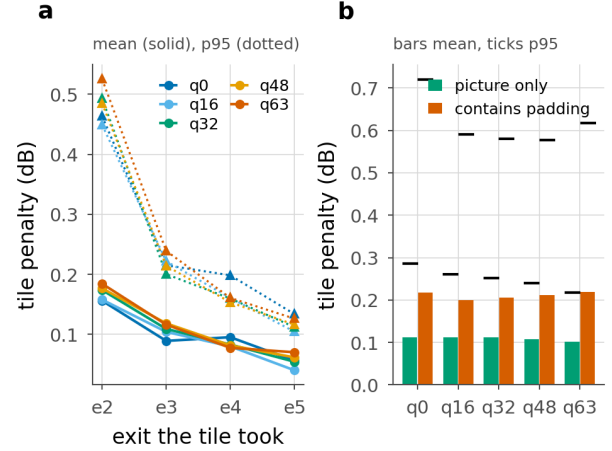
| Rate | Tiles | Mean  | Median | p95   | p99   | Max   | Over 0.25 | Over 0.5 | Over 1.0 |
|------|-------|-------|--------|-------|-------|-------|-----------|----------|----------|
| q0   | 1765  | 0.145 | 0.106  | 0.429 | 0.945 | 1.379 | 244       | 66       | 15       |
| q16  | 1765  | 0.139 | 0.109  | 0.388 | 0.660 | 1.299 | 211       | 50       | 5        |
| q32  | 1765  | 0.141 | 0.107  | 0.393 | 0.640 | 1.032 | 218       | 48       | 1        |
| q48  | 1765  | 0.140 | 0.109  | 0.374 | 0.722 | 1.423 | 191       | 47       | 4        |
| q63  | 1765  | 0.139 | 0.106  | 0.386 | 0.744 | 1.421 | 195       | 48       | 4        |

**Table 64.** How the 0.1 dB is distributed over the 1765 tiles of one pass over the test set. The last three columns count tiles losing more than a quarter, a half and a whole decibel. The median tile is at or below the budget at every rate, and the tail is long: at q0, 302 tiles lose more than 0.25 dB, 79 lose more than 0.5 dB, and the worst loses 1.97 dB. A budget stated as a mean says nothing about the last three columns, which is why they are printed.

results/supp\_opquality\_PAPER.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar, at the same operating points as the metric table above. A tile penalty is the decibel between the released decode of that tile and ours.



**Figure 46.** What the set mean hides. Every test sequence as a point, at the 0.1 dB budget; the bar is the median. At the lowest rate the saving runs from 24.2% to 39.1% across the 53 sequences and at the highest from -1.0% to 39.1%. The spread within a rate is larger than the difference between rates, and it is content, not noise: the same sequences sit at the same end of it at every rate.



**Figure 47.** Where the tail comes from. **a**, mean and 95th percentile tile penalty against the exit the tile took. The tail sits on e2, the shallowest exit the clamp allows: at q0 its mean is 0.202 dB against 0.054 dB at e5, and all 79 tiles above 0.5 dB took it. **b**, the same penalty split by whether the tile contains replicated padding. A 1080p frame is padded to 2048x1280, so 200 of the bottom tile row's 256 rows are invented by replication, and those tiles carry roughly double the mean penalty at every rate: 0.261 dB against 0.121 dB at q0, and 65 of the 79 tiles above half a decibel are padded ones.

Drawn by scripts/supp\_results\_figs.py from the tail\_by\_exit and tail\_by\_padding blocks of results/supp\_opquality\_PAPER.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar. A tile counts as padded if any of its pixels came from the replicate pad, recorded per tile as pad\_fraction. Padding geometry from results/tile\_definition.json, same checkpoint.

## D.7. Composition with weight quantisation

Early exiting removes work and quantisation makes the remaining work cheaper, so the two ought to compose. Whether they do is a question about the same weights, and it is measured here rather than assumed.

| Weights | Rate | e2    | e3    | e4    | e5    | Layers |
|---------|------|-------|-------|-------|-------|--------|
| float32 | q0   | 0.132 | 0.058 | 0.039 | 0.021 | 0      |
| float32 | q32  | 0.284 | 0.144 | 0.100 | 0.037 | 0      |
| float32 | q63  | 0.650 | 0.371 | 0.265 | 0.085 | 0      |
| int8    | q0   | 0.140 | 0.065 | 0.045 | 0.026 | 86     |
| int8    | q32  | 0.315 | 0.171 | 0.129 | 0.062 | 86     |
| int8    | q63  | 0.776 | 0.496 | 0.394 | 0.207 | 86     |
| int6    | q0   | 0.297 | 0.217 | 0.206 | 0.169 | 86     |
| int6    | q32  | 1.046 | 0.877 | 0.845 | 0.736 | 86     |
| int6    | q63  | 2.794 | 2.535 | 2.506 | 2.287 | 86     |
| int4    | q0   | 1.835 | 1.594 | 1.555 | 1.390 | 86     |
| int4    | q32  | 4.390 | 4.120 | 4.138 | 3.907 | 86     |
| int4    | q63  | 8.356 | 8.039 | 8.139 | 7.898 | 86     |

**Table 65.** Quality given up against the released decoder with the decoder weights quantised, at four exits and three rates. The float32 rows are the ladder itself and are the baseline each integer row should be read against. At q0 int8 is nearly free, moving the deepest exit from 0.027 dB to 0.032 dB, so the two levers compose. At q63 they compete: the deepest exit moves from 0.108 dB to 0.223 dB, which is more than a whole 0.1 dB budget spent before any tile has exited early. Six-bit costs 0.13 dB at the deepest exit at q0 and 2.18 dB at q63, and four-bit is unusable at every rate. The last column is the number of layers quantised, the same 86 in every integer row.

results/supp\_quant\_PAPER.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar, 32 held-out images at 512 px. Weight-only, symmetric, per output channel, no calibration, on dec.\* and router\_head.\* with the encoder asserted unchanged. Nothing here measures an integer kernel: the file records the bit width and the implied bit-operation ratio, and the arithmetic ran in floating point on quantised weights.

## D.8. Integrated figures, and what they depend on

A budget is one sample of a curve. Video coding answers this for rate against quality with Bjontegaard's construction [4], integrating one axis over a stated interval of the other, and the same construction applies with compute in place of rate. BD-saving is the mean compute saved over an interval of decibels and BD-quality is the mean decibel paid over an interval of saving. Neither means anything without its interval and its convention, and both are given with theirs.

| Rate               | BD-saving | BD-quality | BD-saving | BD-quality | Points |
|--------------------|-----------|------------|-----------|------------|--------|
| q0                 | 33.2%     | 0.0372     | 34.9%     | 0.0330     | 15     |
| q16                | 29.4%     | 0.0509     | 31.6%     | 0.0437     | 14     |
| q32                | 24.8%     | 0.0652     | 27.7%     | 0.0544     | 13     |
| q48                | 23.0%     | 0.0705     | 25.7%     | 0.0595     | 13     |
| q63                | 20.9%     | 0.0758     | 24.0%     | 0.0636     | 12     |
| Mean where defined | 26.3%     | 0.0599     | 28.8%     | 0.0508     |        |

**Table 66.** BD-saving and BD-quality per rate under both decibel conventions: the first pair of columns is per frame and the second is pooled. BD-saving is defined at only two of the five rates, because at q0, q16 and q32 the frontier saturates below the upper limit of the interval and there is no curve left to integrate. The convention moves the answer by 1.9 points of BD-saving and 0.018 dB of BD-quality on identical allocations, and it moves it consistently, since per-frame averaging raises every floor and pooling does not. Points is the number of measured frontier points inside the interval.

results/supp\_bd\_PAPER\_per\_frame.json and results/supp\_bd\_PAPER\_pooled.json, both on runs/RECIPE512/ckpt\_PAPER.pth.tar over 53 sequences, integrating results/supp\_paper\_curve\_PAPER.json. BD-saving is over 0.066 to 0.300 dB per frame and 0.059 to 0.300 dB pooled, the lower limit being the highest floor any rate has under that convention. BD-quality is over 10% to 30% saving. The floors that set those lower limits are recorded per rate in both files as floor\_db, and appear directly in the frontier table below at the two rates where the 0.05 dB budget is unreachable.

Two sensitivities sit around those numbers and are worth stating beside them. The first is the anchor. Our deepest exit is not bit-exact with the released decoder, since it carries seam repair and a fine-tuned trunk, and it starts every budget slightly behind: the drift runs from 0.005 dB at q0 to 0.036 dB at q63, which at q63 is a third of the whole budget. Setting it to zero would raise BD-saving at q63 from 28.7% to 34.9%, a gain of 6.2 points. The second is the interval. Over six defensible intervals the mean BD-saving moves from 22.8% to 29.2%, a range of 6.3 points, while the gap between the lowest and highest rate stays inside 15.3 to 16.8 points

on every one of them; the level depends on the integration and the rate effect does not.

Anchor drift from results/supp\_anchor\_PAPER.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar, 53 frames, per-frame convention; the zero-drift figure is the bd\_saving\_pct\_zero\_drift field of results/supp\_bd\_PAPER\_per\_frame.json. Interval sensitivity from results/bd\_sensitivity.json, which is not pinned: scripts/bd\_sensitivity.py names its curve as results/paper\_curve\_grid128.json, on runs/wdec\_j2\_p128\_grid/ckpt\_epo0.pth.tar at 128 px tiles over 40 sequences, so its levels are not the levels above and only its robustness conclusion is quoted.

| Budget | A BD-Rate | B BD-Rate | Difference | A saved | B saved |
|--------|-----------|-----------|------------|---------|---------|
| 0.1 dB | 1.032%    | 1.007%    | 0.025 pt   | 28.3%   | 25.3%   |
| 0.3 dB | 2.339%    | 2.318%    | 0.020 pt   | 39.1%   | 39.0%   |
| 0.5 dB | 2.339%    | 2.318%    | 0.020 pt   | 39.1%   | 39.0%   |

**Table 67.** BD-Rate cost of the two configurations: the rate a reader would have to spend to buy back the quality the compute saving costs, integrated over the five rate points, which is the number that makes this trade-off comparable with a published codec result. Configuration A signals the exit map in the bitstream and pays 64 to 91 bits a frame for it at the 0.1 dB budget, depending on the rate, configuration B infers it from what the decoder already holds and pays none. Their BD-Rate costs differ by at most 0.025 of a point, so the map is close to free. What A buys with it is compute: the gap in the last two columns is 3.0 points at the tight budget and 0.2 at the loose one, where both configurations are at the ceiling.

results/bdtrate.json, every row on runs/RECIPE512/ckpt\_PAPER.pth.tar, which its sources block records file by file.

## D.9. Raw values behind the curves

Everything above is an integral, a mean or a picture. What follows is the measured numbers themselves, so that a reader can compare against them without digitising a plot or rerunning a decode.

| Rate | Budget | lambda    | dB     | Model | Meter | Map bits |
|------|--------|-----------|--------|-------|-------|----------|
| q0   | 0.1    | 7.280e-05 | 0.0999 | 34.5  | 33.7  | 64       |
| q0   | 0.2    | 1.000e+00 | 0.1391 | 39.1  | 38.3  | 48       |
| q0   | 0.3    | 1.000e+00 | 0.1391 | 39.1  | 38.3  | 48       |
| q0   | 0.5    | 1.000e+00 | 0.1391 | 39.1  | 38.3  | 48       |
| q16  | 0.1    | 2.912e-05 | 0.1000 | 30.8  | 30.1  | 75       |
| q16  | 0.2    | 1.000e+00 | 0.1720 | 39.1  | 38.3  | 48       |
| q16  | 0.3    | 1.000e+00 | 0.1720 | 39.1  | 38.3  | 48       |
| q16  | 0.5    | 1.000e+00 | 0.1720 | 39.1  | 38.3  | 48       |
| q32  | 0.1    | 1.305e-05 | 0.0999 | 26.7  | 26.1  | 84       |
| q32  | 0.2    | 8.293e-05 | 0.2000 | 38.3  | 37.5  | 52       |
| q32  | 0.3    | 1.000e+00 | 0.2178 | 39.1  | 38.3  | 48       |
| q32  | 0.5    | 1.000e+00 | 0.2178 | 39.1  | 38.3  | 48       |
| q48  | 0.1    | 7.315e-06 | 0.0999 | 25.4  | 24.8  | 89       |
| q48  | 0.2    | 3.485e-05 | 0.2000 | 36.4  | 35.6  | 57       |
| q48  | 0.3    | 1.000e+00 | 0.2526 | 39.1  | 38.3  | 48       |
| q48  | 0.5    | 1.000e+00 | 0.2526 | 39.1  | 38.3  | 48       |
| q63  | 0.1    | 4.534e-06 | 0.1000 | 24.1  | 23.5  | 91       |
| q63  | 0.2    | 1.972e-05 | 0.1997 | 35.0  | 34.2  | 61       |
| q63  | 0.3    | 1.000e+00 | 0.2880 | 39.1  | 38.3  | 48       |
| q63  | 0.5    | 1.000e+00 | 0.2880 | 39.1  | 38.3  | 48       |

**Table 68.** The deployed operating points: four budgets at five rates. Model is the saving the cost model predicts for the chosen map and Meter is what a forward hook on every convolution and linear layer counted on the same decode, so the two columns are the cost model checked against itself; the model reads high by 0.4 to 0.8 points throughout. Map bits is the entropy of the per-frame exit histogram, charged to the bitstream before any saving is computed, and it falls as the budget loosens because the histogram concentrates.

results/signalled\_RECIPE512\_ctc53.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar, 53 sequences, per-frame convention.  $\lambda$  is the multiplier the two-level bisection settled on, and a  $\lambda$  of exactly 1 is the top of its bracket, which marks a budget the ladder saturated below rather than a price that was chosen. Saving is against the released decoder.

| Rate | Budget | lambda    | dB pooled | dB per frame | Saved | e2/e3/e4/e5 |
|------|--------|-----------|-----------|--------------|-------|-------------|
| q0   | 0.05   | 2.742e-05 | 0.0500    | 0.0611       | 29.0  | 51/31/11/6  |
| q0   | 0.1    | 1.077e-04 | 0.1000    | 0.1157       | 37.8  | 92/7/1/0    |

| Rate | Budget | lambda    | dB pooled | dB per frame | Saved | e2/e3/e4/e5 |
|------|--------|-----------|-----------|--------------|-------|-------------|
| q0   | 0.15   | n/a       | 0.1239    | 0.1391       | 39.1  | 100/0/0/0   |
| q0   | 0.2    | n/a       | 0.1239    | 0.1391       | 39.1  | 100/0/0/0   |
| q0   | 0.25   | n/a       | 0.1239    | 0.1391       | 39.1  | 100/0/0/0   |
| q0   | 0.3    | n/a       | 0.1239    | 0.1391       | 39.1  | 100/0/0/0   |
| q0   | 0.5    | n/a       | 0.1239    | 0.1391       | 39.1  | 100/0/0/0   |
| q16  | 0.05   | 1.019e-05 | 0.0500    | 0.0605       | 24.3  | 40/28/15/17 |
| q16  | 0.1    | 4.165e-05 | 0.0999    | 0.1148       | 35.4  | 80/15/3/2   |
| q16  | 0.15   | 1.336e-04 | 0.1500    | 0.1646       | 38.9  | 98/2/0/0    |
| q16  | 0.2    | n/a       | 0.1585    | 0.1720       | 39.1  | 100/0/0/0   |
| q16  | 0.25   | n/a       | 0.1585    | 0.1720       | 39.1  | 100/0/0/0   |
| q16  | 0.3    | n/a       | 0.1585    | 0.1720       | 39.1  | 100/0/0/0   |
| q16  | 0.5    | n/a       | 0.1585    | 0.1720       | 39.1  | 100/0/0/0   |
| q32  | 0.05   | 4.037e-06 | 0.0500    | 0.0608       | 18.6  | 25/26/21/28 |
| q32  | 0.1    | 1.772e-05 | 0.1000    | 0.1157       | 32.5  | 69/18/8/4   |
| q32  | 0.15   | 4.134e-05 | 0.1500    | 0.1647       | 37.2  | 89/9/2/0    |
| q32  | 0.2    | 1.435e-04 | 0.1998    | 0.2152       | 39.1  | 100/0/0/0   |
| q32  | 0.25   | n/a       | 0.2019    | 0.2178       | 39.1  | 100/0/0/0   |
| q32  | 0.3    | n/a       | 0.2019    | 0.2178       | 39.1  | 100/0/0/0   |
| q32  | 0.5    | n/a       | 0.2019    | 0.2178       | 39.1  | 100/0/0/0   |
| q48  | 0.05   | 1.884e-06 | 0.0500    | 0.0599       | 15.8  | 22/18/24/35 |
| q48  | 0.1    | 9.772e-06 | 0.1000    | 0.1148       | 31.0  | 64/19/12/6  |
| q48  | 0.15   | 2.126e-05 | 0.1500    | 0.1666       | 36.1  | 84/11/4/1   |
| q48  | 0.2    | 5.103e-05 | 0.1999    | 0.2191       | 38.5  | 96/3/1/0    |
| q48  | 0.25   | n/a       | 0.2305    | 0.2526       | 39.1  | 100/0/0/0   |
| q48  | 0.3    | n/a       | 0.2305    | 0.2526       | 39.1  | 100/0/0/0   |
| q48  | 0.5    | n/a       | 0.2305    | 0.2526       | 39.1  | 100/0/0/0   |
| q63  | 0.05   | 1.054e-06 | 0.0500    | 0.0598       | 13.7  | 16/18/25/40 |
| q63  | 0.1    | 6.072e-06 | 0.1000    | 0.1154       | 29.7  | 58/21/15/6  |
| q63  | 0.15   | 1.282e-05 | 0.1499    | 0.1658       | 35.1  | 80/12/6/2   |
| q63  | 0.2    | 2.536e-05 | 0.1998    | 0.2180       | 37.8  | 93/6/2/0    |
| q63  | 0.25   | 6.566e-05 | 0.2498    | 0.2784       | 39.0  | 99/1/0/0    |
| q63  | 0.3    | n/a       | 0.2613    | 0.2880       | 39.1  | 100/0/0/0   |
| q63  | 0.5    | n/a       | 0.2613    | 0.2880       | 39.1  | 100/0/0/0   |

**Table 69.** The frontier itself: every operating point behind the per-class figure, at seven budgets and five rates, with the exit shares that produced each one. Both decibel conventions are printed for the same allocation, and they differ by 0.010 to 0.029 dB over the table and by 0.015 to 0.016 dB at the 0.1 dB budget, which is a fifth of that budget. A row reading n/a in the saving column is a budget below the floor at that rate, where no allocation meets the target. A row reading 100/0/0/0 is past saturation: the file stores neither a multiplier nor a histogram there, and the shares are filled in from what saturation means, every tile at the shallowest selectable exit.

results/supp\_paper\_curve\_PAPER.json, on runs/RECIPE512/ckpt\_PAPER.pth.tar. Saving is converted to the released-decoder denominator by charging the retained cost the price of our deepest exit, 1.0095 released decodes, which is 1.0 at the precision the paper prints and is derived here from the ceiling in results/saturation\_RECIPE512\_ctc53.json and the saturated saving in this file. One caution about that denominator: this file bills the shared stem inside every tile's cost, while results/supp\_latency\_batch\_1920x1080.json bills it once per frame through flexuf/cost.py frame\_relative\_cost, and the two accountings differ by about a third of a point on an identical exit map at q0.

## D.10. Reusing an exit map

The signalled configuration pays one extra decode per frame at the encoder to find its map. How much that matters depends on how often the map has to be found again, so we probed two kinds of reuse: across frames of one sequence, and across the quality index. The probe is narrow and is reported as one.

| reuse  | map found at | applied at | in place, dB | reused, dB | in place, saved % | reused, saved % |
|--------|--------------|------------|--------------|------------|-------------------|-----------------|
| frames | q0 frame 0   | frame 1    | 0.0998       | 0.1014     | 34.20             | 34.20           |
| frames | q0 frame 0   | frame 2    | 0.0991       | 0.0994     | 34.40             | 34.20           |
| frames | q0 frame 0   | frame 4    | 0.0991       | 0.1007     | 34.40             | 34.20           |
| frames | q0 frame 0   | frame 8    | 0.0997       | 0.1024     | 34.52             | 34.20           |
| frames | q32 frame 0  | frame 1    | 0.0976       | 0.1012     | 30.43             | 30.89           |
| frames | q32 frame 0  | frame 2    | 0.0955       | 0.0992     | 30.66             | 30.89           |
| frames | q32 frame 0  | frame 4    | 0.0957       | 0.0970     | 30.94             | 30.89           |
| frames | q32 frame 0  | frame 8    | 0.0999       | 0.0976     | 31.76             | 30.89           |
| frames | q63 frame 0  | frame 1    | 0.0905       | 0.0930     | 26.35             | 26.46           |
| frames | q63 frame 0  | frame 2    | 0.0905       | 0.0933     | 26.24             | 26.46           |
| frames | q63 frame 0  | frame 4    | 0.0916       | 0.0939     | 26.33             | 26.46           |
| frames | q63 frame 0  | frame 8    | 0.0926       | 0.0934     | 26.71             | 26.46           |
| rate   | q0           | q32        | 0.0982       | 0.1276     | 30.89             | 34.20           |
| rate   | q0           | q63        | 0.0925       | 0.1586     | 26.46             | 34.20           |
| rate   | q32          | q0         | 0.0998       | 0.0947     | 34.20             | 30.89           |
| rate   | q32          | q63        | 0.0925       | 0.1228     | 26.46             | 30.89           |
| rate   | q63          | q0         | 0.0998       | 0.0824     | 34.20             | 26.46           |
| rate   | q63          | q32        | 0.0982       | 0.0874     | 30.89             | 26.46           |

**Table 70. What a reused exit map delivers.** Each row applies a map found in one place to a decode somewhere else and reports what that decode actually delivered, beside the map recomputed in place. Read the two decibel columns first: a reused map is only usable if the decode it produces still meets the budget, and the saving column means nothing on a row where it does not.

results/map\_transfer.json, budget 0.1 dB, on the checkpoint the file records. The offsets and the quality pairs in the file are the whole of the probe; nothing here is a statement about reuse in general.

Across frames of one sequence, reuse is close to free. Taking the map found on frame 0 and applying it eight frames later raises the delivered distortion by 0.003 dB at q0, about five percent of the budget, and the saving does not move at all: 34.20% transferred against 34.20–34.52% recomputed. The allocation is a property of where the content is hard, and that moves slowly. An encoder that searched once per group of pictures rather than once per frame would divide its extra cost by the group length, on the offsets we probed.

Across the quality index the map does have to be recomputed, and the direction decides how badly. Found at q0 and applied at q63 it delivers 0.159 dB against a 0.1 dB budget, claiming the low-rate saving of 34.20% while spending nearly twice the quality it is allowed. The reverse is safe and wasteful. Shallow exits are cheap in quality at low rate and expensive at high rate, so a map is calibrated to the rate it was found at, and reusing one upward breaks the quality guarantee with nothing in the decode reporting that it has. In our reuse probe, then, an encoder can search once per rate and reuse that search across the frames we tested; how far that carries beyond the offsets and sequences probed here we have not measured.

## D.11. Failure cases

Five cases are worth naming, all of them measured in the files above rather than inferred.

- **Sequences that cost more than they save.** At q63 four sequences finish at minus 0.95%, which is every tile at full depth and our deepest exit costing that much more than a released decode: RaceHorses at

832x480 and at 416x240, BlowingBubbles and BQSquare. One more sits under 3% and ten in all save less than a tenth of a decade. On this content the method should be switched off rather than run at a budget, and the exit histogram is what tells the encoder so.

- **The bottom tile row of a padded frame.** The six worst tiles in the test set are all in the last row of a 1080p frame at 78% padding, each losing between 1.72 and 1.97 dB. The allocation is behaving correctly given its table, and the table is measuring a region that will be cropped away before anyone looks at it, so the budget is being spent on invented pixels.
- **The two smallest classes at high rate.** HEVC C and HEVC D save 2.5% at q63, which is one tile in four moved one exit up and nothing else. With two tiles on a 416x240 frame there is no allocation worth making, and the honest reading is that this method needs a frame large enough to hold a useful number of tiles.
- **Quantisation and early exit compete at high rate.** Int8 weights cost 0.115 dB at q63 at the deepest exit, more than the whole budget the allocation is then asked to work inside, while at q0 they cost 0.005 dB. A deployment cannot assume the two savings multiply.
- **A 0.5 dB budget has no per-sequence breakdown to report.** At 0.5 dB every rate is past saturation, the frontier stops at the ceiling of 39.1%, and a saturated operating point carries no allocation to break down, which is why the per-sequence sweep runs from 0.10 dB to 0.30 dB.

Worst tiles from the worst\_tiles block of results/supp\_opquality\_PAPER.json and negative sequences from results/supp\_per\_sequence\_PAPER\_b010.json, both on runs/RECIPE512/ckpt\_PAPER.pth.tar; the saturation points from results/saturation\_RECIPE512\_ctc53.json.



**Figure 48. What the saving looks like.** The same bitstream decoded by the released decoder and by ours at the 0.1 dB operating point. The crop is the tile that gave up the most quality on this frame, chosen automatically rather than by eye, so it shows the method's worst case here and not a flattering one.



**Figure 49. The same frame at the top of the rate range.** Bosphorus at q63, 0.4653 bpp, the same 0.1 dB budget and the same crop as the figure above. The released decoder reaches 45.26 dB and the routed decode 45.16 dB for 0.099 dB delivered, and the saving is 17.4% of the arithmetic against 28.1% at q32.

The allocation is what moved between the two pictures. On this frame 15 of 40 tiles take the shallowest exit at q32 and 1 does at q63, while the deepest exit goes from 0 tiles to 7. The picture is as hard to tell apart at either rate; what changes is how much of the decoder the budget will release, which is the rate dependence of the main paper's operating-point section arriving in a single frame.

## E. The operating window

A budget D is the peak-signal-to-noise ratio the decode is allowed to give up, and an allocation is admissible if the loss it incurs stays at or below it. Between a floor, below which no allocation is admissible, and a

saturation point, above which every allocation has reached the same ceiling, lies the window of budgets in which the decoder has anything to trade. Section C derives the two ends and Section D tabulates them. This section is about the window itself.

It covers how the two ends are measured and how far the measurements agree, what has to be stated with every decibel before any of it means anything, the sweep across the whole window, and how much of the rescaling that collapses the five rates onto one curve survives a change of sweep, of convention or of checkpoint.

### E.1. Floor, saturation point and ceiling

The floor  $D_{\min}$  is the loss of a tiled decode with every tile at full depth. No early exiting has happened, so it is the tiling penalty alone: the border of every tile has been convolved against padding rather than against its neighbour. The saturation point  $D_{\text{sat}}$  is the loss when every tile takes exit j, the shallowest the split depth allows. Neither is a quantity to search for. Each is one forward pass per rate with the exit map held constant, which is what makes the rescaling in Section E.4 cheap enough to be worth doing in deployment.

Between them the saving is fixed by the ladder rather than by the budget. With  $c_k$  the cost of exit k in units of one released full-frame decode, j the split depth and K the number of exits, the ceiling is closed form under either compute denominator:

$$S_{\max}^{\text{rel}} = 100(1 - c_j), \quad S_{\max}^{\text{self}} = 100(1 - c_j/c_{K-1}), \quad (30)$$

With  $c_j = 0.6088$ , recorded in results/saturation\_RECIPE512\_ctc53.json, and  $c_{K-1} = 1.0095$ , recorded as deepest\_exit\_cost in results/router\_RECIPE512\_b01.json, the first reads 39.1% and the second 39.7%. The deepest exit costs more than one released decode because it carries the seam repair, which the released decoder does not run, and that difference is the whole of the gap between the two ceilings.

Section D tabulates both ends at all nine measured rates, and that table is not repeated here. The one property of it this section needs is that a budget in decibels means something different at each rate: both ends widen as the rate falls, the band widens faster than the floor rises, and the 0.1 dB budget that sits 65% of the way up the band at q0 sits 20% of the way up at q63. The rescaling of Section E.4 turns out to need that position rather than the decibel figure.

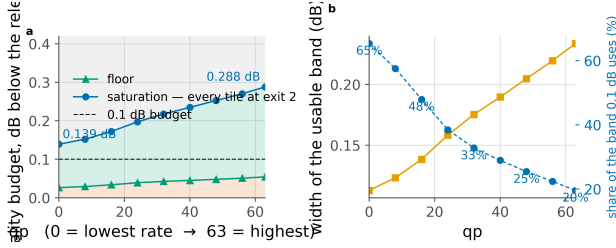
A budget below the floor admits no allocation at all, which is a different failure from a budget falling in a gap of the Lagrangian hull, where the sweep returns the nearest reachable point and Everett's argument still makes it optimal for the compute it consumes [26, 27]. Below the floor the feasible set is empty and there is nothing to return. No cell of the grid in Section E.3 is in that state on the reported weights: the smallest budget swept, 0.05 dB, is above the floor at every rate, so every cell has an allocation to return. On an earlier checkpoint the three highest rates were below the floor at that budget and the grid recorded them as unreachable, which is what the machinery for an empty feasible set is there for.

Two estimates of the floor are in play and they are not interchangeable. Table 71 gives them, because the rescaling below divides by the distance between the floor and the saturation point, and near the floor that is a small difference divided by a small difference.

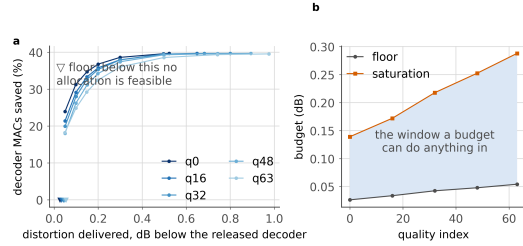
| source              | what it measures                                                | q0    | q63   |
|---------------------|-----------------------------------------------------------------|-------|-------|
| the saturation file | a deployed decode with every tile at full depth, over 53 frames | 0.026 | 0.054 |
| the dense grid      | the same quantity, over 106 frames                              | 0.022 | 0.048 |

**Table 71. Two measurements of the floor, in decibels.** They differ by three to six thousandths of a decibel because they are the same quantity over a different number of frames. A third estimate, the per-tile table at a zero multiplier, is read off an operating point the sweep could not meet, and on the reported weights there is no such point. Everything downstream of here takes both ends of the window from the first row, so that the floor and the saturation point are on one measurement.

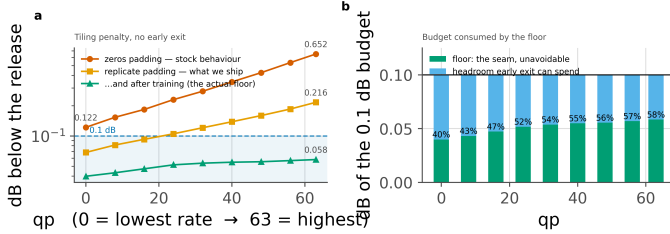
The three rows are results/saturation\_RECIP512\_ctc53.json, results/signalled\_RECIP512\_grid.json and results/supp\_paper\_curve\_PAPER.json, all on the pinned checkpoint and all with decibels averaged per frame. The third records a floor only where it found the 0.05 dB budget unreachable, which is why q0 is empty.



**Figure 50. Three regions, and only the middle one is a design choice.** Below the floor no allocation meets the budget; above saturation every tile already takes the cheapest exit and a looser budget buys nothing. The 0.1 dB budget uses 65% of the band at the lowest rate and 20% at the highest, which is why the same budget behaves so differently at the two ends of the rate range.



**Figure 51. The window a budget works in.** **a**, saving against the distortion actually delivered, per rate. Every curve begins at a floor, below which no allocation meets the budget, and flattens at a saturation point, above which a looser budget buys nothing. **b**, those two limits against rate. The shaded band is the only region in which a budget is a design choice rather than a formality. The main paper states both limits and reads them off this picture.



**Figure 52. The tiling penalty across the whole rate range, every tile at full depth, so this is the floor and nothing else.** **a**, the steps that reduce it, of which the two that matter cost no arithmetic; the largest single factor is training the ladder with the seam present. **b**, the floor as a share of the 0.1 dB budget: it is charged *inside* the budget and consumes a growing part of it as the rate rises, which is one of the two reasons the saving falls with rate.

## E.2. Three conventions, and the conversions between them

Three separate choices have to be pinned before a figure in this section means anything, and the project makes all three in two ways in different files. The loss can be measured against the released decoder's full-frame decode of the same latent, which puts the cost of cutting the frame into tiles inside the budget, or against this model's own deepest exit decoded tile by tile, which cancels it. Decibels can be averaged per frame and then over frames, which is what the released codec's own evaluation does, or pooled into one mean squared error first. And the saving can be divided by the released decode or by our own deepest exit. Two of the three conversions are exact:

$$D^{\text{rel}} = D^{\text{self}} + D_{\text{min}}, \quad S^{\text{rel}} = 100 - c_{K-1}(100 - S^{\text{self}}). \quad (31)$$

| choice                | the two options                                                 | what separates them                                                                     |
|-----------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| loss measured against | the released full-frame decode, or our own tiled deepest exit   | the floor: 0.026 dB at q0, 0.054 dB at q63                                              |
| decibels averaged     | per frame and then over frames, or pooled over every tile first | measured, not closed form: a pooled 0.1 dB is 0.116 dB per frame at q0 and 0.115 at q63 |
| saving divided by     | the released decode, or our deepest exit at 1.0095 of it        | a self-referenced 30% is 29.33% release-referenced, and a self-referenced 0% is -0.95%  |

**Table 72. What has to be stated with every decibel and every saving in this section.** Take from it that a number quoted without its convention is uncertain by about a third of a 0.1 dB budget on the quality axis and by about two thirds of a point on the compute axis. The first and third rows convert exactly, by the identity above; the second does not, because a mean of logarithms is not the logarithm of a mean.

The floor is results/saturation\_RECIP512\_ctc53.json; the two averaging figures are the 0.1 dB operating points of results/supp\_paper\_curve\_PAPER.json, which bisects on the pooled decibel and records the per-frame decibel of the same allocation beside it; the deepest-exit cost is deepest\_exit\_cost in results/router\_RECIP512\_b01.json. The averaging convention itself is defined in Section A.

## E.3. The grid

Table 73 is the sweep the main paper quotes: nine budgets across five rates, measured against the released decoder on both axes, with decibels averaged per frame, two frames from each of the 53 test sequences. Allocation is by the exact per-tile losses, so every entry is an upper bound on what a decoder-side router can reach.

| budget dB | q0   | q16  | q32  | q48  | q63  |
|-----------|------|------|------|------|------|
| 0.050     | 23.3 | 17.4 | 11.4 | 8.2  | 5.1  |
| 0.075     | 29.9 | 25.5 | 20.9 | 19.6 | 18.1 |
| 0.100     | 34.5 | 30.8 | 26.7 | 25.4 | 24.1 |
| 0.150     | 39.1 | 37.1 | 33.9 | 32.3 | 30.9 |
| 0.200     | 39.1 | 39.1 | 38.3 | 36.4 | 35.0 |
| 0.250     | 39.1 | 39.1 | 39.1 | 39.1 | 37.8 |
| 0.300     | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |
| 0.400     | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |
| 0.500     | 39.1 | 39.1 | 39.1 | 39.1 | 39.1 |

**Table 73. Compute saved against budget, release-referenced.** *none* marks a budget below that rate's floor, where no allocation is admissible. Reading down a column, the return on a looser budget falls away long before the ceiling is reached. Reading across a row, the rate dependence is largest at the tightest budget and vanishes once every rate has saturated. Cells are the cost model against the release, the convention the rest of this supplement compares in; hook-counted, the 0.1 dB row reads 33.7% at q0 and 23.5% at q63, which reproduces the headline of the main paper, 33.7% and 23.5%, to within a tenth of a point on a different frame count.

results/signalled\_RECIP512\_grid.json, pinned checkpoint, two frames per sequence, budget bisected on the delivered per-frame decibel of a real decode of the mixed map. The headline file results/signalled\_RECIP512\_ctc53.json uses one frame per sequence and reports 33.7% and 23.5% at the same budget.

A second sweep of the same checkpoint prices the same window differently, and Section D prints its operating points in full. It bisects on the pooled decibel rather than the per-frame one and reads 53 sequences at one frame each rather than two, and at a nominal 0.1 dB it saves several points more at every rate than the grid above. The conversions of Section E.2 account for part of that and not for all of it, which is the first sign of what Section E.5 measures. Its own saturation points are the useful thing here: the budget at which a rate runs out of allocation is 0.15 dB at q0, 0.20 dB at q16, 0.25 dB at q32, 0.25 dB at q48, 0.30 dB at q63. The three budgets the paper reports are therefore not three points on one curve. At 0.1 dB every rate still has an allocation to make, at 0.3 dB only q48 and q63 do, and at 0.5 dB none does.

results/supp\_paper\_curve\_PAPER.json, pinned checkpoint, 53 sequences at one frame each, 35 operating points at seven budgets. A saturated point is flagged in the file rather than inferred here, and carries no exit histogram, because the allocation has stopped changing.

Budgets in [0.139, 0.152] dB saturate the lowest rate and no other, a window 13 thousandths of a decibel wide. Above 0.288 dB every rate

has saturated, so the binding constraint is the ladder rather than the budget. Both figures are read off the nine-rate table in Section D rather than off either sweep.

#### E.4. Rescaling onto the band

With both ends of the window measured, a budget can be quoted as a position in the window rather than as a decibel figure. Write

$$u = \frac{D - D_{\min}}{D_{\text{sat}} - D_{\min}}, \quad (32)$$

so that  $u = 0$  is the floor and  $u = 1$  the saturation point. At a matched decibel the five rates are 10.2 points apart in saving. At a matched  $u$  they are 3.2 points apart on average and 8.0 at worst, so the window accounts for most of the rate dependence of the trade-off and a couple of points are left over.

| u   | q0   | q16  | q32  | q48  | q63  | spread |
|-----|------|------|------|------|------|--------|
| 0.1 | 23.3 | 17.4 | 15.2 | 16.5 | 18.7 | 8.17   |
| 0.2 | 23.3 | 21.1 | 21.5 | 22.8 | 24.2 | 3.12   |
| 0.3 | 26.0 | 25.6 | 25.5 | 26.7 | 27.4 | 1.88   |
| 0.4 | 29.0 | 28.5 | 28.5 | 29.5 | 30.6 | 2.12   |
| 0.5 | 31.3 | 31.2 | 31.0 | 32.3 | 32.6 | 1.61   |
| 0.6 | 33.4 | 32.9 | 33.6 | 34.0 | 34.5 | 1.60   |
| 0.7 | 34.5 | 34.7 | 35.2 | 35.7 | 36.0 | 1.54   |

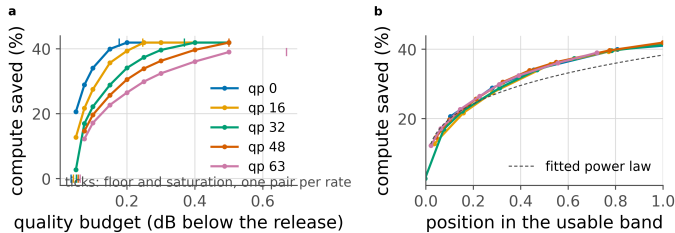
**Table 74. The five rates at matched positions in their own windows.** Read across a row: five rates whose raw savings at a matched decibel differ by 10.2 points agree here to about a point and a half. The spread is largest at the bottom of the window, where the floor subtraction is a large fraction of a small number and the two files disagree about the floor by a few thousandths of a decibel.

Computed in the build from results/signalised\_RECIPES12\_grid.json and results/saturation\_RECIPES12\_ctc53.json by the rescaling in scripts/tradeoff\_figure.py, linearly interpolated between measured points. Every rate has measured points spanning  $u = 0.10$  to  $0.70$ , so no entry is an extrapolation. Mean spread over these seven positions 2.86 points, worst 8.17, against the 3.2 and 8.0 recorded in results/band\_collapse.json, which interpolates on a wider grid and holds the last measured value beyond each rate's last point.

The collapsed curve is then described by one free number. With  $C$  the architectural ceiling, which is not fitted,

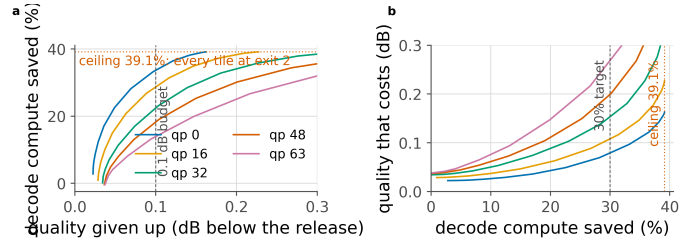
$$S(u) \approx C u^\beta, \quad (33)$$

and fitting  $\ln(S/C)$  against  $\ln u$  by least squares through the origin gives 0.35 on the pinned checkpoint, with  $R^2 = 0.984$  and a worst residual of 2.2 points. The same procedure on the second training run gives 0.28 with  $R^2 = 0.907$  and a worst residual of 4.1 points – the collapse replicates, the quality of the one-parameter description of it does not. The sensitivity table at the end of this section is the reason to expect that: the assumed ceiling is one of the things this exponent is most sensitive to, and the second run's ceiling is its own. Figure 53 draws it and Section E.5 takes it apart.



**Figure 53. The collapse on the second training run.** **a**, saving against budget, one line per rate, with each rate's floor and saturation point ticked. **b**, the same with the budget axis rescaled onto each rate's own window; dashed is the one-parameter power law. The main paper shows this pair for the pinned checkpoint.

docs/figures/budget\_band\_BEST.png, drawn by scripts/tradeoff\_figure.py from results/signalised\_BEST\_grid.json and results/saturation\_BEST\_ctc53.json, both on runs/BEST/ckpt\_eval.pth.tar rather than on the pinned checkpoint.



**Figure 54. The trade-off, whole.** **a**, what a budget buys, per rate; the ceiling is 38.3% and  $q_0$  reaches it at 0.139 dB. **b**, the same relation inverted, so it reads as what a saving target costs in quality. The budgets the main paper reports are three points on this curve, and the flat right-hand end of each trace is that rate sitting on the ceiling.

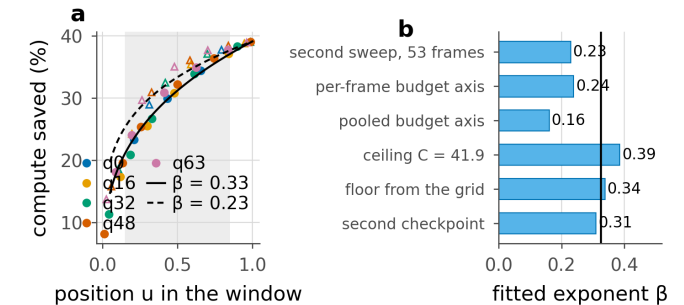
#### E.5. How weak the replication is

The main paper says the collapse replicates and the exponent does not. Two checkpoints is a thin basis for either half of that sentence, and Table 75 is meant to show how thin. Each row refits the same one-parameter law after changing exactly one thing, over the window of positions where every sweep has points.

| what is changed                                           | $\beta$<br>before | $\beta$<br>after | $\Delta$ |
|-----------------------------------------------------------|-------------------|------------------|----------|
| the checkpoint, pinned to BEST, each with its own ceiling | 0.325             | 0.310            | -0.015   |
| the sweep, 106 frames to 53, one checkpoint               | 0.325             | 0.229            | -0.096   |
| the budget axis, per-frame decibels to pooled, one sweep  | 0.239             | 0.160            | -0.078   |
| the assumed ceiling $C$ , 39.1 to 41.91                   | 0.325             | 0.385            | 0.060    |
| the floor, the saturation file to the grid's own          | 0.325             | 0.338            | 0.013    |
| the rate, fitted one at a time, $q_{63}$ against $q_{32}$ | 0.285             | 0.360            | 0.075    |
| the fit form, $1/b$ from the inverse fit, per rate        | 0.326             | 0.459            | 0.133    |

**Table 75. What moves the exponent.** Take from it that the exponent describes a sweep rather than a decoder. Every row except the floor moves it by as much as changing the checkpoint does, re-sweeping the same checkpoint moves it twice as far, and reading the budget axis in the other averaging convention moves it almost as far again. The comparison the main paper draws between 0.35 and 0.28 is inside the noise of every other choice on this list.

Computed in the build. Rows 4, 5 and 6, and the before column of rows 1 and 2, refit results/signalised\_RECIPES12\_grid.json against results/saturation\_RECIPES12\_ctc53.json; the after column of row 1 uses results/signalised\_BEST\_grid.json and results/saturation\_BEST\_ctc53.json on runs/BEST/ckpt\_eval.pth.tar, whose ceiling is its own; the after column of row 2 uses the operating points of results/supp\_paper\_curve\_PAPER.json and row 3 uses its raw sweep; row 7 is results/frontier\_low.json, fitted to results/curve\_BEST.json on the same second run over 40 sequences. Every fit is over  $u$  in  $[0.15, 0.85]$  and through the origin in log space, on 16 to 22 points.  $R^2$  stays between 0.897 and 0.969 across the rows, so no row is a bad fit to its own data.



**Figure 55. The collapse, and what the exponent is sensitive to.** **a**, the rescaled points from two sweeps of the pinned checkpoint, filled circles for the 106-frame grid and open triangles for the 53-frame frontier, with the power law fitted to each over the shaded window. The two sweeps of one checkpoint do not lie on one curve. **b**, the exponent under one change at a time, with the vertical line at the exponent of the first row of Table 75. The change of checkpoint, which the main paper reads as a finding, is the smallest of the six.

docs/figures/band\_sensitivity\_PAPER.png, drawn by scripts/supp\_fig\_band\_sensitivity.py, which imports the fit from this section's module so

that the figure and the table cannot disagree. Sources as for the table above.

The floor is the one input the exponent is insensitive to over this window, and only over this window. Refitting over the whole range instead, where the points below  $u = 0.15$  sit, moves it from 0.347 to 0.387 when the floor is taken from the grid rather than from the saturation file. A rescaling is only as good as the two numbers it divides by, and near the floor it is dividing a small difference by a small difference.

Both fits are re-run in this build from the files they name, so the exponent quoted here is the exponent those inputs now support: results/band\_collapse\_BEST.json reproduces at 0.3253 with  $R^2 = 0.9843$  against the 0.2827 and 0.9068 it records, and the two digits the paper quotes, 0.35 and 0.984, are unchanged by the exercise.

## E.6. Which ladder a budget wants

Once a budget saturates a ladder, the only way to spend more quality is an exit that does not exist, so the ladder itself is a choice made at the operating point rather than a hyperparameter tuned once. Four ladders have been run far enough to compare, and the two orderings cross between 0.1 and 0.3 dB.

| Ladder    | Config       | Ceiling | 0.1 dB      | 0.2 dB      | 0.3 dB       | 0.5 dB       |
|-----------|--------------|---------|-------------|-------------|--------------|--------------|
| RECIPE512 | K6 j2 256px  | 38.3    | <b>27.6</b> | <b>36.8</b> | 38.3         | 38.3         |
| BEST      | K6 j2 256px  | 38.3    | 24.2†       | –           | 38.5†        | 41.3†        |
| BEST128   | K6 j2 128px  | 38.3    | 19.3†       | –           | 36.6†        | 40.6†        |
| FINE12    | K12 j4 128px | 49.5    | 19.0*†      | –           | <b>42.0†</b> | <b>48.6†</b> |

**Table 76. Ladder settings**, mean saving over the five rates, on each run’s own latest checkpoint. Ceilings are  $100(1 - c_j)$  for that ladder, corrected by the offset the hook count shows. The asterisk marks a setting where some rate did not reach the budget, so the mean is over the rest; the dagger marks a saving taken from the arithmetic model rather than counted off the decode, which reads 0.4 to 0.8 points optimistic. Only RECIPE512 has been re-measured with hooks at every budget, and it is the only run the main paper reports.

The four signalled \*.json curve files named in scripts/make\_paper\_tables.py, generated into paper/tables/runs.tex. The four runs are at different epochs, so no row here separates a ladder from the run that trained it; section H measures how far two runs of one recipe land apart.

At 0.1 dB the coarse ladder wins and the fine one ( $K = 12, j = 4$ ) cannot reach the budget at the highest rate at all. Splitting later puts twice as many blocks in the per-tile section, and the seam penalty grows with that count, so a finer ladder raises the floor it has to clear before it can spend anything. At 0.5 dB the fine ladder wins, 48.6% against 38.3%, because the coarse one has been pinned at its ceiling since 0.3 dB and has nothing left to spend. The band of E.1 says which side of that crossover a given budget sits on, which is the practical use of measuring the two ends.

## E.7. What would raise the ceiling

The four ladders above were trained; the ceiling is not, and it can be read off the cost model for any split without running anything. That matters because the ceiling is the binding constraint at a loose budget – at 0.3 dB the shipped ladder reaches 91 to 100% of its own ceiling, so nothing but a higher ceiling helps – and because the obvious way to raise it, splitting earlier, is paid for in seam. Both halves are measured here, against each other.

| split j     | blocks per tile | ceiling | seam q0 | seam q32 | seam q63 |
|-------------|-----------------|---------|---------|----------|----------|
| 4           | 4               | 13.0%   | 0.009   | 0.018    | 0.038    |
| 3           | 6               | 24.2%   | 0.032   | 0.053    | 0.089    |
| 2 (shipped) | 8               | 39.1%   | 0.065   | 0.107    | 0.194    |
| 1           | 10              | 54.0%   | 0.100   | 0.149    | 0.234    |
| 0           | 12              | 68.9%   | 0.175   | 0.314    | 0.606    |

**Table 77. The ceiling against what it costs**. Ceiling is  $100(1 - c_j)$  from flexuf/cost.py on a 40-tile 1080p frame, frame-relative, for a  $K = 6$  ladder over the same twelve blocks. The seam columns are the measured penalty at that split on the *untrained* ladder, from results/supp\_seam\_vs\_split.json; training absorbs most of it – the trained floor at the shipped split is 0.054 dB at q63 against the untrained number in this table – so read the seam columns as a ratio between splits rather than as decibels a trained model would pay. Halving the split depth is worth about fifteen points of ceiling and costs about a fifth of a decibel of seam at the highest rate, which at a 0.1 dB budget is the whole budget and at 0.3 dB is not.

Ceilings computed in the build from flexuf/cost.py; seam from results/supp\_seam\_vs\_split.json, 20 frames at 256 px tiles. Nothing in this table is a trained result: it is the design space, and the four rows of the previous table are four points in it.

## E.8. What the window claims, and what it does not

- The floor and the saturation point are measured, cheap and unambiguous. Each is one forward pass per rate, and the ceiling between them is closed form in the per-exit costs.
- The collapse is the claim. Rescaling a budget onto its own window takes the spread across rates from 10.2 points to under two at matched positions, on both checkpoints.
- The exponent is an empirical fit and a description of one sweep, not a law. It moves further when the same checkpoint is swept again, or when the budget axis is read in the other averaging convention, than it moves between the two training runs the main paper compares.
- Everything here is the oracle allocation, obtained with the per-tile losses known. It bounds any router and says nothing about how much of the window a decoder-side predictor reaches, which is what Section F measures.

## F. The router and the calibrated bit rule

An exit map assigns one exit index to every tile of a frame, and this work produces one in three ways. Configuration A searches all  $K$  exits for every tile at the encoder, which holds the source, and signals the answer. Configuration B predicts the map at the decoder from data the decoder already holds, and signals nothing. The third way computes it at the decoder from one number the entropy coder has already produced, with nothing learned. The main paper states the rule by which the head turns logits into an exit and where its tilt comes from, and that is not repeated. What is here is the head itself, what each of its inputs is worth, what it costs to run, and how it fares against the third way.

Write  $D(t, k)$  for the mean squared error of tile  $t$  decoded at exit  $k$  and  $c_k$  for the cost of exit  $k$  in units of one released decode. At a price  $\lambda$  on compute, the Lagrangian oracle gives each tile the exit that minimises

$$\ell(t, k) = D(t, k) + \lambda c_k, \quad k^*(t) = \arg \min_{k \geq j} \ell(t, k). \quad (34)$$

The split depth  $j$  is the number of trunk groups every tile runs before any tile may leave, so exits below  $j$  do not exist and the minimisation starts there. Throughout this section, *agreement* means the fraction of tiles on which a predictor picks the same exit as this oracle at the same  $\lambda$ . It is the quantity the head is trained on. Part of what F.4 reports is that it is not the quantity that decides which allocation is cheaper.

### F.1. What each input is worth

The head reads four per-tile signals and one per-frame signal. Since F.4 finds that a rule reading a single number beats it, it is fair to ask which of the five carries anything. The ablation trains the same head once per input group, with every other group zeroed after its projection rather than deleted. Architecture, parameter count, optimiser, seed, image order and

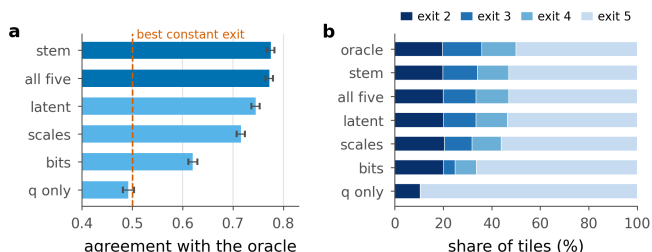
quality draws are then identical across the six runs, and the only thing that differs is how much the head is allowed to know. All six carry 148,176 parameters, which is the deployed head's 144,024 plus the projection for the bit count that the deployed head does not use.

The quality index  $q$  stays live in every single-group run. It is one number per frame, identical for every tile in that frame, so it can say which operating point the decoder is at and cannot, even in principle, tell two tiles of one frame apart. Holding it live keeps the runs comparable on per-tile information alone. The run with  $q$  on its own is then the floor that choice implies, which is the agreement reachable with no per-tile information whatever.

| live inputs                | agreement | $\pm$ s.e. | last 250 | over floor |
|----------------------------|-----------|------------|----------|------------|
| stem,qp                    | 0.7750    | 0.0074     | 0.7607   | +0.2744    |
| stem,latent,scales,bits,qp | 0.7721    | 0.0075     | 0.7543   | +0.2715    |
| latent,qp                  | 0.7448    | 0.0080     | 0.7433   | +0.2442    |
| scales,qp                  | 0.7156    | 0.0083     | 0.7170   | +0.2150    |
| bits,qp                    | 0.6202    | 0.0091     | 0.6137   | +0.1196    |
| qp                         | 0.4923    | 0.0114     | 0.4900   | -0.0083    |

**Table 78. What each router input is worth.** Read the first column against the last: only the stem moves agreement far from what a single constant exit already achieves. Agreement with the oracle's exit choice on 1,200 frames and 4,800 tiles the head never trained on, with one standard error taken across frames rather than across tiles, since tiles of one image are not independent draws. *last 250* is the running held-out agreement over the final 250 training steps, on 12 tiles a step, and confirms that the end-of-run measurement is not a fluctuation. *over floor* is the margin over the best single constant exit, which scores 0.5006 on these tiles.

results/router\_ablation.json, and the six per-variant training records results/router\_ablation\_stem.json and its five siblings. All on the pinned checkpoint, all at  $\lambda = 1.3 \times 10^{-5}$ , all 3,000 steps from seed 0.



**Figure 56. The input ablation, both halves of it.** **a**, agreement with the oracle per variant, with one standard error across frames; the dashed line is the best single constant exit. The two dark bars are the two heads that see the stem. **b**, where each variant sends its tiles, against the oracle's own distribution. Every variant reproduces the oracle's use of the shallowest exit to within a point, and what degrades as inputs are removed is the separation of the middle of the ladder: the bit count alone nearly empties exit 3, and  $q$  alone uses two of the four exits available to it.

Drawn by scripts/supp\_router\_figs.py from results/router\_ablation.json, fields agree, stderr, pred\_hist and oracle\_hist.

**The stem alone.** Reading the stem and  $q$  gives 0.7750; reading the stem, the latent, the scales, the bit count and  $q$  gives 0.7721. The difference is 0.0029 against a standard error of 0.0105 on the difference, so it is under a third of one standard error and its sign is not determined. Four of the five inputs add nothing this measurement can resolve. That result is about those four inputs in this head, and it does not show that the latent, the scales or the bit count are uninformative about a tile. What it shows is that whatever they carry is already carried by the stem, which is computed downstream of all of them.

**The single-group ordering.** The stem at 0.7750 is 0.0302 above the latent, against a standard error of 0.0109 on the difference; the latent is 0.0292 above the scales; and the scales are 0.0954 above the bit count, which is more than seven standard errors. The bit count is the weakest per-tile input the head has, and F.4 shows a rule that reads the bit count and nothing else beating the head at every rate. The input the head learns least from is the input that wins when it is used differently.

**The floor.** The quality index on its own reaches  $0.4923 \pm 0.0114$  where the best single constant exit scores 0.5006. A head with no per-tile information cannot beat the best constant in expectation, and this one does not. The floor belongs beside every other row: an agreement of 0.62 cannot be read at all until it is known that 0.50 is free.

**The ablation's own cost.** The six trainings took 6.4 hours of one NVIDIA RTX A6000 between them, from 55 to 66 minutes each (the ablation's own records, field wall\_s). The decoder is frozen for all of them, so none of that is decoder training. It is the price of the question, and it is worth stating beside a negative answer.

One head covers every operating point, and the quality index is what makes that possible. It is an input rather than a selector: during training it is drawn uniformly at random for each batch, so one set of weights sees all 64 indices, and at test time the same weights run at every rate with the offline  $\beta$  table of A.12 in front of them. No head per rate is proposed anywhere in this work, since that multiplies the stored parameters by the number of operating points a deployment offers, and F.5 is the measurement of what the single head gives up for that.

## F.2. What the head costs

| $q$ | decode (ms) | stem (ms) | router (ms) | share (%) |
|-----|-------------|-----------|-------------|-----------|
| 0   | 114.29      | 41.25     | 0.571       | 0.500     |
| 32  | 114.41      | 41.32     | 0.568       | 0.497     |
| 63  | 114.44      | 41.31     | 0.568       | 0.496     |

**Table 79. The head against the decode it decides for.** The share column is the number to take away, and it is three times what the operation count predicts. Three timings on one latent, interleaved so that a co-tenant's load drift lands on all of them equally. *decode* is the tiled decoder at the deepest exit, which is what the head's share is a share of; *stem* is the first  $j$  groups, which the head does not pay for because the decoder runs them anyway before the split. The head takes 0.50% of the decode where its operation count predicts 0.163%, a factor of 3.0.

results/router\_latency.json: 1920 $\times$ 1080 padded to 2048 $\times$ 1280, 40 iterations after warm-up, one NVIDIA RTX A6000, the 144,024 parameter head runs/RECIPE512/routers2/v2\_lam1.3e-5.pth. Measured on runs/RECIPE512/ckpt\_eval.pth.tar rather than on the pinned checkpoint; the quantity is a ratio of two timings of the same weights, so it does not depend on which epoch they came from.

The extra 0.33 points are launch overhead. A head of 144,024 parameters evaluated on 40 vectors is small enough that its wall-clock is dominated by the cost of starting its kernels, and an operation count does not contain that. Every configuration-B saving in this work charges the head at its operation share, which is the smaller of the two numbers and therefore the more flattering to us; the honest reading is that the head costs about half a percent of the decode in time. It stays small against what it decides about, since one trunk block is 7.5% of the decode by the cost vector below.

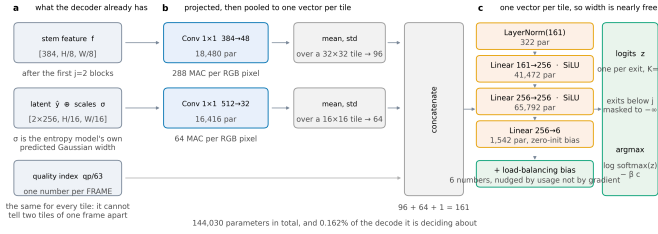
**Two heads, two shares.** The pinned checkpoint carries a head that was trained jointly with the decoder, and a second head was trained afterwards against that decoder once it was frozen. They are different objects and they are priced differently: the jointly trained one is charged 0.044% of the decode and the frozen-decoder one 0.163%, because the second reads the decoded latent and the entropy model's scales as well as the stem. Both shares are measured by hooks on the head's own convolutions and linear layers rather than assumed, in the same run that produced the saving. F.4 reports both heads.

results/router\_RECIPES12\_b01\_jointhead.json and results/router\_RECIPES12\_b01\_e4head.json, field router\_compute\_share\_pct in each.

**For scale, what the search costs instead.** Configuration A does not run a head; it runs the argmin itself, which needs the true error of every exit, and section G prices that at 4.52 of a tiled decode per frame. A decoder-side head at half a percent of one decode, and a rule at nothing, are both several orders below it, which is the whole reason for wanting one.

**The head, exactly.** Three things enter. The first is the feature at the split point, 384 channels at one eighth of frame resolution, which is the last

thing every tile shares. The second is the decoded latent concatenated with the entropy model's scales, the predicted Gaussian width used to code each latent position, which is already computed during the decode and is, position by position, an estimate of how hard that position was to code. The third is the quality index. Each of the first two is projected by a  $1 \times 1$  convolution, to 48 and 32 channels, then reduced to one vector per tile by taking the mean and the standard deviation over the tile. That gives  $96 + 64 + 1 = 161$  numbers per tile, which pass through a LayerNorm and a three-layer perceptron of width 256 with SiLU activations, ending in K logits. Almost all of the 144,024 parameters sit in the  $1 \times 1$  on the stem, which is the only part that runs per pixel; the perceptron runs once per tile, forty times for a 1080p frame, so its width is nearly free.



**Figure 57. The router head.** **a**, the three things it reads, all already held by the decoder at the moment the decision is needed. **b**, each projected by a  $1 \times 1$ , then reduced to one vector per tile by its mean and standard deviation. **c**, a three-layer perceptron, one pass per tile. Every shape and parameter count in the figure is read off the module when the figure is drawn.

**What it is trained against.** The head is fitted against a frozen decoder, so the exits it chooses between do not move while it learns. Its target is the Lagrangian oracle's choice and the loss is a cross-entropy to that target, with each tile weighted by its *regret*, meaning by how much the oracle's own bracket grows if the head's exit is used in place of the oracle's. A tile whose two best exits are nearly tied then counts for less than one where the wrong choice is expensive, which is the same asymmetry that makes agreement a poor score in F.4. A router of this shape can collapse onto a single exit during training, so it carries a balancing term: a per-exit bias added to the logits before the decision and updated by usage rather than by a gradient, which is what makes it loss-free [14]. It is biased toward the oracle's own exit distribution rather than toward uniform, because at a high multiplier the oracle genuinely does send every tile to one exit and forcing spread there would force mistakes.

### F.3. The calibrated bit rule

The entropy model produces one number per tile before the trunk runs, at no cost, and then discards it: how many bits that tile's latents took. Per-block bit allocation is a standard quantity in learned compression, where it is something to choose, and block-level rate control sets it so that complex regions get more bits [40]. Here it is read in the other direction, after the fact and at the decoder, as a statement about how hard the region was. The results files call the rule *rate-rank*; the paper calls it the calibrated bit rule. What follows is all of it.

**The statistic.** Sum the entropy coder's estimated bits  $r_i$  over the latent positions  $i$  falling inside tile  $t$ , and divide by the mean over the  $N$  tiles of that frame. Normalising per frame rather than globally is deliberate: the absolute rate level is a property of the frame, and what decides a tile is how it compares with the rest of its own frame.

$$b(t) = N \frac{\sum_{i \in t} r_i}{\sum_i r_i}. \quad (35)$$

**The surrogate.** Assume every tile has the same shape of decay across the ladder, up to a scale that depends only on  $b(t)$ .

$$\log D(t, k) \approx \alpha \log b(t) + c + \log \varphi_k. \quad (36)$$

$\phi$  has one entry per reachable exit and says what that exit costs on an average tile;  $\alpha$  is an exponent fitted rather than assumed, which is what lets the rule find the direction of the relation instead of asserting one. The fit is offline and leave-one-sequence-out, so no sequence contributes to the profile that routes it, and its whole output is six numbers per rate.

**The decision.** Run the oracle's own Lagrangian on the surrogate table in place of the true one, and bisect  $\lambda$  against the budget as configuration A does.

$$\hat{k}(t) = \arg \min_{k \geq j} [e^c b(t)^\alpha \varphi_k + \lambda c_k]. \quad (37)$$

#### the decoder, once per frame

- 1 *input*  $r$ , the entropy coder's estimated bits per latent position; the cost vector  $c$ ; the price  $\lambda$ ; the calibration  $(\alpha, c_0, \phi)$  for this quality index
- 2 *for* each tile  $t$  of the  $N$  tiles of the frame
- 3  $b \leftarrow N \cdot (\text{sum of } r_i \text{ inside } t) / (\text{sum of } r_i \text{ over the frame})$
- 4 *for*  $k = j \dots K-1$
- 5  $L_k \leftarrow e^{c_0} \cdot b^\alpha \cdot \phi_k + \lambda \cdot c_k$
- 6  $\text{exit}(t) \leftarrow \text{the } k \text{ that minimises } L_k$
- 7 *return*  $\text{exit}$

#### offline, once per quality index, leave-one-sequence-out

- 8  $d(t) \leftarrow \text{mean over } k \geq j \text{ of } \log D(t, k)$
- 9  $(\alpha, c_0) \leftarrow \text{least squares of } d(t) \text{ on } \log b(t)$
- 10  $\log \phi_k \leftarrow \text{mean over } t \text{ of } [\log D(t, k) - \alpha \log b(t) - c_0]$

**Table 80. The calibrated bit rule, in full.** Lines 1 to 7 are what the decoder runs: one power and  $K$  products per tile, nothing learned, nothing signalled. Lines 8 to 10 are the offline calibration, whose entire output is the six numbers per rate printed in the next table.  $\lambda$  is bisected in two levels, on the cheap per-tile table first and then against a real decode of the resulting map, because the table and the decode differ slightly at the tile seams. Sweeping  $\lambda$  traces the lower convex hull of the achievable set in the sense of [26], so the point returned is the best one on that hull at the budget it consumes.

| q  | delivered dB | $\alpha$ | $\phi_2$ | $\phi_3$ | $\phi_4$ | $\phi_5$ |
|----|--------------|----------|----------|----------|----------|----------|
| 0  | 0.1000       | 10.01    | 1.0216   | 0.9965   | 0.9932   | 0.9890   |
| 16 | 0.1007       | 4.60     | 1.0229   | 0.9966   | 0.9930   | 0.9879   |
| 32 | 0.1000       | 2.75     | 1.0262   | 0.9978   | 0.9910   | 0.9855   |
| 48 | 0.1000       | 1.95     | 1.0282   | 0.9979   | 0.9902   | 0.9842   |
| 63 | 0.0996       | 1.43     | 1.0302   | 0.9991   | 0.9882   | 0.9832   |

**Table 81. The rule's entire state**, at the 0.1 dB budget. There is nothing else to store, and it is printed here so that the rule can be reproduced without refitting it.  $\alpha$  is positive everywhere, so a tile whose latents cost more bits is modelled as harder at every exit and is sent deeper, which is what the correlations in F.4 confirm.  $\phi$  is nearly flat, spanning 3.3% at q0 and 4.8% at q63 between its shallowest and deepest entry, against costs that span 0.61 to 1.01, so what moves a tile along the ladder is its own  $b(t)$  rather than the shape of the profile. *delivered dB* is what the two-level bisection actually landed on.

results/raterank\_RECIFE512\_b01.json, pinned checkpoint, 53 CTC sequences at one frame each.

The costs  $c_k$  on the pinned checkpoint are 0.6088, 0.7578, 0.8697, 1.0095 for exits 2 to 5, recovered from the frame-level saving of maps that send every tile to one fixed exit. The shallowest of them,  $c_j = 0.6088$ , sets the architectural ceiling of 39.1%. The deepest is above 1 because our ladder carries seam repair and an adapter that the released decoder does not.

results/static\_RECIFE512\_b01.json, field uniform, and results/saturation\_RECIFE512\_ctc53.json, fields  $\text{cost}_j$  and  $\text{ceiling\_pct}$ .

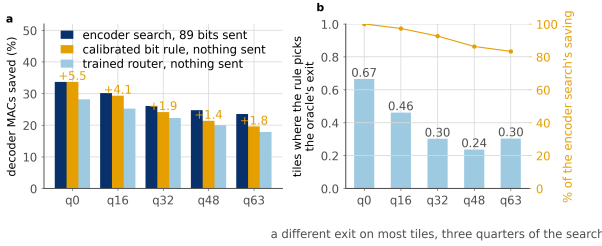
### F.4. The rule against the trained head

Two trained heads exist on this checkpoint and both are reported, since averaging them would hide the only spread this work has. *joint* is the head the checkpoint carries, trained alongside the decoder and read straight out of it. *frozen* is the 144 K head trained afterwards, against the decoder once it had stopped moving. Both were evaluated on the pinned checkpoint over 53 sequences at one frame each, by one script, at the same budgets, and both are charged for their own compute.

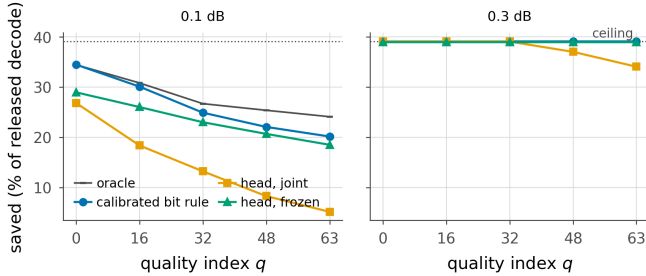
| q  | 0.1 rule | 0.1 joint | 0.1 frozen | 0.3 rule | 0.3 joint | 0.3 frozen |
|----|----------|-----------|------------|----------|-----------|------------|
| 0  | 34.48    | 26.84     | 33.31      | 39.12    | 39.08     | 38.96      |
| 16 | 30.10    | 18.36     | 29.06      | 39.12    | 39.08     | 38.96      |
| 32 | 24.90    | 13.21     | 24.76      | 39.12    | 39.08     | 38.96      |
| 48 | 22.06    | 8.26      | 21.16      | 39.12    | 37.03     | 38.96      |
| 63 | 20.14    | 5.13      | 18.21      | 39.12    | 34.07     | 38.96      |

**Table 82. The calibrated bit rule against two trained heads**, as the percentage of the released decoder's operations saved, at two budgets and five rates. The rule is ahead of both heads in every cell of this table. Every column is charged for what it costs: each head pays its own compute share and the rule pays nothing. The two heads are further apart from each other than either is from the rule at several rates, which F.5 takes up.

Rule: results/raterank\_RECIP512\_b01.json and b03. Joint head: results/router\_RECIP512\_b01\_jointhead.json and its b03 sibling. Frozen head: results/router\_RECIP512\_b01\_e4head.json and b03. All on runs/RECIP512/ckpt\_PAPER.pth.tar.



**Figure 58. Three ways to choose an exit.** a, saving at 0.1 dB. The encoder search sees the source and sends 89 bits a frame; the other two send nothing. The bit rule beats the trained head at every rate, with no learned parameters against the head's 144,030. b, bars are the tiles where the rule picks the search's exit, the line the fraction of its saving it captures anyway.



**Figure 59. The rule sits between the two heads and the oracle.** Saving at each rate, against the Lagrangian oracle measured inside the same run and the architectural ceiling (dotted). At 0.1 dB the rule is above both heads at every rate, and the gap between the two heads is wider than the gap between the rule and either of them. At 0.3 dB the three lowest rates saturate and the three curves separate only at q48 and q63.

Drawn by scripts/supp\_router\_figs.py from the six files named in the table above and results/saturation\_RECIP512\_ctc53.json.

**At the working budget.** Against the joint head the rule is ahead by +7.65 points at its narrowest and +15.02 at its widest, the latter at q63. Against the frozen head it is ahead by +0.13 to +1.94 points. The rule reaches 33.7% at q0 and 19.6% at q63, and it is ahead at all 5 measured rates whichever head it is measured against. A head trained on this decoder against this oracle returns nothing over a rule with no learned parameters, while carrying a compute share that the rule does not.

**At 0.3 dB the low rates saturate and the arithmetic is exact.** At q0, q16 and q32 all three configurations send every tile to the shallowest exit, so the maps are identical and the only difference left is the price of deciding. The joint head falls 0.044 points below the 39.1% ceiling and the frozen head 0.163, which are their compute shares of 0.044% and 0.163% to three decimals. At q48 and q63 the budget still binds and the allocations separate: the rule reaches 39.1% and 38.3% against the joint head's 37.0% and 34.1% and the frozen head's 39.0% and 39.0%. A looser budget gives an allocation more room to be wrong in as well as more room to be right, and the frozen head uses it to be wrong.

**Why 0.5 dB is not in that table.** At 0.5 dB every rate saturates for every configuration, so all three maps are identical and the only thing a comparison could measure is the cost vector each file was written with. The rule's 0.5 dB file is on runs/RECIP512/ckpt\_eval.pth.tar, where the shallowest exit was priced at 0.6088 against the pinned 0.6088, a difference worth -0.00 points of apparent saving. The file is results/raterank\_RECIP512\_b05.json and it records that the rule matches the oracle map exactly at all five rates there, which is the only thing read from it.

| q  | rule agrees | $\rho$ depth | $\rho$ level | $\rho$ spread | oracle |
|----|-------------|--------------|--------------|---------------|--------|
| 0  | 0.666       | +0.41        | +0.83        | +0.61         | 34.46  |
| 16 | 0.462       | +0.45        | +0.85        | +0.68         | 30.83  |
| 32 | 0.302       | +0.47        | +0.86        | +0.73         | 26.70  |
| 48 | 0.237       | +0.51        | +0.84        | +0.80         | 25.38  |
| 63 | 0.303       | +0.45        | +0.88        | +0.71         | 24.10  |

**Table 83. Why the rule works, and it is not by agreeing,** at 0.1 dB. Compare the second column with the fifth: the rule picks the oracle's exit on a minority of tiles and still tracks the oracle's saving. The three Spearman correlations are between a tile's bit count and, respectively, the depth the oracle assigns it, its mean distortion across the ladder, and the spread between its shallowest and its deepest exit. *oracle* is the Lagrangian oracle measured inside the same run, which is the ceiling the rule is chasing.

results/raterank\_RECIP512\_b01.json. The file stores spearman\_bits\_vs\_exit against the negated exit index, as scripts/raterank\_curve.py computes it, so the correlation with depth quoted here is its negative. The oracle column is measured at one frame per sequence and is not charged for the exit map; deployed configuration A, which carries the map, reads 34.5, 30.8, 26.7, 25.4, 24.1% over the same five rates at two frames per sequence (results/signalled\_RECIP512\_ctc53.json).

The rule agrees with the oracle on 0.24 to 0.67 of tiles, well under the frozen head's held-out 0.718, and saves more at every rate. Agreement counts a disagreement between two exits that are within a hair of each other exactly as heavily as one that costs most of the frame's error, and most disagreements are of the first kind. What the rule gets right is the ordering. A tile's bit count correlates with the spread across the ladder, which is how much that tile stands to gain from depth, at 0.61 to 0.80 at every rate, and with the depth the oracle actually assigns at +0.41 to +0.51.

What the rule cannot do is see past that ordering. A rank-1 model gives every tile the same relative profile over exits, so b(t) decides where on the ladder a tile falls and never the shape of its trade-off. That is the ceiling this baseline sits at, and it is the part a learned head would have to earn its parameters on. Neither of our two heads does.

**The two decoder-side signals are not complementary.** Normalising both surrogates to unit mean and blending them with one weight, from the calibrated bit rule at one end to the head's ordering at the other, the rule alone is best at every rate, and every positive weight is worse than none. Trusting the head completely costs 1.0 to 4.9 points, the most at q0. The head reads the entropy model's scales and the bit count is what those scales produce, so blending them mixes a signal with a learned approximation of itself.

An earlier version of this table, measured before the checkpoint was pinned, showed a small gain at the two highest rates, and the paper carried it as evidence of weak complementarity. Re-measuring on the pinned weights removed it. The head in that measurement had been trained against whichever weights runs/RECIP512/ckpt\_eval.pth.tar held at the time, which is a moving pointer, so the first suspicion was that the head had simply been measured against the weights it was fitted to. It had not. Repeating the sweep on the epoch-1 pin with the corrected cost model gives  $w=0$  at every rate as well (results/combined\_RECIP512\_b01\_e1.json), so the absence of complementarity survives a change of checkpoint and the gain was never a property of the two signals. What did change between the two measurements is the cost model: the earlier file predates the correction to the exit adapter's cost, and under the wrong cost an exit assignment can be scored better than it is. Every router number in this document is measured on the pinned checkpoint, after that correction, for this reason.

The same control makes a second point in passing. At  $w=0$  the epoch-1 ladder saves more than the epoch-0 one at every rate – 30.0 against 34.4 at  $q_0$  and 13.8 against 20.2 at  $q_{63}$  – which is the epoch series of Section 5 seen through a different measurement. The checkpoint the paper reports is the first one, and it is the weakest one we have.

| q  | bits        | 0.1  | 0.25 | 0.5  | 0.75 | 0.9  | head |
|----|-------------|------|------|------|------|------|------|
| 0  | <b>34.4</b> | 30.2 | 29.6 | 29.5 | 29.5 | 29.5 | 29.5 |
| 16 | <b>29.9</b> | 26.5 | 26.4 | 26.4 | 26.4 | 26.4 | 26.4 |
| 32 | <b>24.7</b> | 23.4 | 23.0 | 23.0 | 22.9 | 22.9 | 23.2 |
| 48 | <b>21.9</b> | 21.1 | 21.0 | 21.0 | 20.9 | 20.9 | 20.9 |
| 63 | <b>20.2</b> | 18.7 | 18.6 | 18.5 | 18.5 | 18.5 | 18.5 |

**Table 84. Blending the two decoder-side signals** at 0.1 dB, both normalised to unit mean, with the weight running from the calibrated bit rule to the head's ordering. Bold is the best per rate. The two end columns are the two rules measured on their own through the same code path, which is the check that the blend is an interpolation and not a third system.

results/combined\_RECIPES12\_b01.json, on the pinned checkpoint, generated into paper/tables/blend.tex by scripts/make\_paper\_tables.py.

## F.5. How much a trained head varies

No head in this work was trained twice at two seeds, so there is no seed variance to report and none was measured; results/router\_ablation.json records seed 0 for all six of its runs and one\_checkpoint true. What can be reported is the spread between heads that were trained independently of each other, which is a looser quantity than seed variance and a larger one.

**Two heads on one decoder.** The joint and frozen heads differ by -6.47 points at  $q_0$  and by -13.08 at  $q_{63}$ , a range of 6.61 points across the five rates at 0.1 dB, and they cross: the joint head is the better of the two at  $q_0$  and the worse at every other rate. At 5 of the five rates the two heads differ from each other by more than the rule's margin over whichever of them is nearer, so a comparison against one trained head says less than it appears to. That is the reason both are printed in F.4 rather than the better of them.

| q  | oracle | head  | rule  | margin | rule agrees |
|----|--------|-------|-------|--------|-------------|
| 0  | 34.02  | 31.56 | 30.92 | -0.64  | 0.479       |
| 16 | 27.67  | 25.07 | 27.14 | +2.07  | 0.370       |
| 32 | 22.29  | 16.47 | 22.39 | +5.92  | 0.399       |
| 48 | 19.75  | 13.27 | 19.10 | +5.83  | 0.431       |
| 63 | 17.19  | 8.99  | 16.16 | +7.17  | 0.479       |

**Table 85. The same comparison on a second training run.** The margin column is the point: it is positive at four of the five rates and wider than on the pinned run, so the pinned numbers are the conservative ones. BEST is a separate recipe at a different epoch with its own head, trained the same way. The rule is ahead at 4 of 5 rates, by up to 7.2 points, and stays within 3.1 points of the oracle everywhere. The one rate it concedes is the only rate on either run where a trained head finishes ahead, and it does so by under a point.

results/raterank\_BEST\_compare.json: the BEST run, 53 sequences, 0.1 dB, not the pinned checkpoint.

| q  | before | after | change | $\beta$ before | $\beta$ after |
|----|--------|-------|--------|----------------|---------------|
| 0  | 27.18  | 30.27 | +3.10  | 136.6          | 19.0          |
| 16 | 23.27  | 25.82 | +2.54  | 88.8           | 7.7           |
| 32 | 19.40  | 19.43 | +0.03  | 9.1            | -19.1         |
| 48 | 16.06  | 15.54 | -0.51  | -39.7          | -35.9         |
| 63 | 12.94  | 9.58  | -3.36  | -50.3          | -42.2         |

**Table 86. Agreement rises and the saving does not follow.** The change column has both signs. The two heads are one recipe at one  $\lambda$ , trained twice; held-out agreement went from 0.718 to 0.808 while the deployed saving moved by +3.1 points at  $q_0$  and -3.4 at  $q_{63}$ .  $\beta$  is the cost multiplier the bisection applies to move a head trained at one  $\lambda$  onto this operating point, and the retrained head is ahead where that multiplier is small and behind where it is large. This is the same lesson as the rule's low agreement, from the other direction.

results/router\_retrain\_compare.json. The file records no checkpoint, and its before column reads 27.18% at  $q_0$  where the pinned measurement in results/router\_RECIPES12\_b01\_e4head.json reads 33.31%, so the two are not on one

basis. Only the comparison of the two heads within the file is read here.

**Choosing the tilt without the test set.** Every  $\beta$  in the main paper's A-against-B table was bisected against the budget on the test frames themselves, which is the one thing a deployed decoder cannot do. The table below repeats the measurement with the table a deployment would actually ship:  $\beta$  bisected once per quality index on the 512 held-out Open Images validation frames, then applied to the CTC frames without being touched again. Checkpoint, head, frames and budget are identical on both sides, so what separates the two columns is where  $\beta$  came from.

| $\beta$ held out |         | $\beta$ bisected on the test set |       |         |        |       |             |
|------------------|---------|----------------------------------|-------|---------|--------|-------|-------------|
| q                | $\beta$ | saving                           | dB    | $\beta$ | saving | dB    | at equal dB |
| 0                | 173     | 25.3                             | 0.078 | 231     | 28.2   | 0.100 | +0.01       |
| 16               | 148     | 23.8                             | 0.092 | 159     | 25.2   | 0.100 | +0.01       |
| 32               | 133     | 21.5                             | 0.094 | 144     | 22.2   | 0.099 | +0.03       |
| 48               | 33      | 18.3                             | 0.093 | 70      | 20.0   | 0.100 | 0.00        |
| 63               | -34     | 14.8                             | 0.087 | 12      | 17.8   | 0.100 | +0.01       |

**Table 87. Choosing  $\beta$  without the test set.** Left,  $\beta$  bisected on the 512 held-out validation frames and then applied to the test frames unchanged; right,  $\beta$  bisected on the test frames themselves, which is what the main paper's signalled-against-predicted table reports. Savings are hook counts on the routed decode with the router's own compute charged against them. The last column re-bisects on the test set to the quality the held-out  $\beta$  delivered, so the two allocations are differenced at one distortion rather than across two.

results/beta\_calibration.json, generated into paper/tables/beta\_heldout.tex by scripts/make\_paper\_tables.py. The calibration set is the 512 held-out Open Images validation frames the file records, disjoint from the training images and from the test sequences.

The held-out  $\beta$  misses the budget on the low side, at every rate: it delivers between 0.078 and 0.094 dB where 0.1 dB was asked for, 5 of the 5 rates it covers undershoot and 0 overshoot. That is the safe direction: a decoder holding this table never spends quality the budget did not allow, it spends less than it was allowed to. Distortion and saving move together, so those rows report less saving, from 0.7 points at  $q_{32}$  to 3.0 at  $q_{63}$ . The last column shows what is not lost: at equal delivered quality the two allocations agree to within 0.0 points. What moves between the two sets is the decibel a given  $\beta$  delivers, not the ordering it induces.

The highest rate is where this used to fail outright. On the calibration frames the floor, the distortion tiling costs with every tile already at the deepest exit, is 0.079 dB there against 0.054 dB on the test frames, and the gap runs -0.024 to 0.005 dB across the ladder. On an earlier checkpoint that calibration floor sat above the budget: no allocation on 512 px photographs met 0.1 dB at the highest rate, the shipped table had a hole where that rate should be, and a decoder holding it fell back to the deepest allocation, which is our own full-depth path and saves nothing. Four epochs of training pulled the floor below the budget and the hole closed: 0 of the 5 rates are missing from the table now. The gap itself is what remains, and it is what makes the transfer undershoot. A budget written as an absolute decibel sits a different distance above the floor on 512 px photographs than on 1080p video. Calibrating on frames whose tiling penalty matches the ones the decoder will meet is the fix, and for a video decoder that means calibrating on video.

## G. Algorithms, and how each step is implemented

### G.1. What this section covers

Sections A to F state what the system is, what it costs and what it delivers. This one states how it is done: the five procedures the work rests on, each as an algorithm box whose every line names the function that carries it out and the shape it returns, followed by the tensor operations underneath them and the failure each one is written to avoid. A reader who wants to reproduce the system reads A for the recipe and this section for the moves.

The geometry is fixed throughout and is the pinned checkpoint's. The ladder has  $K = 6$  exits over the released decoder's 12 trunk blocks, so  $b = 2$  blocks per exit, and the split depth is  $j = 2$ , which leaves exits 2 to 5

selectable and the first 4 blocks full-frame. One tile is 256 px of RGB, 32 px of the feature grid the per-tile trunk runs on and 16 px of the latent. A 1920×1080 frame is replicate-padded on its bottom and right to 2048×1280, which is a feature grid of 160×256 and 40 tiles in 5 rows of 8. Write T for the tile count of a frame, c for the per-exit cost vector in units of one released decode, and M for the table of per-tile errors with one row per tile and one column per exit. Those three objects are what every procedure below manipulates.

| Procedure                        | File                  | Entry point                   |
|----------------------------------|-----------------------|-------------------------------|
| the deployed decode              | decoder.py            | forward                       |
| the same, tiles ordered by depth | decoder.py            | _suffix_sorted                |
| per-exit price of a tile         | cost.py               | exit_costs                    |
| frame-level price of a map       | cost.py               | frame_relative_cost           |
| MACs a decode performed          | measure.py            | MacMeter                      |
| saving, as the meter counts it   | measure.py            | measured_saving_pct           |
| per-tile error at every exit     | eval.py               | tiled_exit_mses               |
| error a mixed map delivers       | eval.py               | true_frame_mse                |
| the encoder-side search          | signalled_curve.py    | at_lam, bisect_to, true_db    |
| cost of the exit map, in bits    | signalled_curve.py    | map_bits                      |
| the tilt and its bisection       | beta.py               | exit_map, bisect_beta         |
| per-tile logits                  | head2.py              | StemRouterHeadV2.forward      |
| which tiles to override          | hybrid_curve.py       | hybrid_map, h2                |
| one training step                | train_flexuf_image.py | train_one_epoch               |
| the training forwards            | model.py              | forward_random_depth          |
| the objective                    | losses.py             | multi_exit_rd_loss            |
| adapters and the seam gate       | decoder.py            | build_adapter, GridSeamRepair |
| tile-border padding              | padding.py            | wrap_tile_padding             |

**Table 88.** Where each procedure lives. decoder.py and padding.py are under flexuf/backbone/ and head2.py under flexuf/router/; the two curve scripts are under scripts/; train\_flexuf\_image.py is at the repository root and the rest are flexuf/ itself. Take from it the division the rest of the section relies on: everything that decides an allocation is in the scripts and in beta.py, everything that executes one is under backbone/, and the two meet only through the two objects in the middle of the table, a table of per-tile errors and a vector of per-exit prices.

## G.2. The encoder-side search

The encoder holds the source, so it can evaluate the allocation rule exactly rather than predict it. Doing so means building the table M once per rate and then running an argmin over its columns for every candidate multiplier, which is why the expensive part is hoisted out of the search entirely: the per-tile errors, the released decoder's reference error and the bitrate are functions of the latent alone, and only the argmin and the aggregates move with the multiplier. Both loops of the bisection then cost tensor operations on a table of 40 rows by 6 columns.

**input** frames F, budget t in dB, cost vector c of length K

```

1 for each frame in F: once per rate
2   xp = replicate-pad to whole tiles, bottom and right only
3   y, q, aux = net._encode_to_latent(xp, qp)
4   M = tiled_exit_mses(net.dec, y, q, xp, cfg) [T, K]
5   R = reference_frame_mse(ref.dec, y, q, xp) scalar
6   floor = trueDb(0) one decode per frame
7   if floor > t: record budget_reachable = false, stop
8   inner = t
9   repeat at most 6 times:
10    lam = bisectTo(inner) no decode
11    d = trueDb(lam) one decode per frame
12    if |d - t| < 5e-4: leave
13    inner = min(1, max(1e-4, inner + (t - d)))
14   for each frame, at that lam:
15    k = (M + lam · c).argmin(1) [T], the map
16    bits = map_bits(k, K); bpp = bpp + bits / pixels
17    saving = measured_saving_pct(net.dec, ref.dec, y, q, clamp(k, j))
18   report d, and the mean over frames of saving and bits
where
  k(lam) = (M + lam · c).argmin(1)
  tableDb(lam) = 10 log10( meant M[t, k(lam)] / R )
  trueDb(lam) = 10 log10( true_frame_mse(dec, y, q, xp, clamp(k(lam), j)) / R )
  bisectTo(u): lo, hi = 0, 1; 60 times: mid = (lo + hi) / 2;
  if tableDb(mid) ≤ u then lo = mid else hi = mid; return lo

```

**Table 89.** The encoder-side search as scripts/signalled\_curve.py runs it, one line per call. A.12 gives the same search in words and explains why there are two levels; what this adds is the shape at every step and the two places the map is clamped. Lines 4 and 5 are the whole expense and they are outside both loops. Line 15 is the only arithmetic the multiplier touches.

The inner bisection of Table 89 needs a bracket, and the bracket is the unit interval while the answers sit far down inside it: at the 0.1 dB budget the multiplier runs from  $7.3 \times 10^{-5}$  at q0 to  $4.5 \times 10^{-6}$  at q63. The first 18 halvings at the highest rate, and 14 at the lowest, are spent bringing the candidate down to the right order of magnitude before the search resolves anything inside it; the forty or so that remain are what locate the multiplier. The test at the top of the bracket comes first and returns it unchanged when the budget is still not met there, so a multiplier of exactly 1 in the results file is a flag rather than a measurement, and D.8 reads it that way.

Both levels rely on the same monotonicity. Raising the multiplier makes compute expensive relative to error, so tiles fall to shallower exits, the saving rises and the distortion rises with it; neither turns back, so a bisection is valid on either quantity and the code bisects on distortion because that is what the budget is written in. The outer loop exists because the two distortions are not the same quantity. The table measures a tile with every other tile at the same exit, a routed frame is mixed, and a tile's border sees whatever depth its neighbour chose, so the delivered figure sits a little away from the table's. Shifting the inner target by the observed difference and re-bisecting closes it in two or three passes at one decode per frame per pass, against one decode per bisection step if the search were run on the delivered figure directly.

What the table costs is settled twice over. Building it is one tiled decode per exit, and a shallow decode is cheaper than a deep one, so the arithmetic is the sum of the per-exit costs: 4.50 released decodes, from the hook counts in results/ceiling\_measured.json. Timed on a frame rather than added up, it is 4.52 decodes. Of that, 1.23 decodes are two duplicate columns: the clamp makes exits 0, 1 and 2 the same decode, and tiled\_exit\_mses runs the loop over all K exits so that an argmin over all K columns lands on a choice the decoder can carry out. The alternative table, taken from a single full-frame pass with a tap at each exit, costs 1.37 decodes and must not be used to report quality, because it runs the trunk full-frame and the tiling penalty then cancels against a

full-frame reference. It can still be used to rank: on that frame the two searches agree on 82.5% of tiles and land 0.68 points of saving apart.

Sources: results/ceiling\_measured.json for the per-exit hook counts (pinned checkpoint, q32, 2048x1280, 40 tiles); results/supp\_encoder\_cost\_PAPER.json for the timings (pinned checkpoint, Bosphorus at q63, 15 iterations on an NVIDIA RTX A6000, one tiled decode at 114.9 ms); results/signalled\_RECIPES12\_ctc53.json for the multipliers, pinned checkpoint, 53 sequences. The main paper quotes the same ranking comparison from results/encoder\_cost.json, which is the identical measurement on an earlier checkpoint and reads a little differently; the pinned file is the one used here.

Two lines of the box decide what the reported numbers mean. Line 15 runs the argmin against the arithmetic cost model, because a bisection evaluates it for every tile at every candidate multiplier and a decode per candidate is not affordable. Line 17 counts what the chosen decode actually performed, with forward hooks, and that is the figure every table in this paper quotes; B.6 prices the difference between the two and shows it is a closed form rather than a tolerance. A mistake inside the model therefore costs a slightly worse allocation and cannot inflate a headline.

The clamp appears twice and is absent once, which matters where the map is coded in G.5. Line 17 clamps before measuring and trueDb clamps before decoding, so both are priced and measured at the exit that runs. Line 15 does not clamp, and argmin returns the first minimal index, so a tile whose cheapest choice is the shallowest reachable exit is recorded as exit 0. The price is still right, because the cost model gives exits 0, 1 and 2 the same value; the picture is still right, because the decoder clamps; the only quantity that sees six symbols where four exist is the entropy of the map.

### G.3. Routing at the decoder

Configuration B adds no bits, so the decision has to be made from what the decoder already holds, and everything it reads is already in hand at the moment the decision is needed. The stem is the feature after group 1, which the decode has to run whatever the map says; the decoded latent and the entropy model's scales fall out of the pass that produced them; the quality index is one number per frame. Only the head's own convolutions are new arithmetic, and F.2 prices them at 0.163% of a decode.

---

```

input latent y, quant step q, scales, quality index qp, tilt beta
1 stem = dec.upsample(y)
2 for g = 0 ... j-1: stem = dec.groups[g](stem) already needed
3 z = head2(stem, y, scales, qp, 32, 16) [T, K]
4 z = z[:, j:] drop the exits that cannot run
5 lp = log_softmax(z, 1) [T, K-j]
6 optionally lp = lp / mean(|lp|) per frame
7 k = (lp - beta · c[j:]).argmax(1) + j [T], the map
8 saving = measured_saving_pct(..., k) - 100 · rshare
and the budget is met by bisecting the tilt
    bisectTo(u): lo, hi = -2e4, 2e4; 60 times: mid = (lo + hi) / 2;
        if tableDb(mid) ≤ u then lo = mid else hi = mid Table 89's, with k from line 7
    then the same outer correction as Table 89

```

---

**Table 90.** Routing at the decoder, as flexuf/beta.py and scripts/router\_curve.py run it. Take from it lines 4 and 7, which are one operation split in two: the logits are sliced down to the exits that exist, and the column index is put back on the exit axis afterwards. Line 8 charges the decoder for the head it ran, which configuration A never runs.

Line 4 of Table 90 is a slice where the natural spelling is a mask, and the reason is a defect that reached the numbers. The head suppresses the exits below the split by assigning them a large negative constant, which is a suppression only while the head's own logits stay well above that constant. Nothing in a cross-entropy or a regret objective penalises a common offset, one drifted in, and the raw outputs settled at the scale of the mask itself, at which point the masked entries were the largest in every row and the argmax selected an exit that does not exist. The clamp in the decoder then turned that into the shallowest real exit, for reasons having nothing to do with the tile. Every decision site now reads the

sliced logits, the head files were left alone because live training runs import them, and A.6 lists the four places the same bound has to be applied.

The tilt needs a wider bracket than the multiplier does. A confident head has log-probability gaps of hundreds, so a tilt of a few tens moves nothing at all, and the bracket runs to plus and minus  $2 \times 10^4$ ; at the 0.1 dB budget and q0 the bisection settles at 33.

The outer correction against a real decode is the same as in the search above, and it lives in the same two functions. The reason is the calibration experiment: it has to run the identical bisection on held-out images and then apply its answer unchanged to the test frames. A bisection copied into a second caller is a second bisection, and this project has twice paid for a selection rule that was fixed in one copy and not the other.

Line 6 is available to a decoder and is worth stating for that reason. Dividing a frame's log-probabilities by their own mean magnitude is a statistic of the decoder's own output, so it adds no bits and reads nothing from the source; what it buys is that one global tilt means the same thing on a frame the head is confident about and on a frame it is not. Line 8 subtracts the head's own arithmetic from the saving, measured by the same hook count applied to the head alone rather than assumed.

### G.4. Choosing what to override

Configuration C signals the exit of a fraction of the tiles and lets the decoder predict the rest, so it needs a rule for which tiles are worth the bits. The rule is the one the objective itself supplies. At a fixed multiplier the frame's Lagrangian cost is a sum of independent one-tile terms, so replacing the router's choice on a set of tiles removes exactly the sum of those tiles' regrets, and the best set of a given size is the set of largest regrets. The selection is therefore a topk, and the recovered fraction as a function of the signalled fraction is the Lorenz curve of the regret distribution, which is the object C.7 works with.

---

```

input M [T, K], lp [T, K-j], tilt beta, multiplier lam, fraction rho
1 kr = (lp - beta · c[j:]).argmax(1) + j what the decoder would do
2 if rho ≤ 0: return kr, 0 exactly configuration B
3 L = M + lam · c [T, K]
4 ko = L.argmin(1) what the encoder would do
5 if rho ≥ 1: return ko, T exactly configuration A
6 d = L.gather(1, kr) - L.gather(1, ko) regret, [T], never negative
7 S = d.topk(round(rho · T)).indices
8 k = kr.clone(); k[S] = ko[S]
9 return k, |S|
outside the rule
    beta is bisected once, to what configuration B needs for the budget
    lam is re-bisected at every rho, over the signalled tiles
    bits = T · H2(rho) + 3 · |S|
    the router's own cost is charged unless rho = 1

```

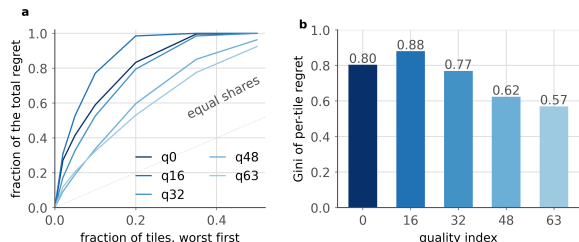
---

**Table 91.** The override rule, as scripts/hybrid\_curve.py runs it. Take from it that the two endpoints are reached exactly rather than approached: at rho = 0 nothing but the router's own argmax is evaluated, and at rho = 1 the router's output is not consulted at all and its compute is refunded, so the columns between them measure an interpolation between two systems rather than three unrelated ones.

One multiplier for both branches was tried first and does not work, for a reason the two bisected values make plain. The router's term is a log-probability and the encoder's is a squared error, so tying them together needs a conversion constant, and the head is confident enough that the constant is of order  $10^{-6}$ : a shared multiplier would have to run to  $10^8$  before the router responded at all, and at that multiplier the oracle branch ignores distortion entirely and sends every signalled tile to the cheapest exit. Measured at the 0.1 dB budget and q0, the tilt lands at 33 and the multiplier at  $7.5 \times 10^{-5}$ . Keeping them separate also matches what each one is: the router is a fixed component whose operating point does not depend on how many tiles are signalled, and the multiplier is the only knob left to absorb the quality the overrides buy back.

results/hybrid\_RECIPES12\_b01\_fixed.json, pinned checkpoint, 53 sequences at one frame each, with the router head runs/RECIPES12/routers2/v2\_lam1.3e-5.pth, itself trained against runs/RECIPES12/ckpt\_eval.pth.tar and not against the pinned checkpoint. F.2 carries that provenance in full.

Two properties of the rule are asserted on synthetic tables in tests/test\_hybrid\_map.py rather than argued: that the overridden set is exactly the tiles of largest regret, with the smallest overridden regret at least the largest untouched one, and that the objective is monotone in the signalled fraction. The endpoint checks are there too, at the level of the returned map rather than of the summary statistics, so a rho of 0 that merely happened to score like configuration B would not pass.



**Figure 60. The loss is concentrated.** **a**, the share of the total regret carried by the worst fraction of tiles, per rate; the dotted line is what an even spread would look like. **b**, the same as a Gini coefficient. At every rate a small minority of tiles carries most of what staying silent costs, which is why signalling a fraction of the map recovers most of the gap.

## G.5. What the map costs to send

The exit map is the one thing configuration A adds to the bitstream, and it is charged before any saving is computed. What is charged is an entropy and a fixed side cost, in four lines.

---

**input** exit map  $k$  of length  $T$ , symbol count  $K$

```

1  $p = \text{bincount}(k, \text{minlength} = K) / T$  the frame's own histogram
2  $H = -\sum \text{over symbols of } p \log_2 p$  bits per tile
3  $\text{bits} = H \cdot T + 8K$  payload, then the histogram
4  $\text{bpp} = \text{bpp} + \text{bits} / \text{pixels}$  before any saving is computed
and for configuration C, which sends a different object
 $\text{bits} = T \cdot H_2(\text{rho}) + 3 \cdot |S|$  the mask, then one symbol per override

```

---

**Table 92.** The map coder, in scripts/signalled\_curve.py and scripts/hybrid\_curve.py. Take from it that line 3 charges the histogram at a flat byte a symbol rather than coding it, and that no arithmetic coder is run at any point: the payload is the entropy, which is what an ideal coder would spend and a real one a little more of.

At the 0.1 dB budget that comes to 64 bits a frame at q0 and 91 at q63, of which 48 bits are the histogram in both cases; the payload itself is 16 and 43 bits. Against a mean of 33.3 tiles a frame over the same test set, that is 0.48 and 1.28 bits a tile, where a fixed-length code over the 4 exits that exist would spend 2. The map grows with the rate because the allocation spreads out: at low rate most tiles take the same exit and the histogram is concentrated, at high rate the mass is spread over the ladder and the entropy rises. As a share of the bitstream it runs to at most  $8.9 \times 10^{-5}$  bpp, and every row of the results file records both the bits and the bpp they became.

results/signalled\_RECIPES12\_ctc53.json, pinned checkpoint, 53 sequences, rows at the 0.1 dB budget. The tile count is the tiles\_per\_frame field of results/hybrid\_RECIPES12\_b01\_fixed.json, measured on the same frames.

Two further things make the reported cost an upper bound, and they are worth naming, because understating our own overhead would be the same class of error as overstating our saving. Line 1 counts the unclamped argmin, so the cheapest allocation is spread over the three indices that share one cost and counted as three symbols where a codec would signal one; the comment in signalled\_curve.py records the effect as at most 7 bits a frame at q0 and 0 at q63. And the three raw bits a configuration C override pays would code one of 4 exits in two.

Configuration C sends a mask saying which tiles were overridden and a symbol for each override, so its bill is the binary entropy of the overridden fraction over the tiles, plus the overrides themselves. At q0

and a signalled fraction of 5% that is 14 bits a frame for 1.6 overrides, against nothing at all at a signalled fraction of zero, which is what makes that column exactly configuration B. The binary entropy is checked against its closed form in the test file rather than trusted.

## G.6. One training step

The training step is where the pinned checkpoint's behaviour is decided, and the shape of it is not the shape the objective is usually written in. A step draws one exit per tile uniformly from the reachable ones, decodes the batch once at that mixed depth, and measures the reconstruction against the source. The per-exit auxiliary losses are not evaluated within a step at all.

---

```

input crop  $x$  [B, 3, 512, 512], quality index  $qp$  [B], multipliers  $\text{lam}$  [B]
1  $\text{lr} = \text{schedule}[\text{epoch} + 99]$ ; new modules at  $20 \times$  per epoch, A.11
2  $y, q, \text{aux} = \text{encode}(x, qp)$  analysis, pass 1
3  $m \sim \text{uniform over } j \dots K-1, \text{ shape } [B \cdot T]$  fresh every step
4  $\text{xhat} = \text{dec}(y, q, \text{exit\_map} = m)$  ONE mixed-depth decode
5  $L = \text{mean}(\text{lam} \cdot \text{mse}(x, \text{xhat})) + \text{bpp}(\text{aux})$ 
6  $y_2, q_2 = \text{encode}(x, qp)$ ;  $\text{ref} = \text{frozen.dec.forward\_full}(y_2, q_2)$  pass 2, no grad
7  $y_3, q_3 = \text{encode}(x, qp)$ ;  $\text{deep} = \text{dec.forward\_full}(y_3, q_3)$  pass 3
8  $L = L + w_a \cdot \text{mean}(\text{lam}) \cdot \text{mse}(\text{deep}, \text{ref})$ 
9  $y_4 = \text{encode}(x, qp)$ ;  $f = \text{dec.exit\_features}(y_4)$  pass 4
10  $L = L + w_d \cdot \text{distill}(f)$ 
11  $(L / \text{accum}).\text{backward}()$ 
12 if the window closes:  $\text{norm} = \text{clip\_grad\_norm}_{(0.1)}$ ;
    if  $\text{norm}$  is not finite: skip the step else  $\text{optimizer.step}()$ 
13 every 200 steps: one full-frame forward_all_exits, for the per-exit log

```

---

**Table 93.** One training step of the pinned run, as train\_flexuf\_image.py executes it. A.10 and A.11 give the recipe and the value of every constant; what this adds is the order, the four analysis passes and the two lines that are diagnostics rather than objective. Line 3 draws from the reachable exits only, which is the same bound the decoder's clamp enforces.

Line 4 of Table 93 is the choice that separates this recipe from a plain multi-exit one. Decoding every exit separately trains a condition that never occurs, because a deployed frame is mixed: a tile at exit 2 sits next to one at exit 5 and the head has to stitch across that discontinuity. One decode at a random mixed depth trains the condition that always occurs, and it costs one decode a step instead of  $K$ . The relation to the weighted objective is exact enough to state. With the auxiliary weights all equal, which is what the pinned run sets, the random draw makes the step's distortion term an unbiased single-sample estimate of the equally weighted mean of the per-exit errors. The difference from evaluating that mean directly is what the neighbours are doing: the estimator measures each exit with its neighbours at random depths, and the direct form measures it with every neighbour at the same depth. That difference is the reason for the design, and the main paper's training subsection measures what closing it is worth.

Lines 2, 6, 7 and 9 are four analysis passes over the same batch. The quantiser is a rounding with a straight-through gradient and the noise the rate estimate uses is drawn separately from it, so the four passes compute the same latent, and with the encoder frozen none of them can diverge from the others. The model carries a single entry point that produces every term from one pass and exists so that DistributedDataParallel sees each parameter inside one forward call; the runs here are one GPU each, the trainer calls the pieces directly, and the redundancy is what that leaves behind. It costs wall clock and nothing else, and no file in results/ measures how much.

Line 12 is the released recipe's guard, applied to the parameters that are actually training rather than to all of them: the norm is clipped at 0.1 and a batch whose norm is not finite is dropped instead of being allowed to reach the weights. Gradients accumulate over a window and the loss is divided by the window length, so the accumulated gradient equals what a single large batch would produce; the pinned run uses a window of one. Line 13 exists because line 4 returns a single mixed-depth reconstruction

and therefore no per-exit breakdown, and that breakdown is the health signal for the failure this literature reports most often, a ladder whose exits converge on each other until only two branches are alive. One extra full-frame pass every 200 steps recovers it for the log at a fraction of a percent of training time.

The joint path exists in the same file and is not what the pinned checkpoint trained under. It replaces the random draw with the router's own sample, taken through a Gumbel-softmax with a hard argmax, and multiplies the chosen tile's output by the selected probability divided by its own detached copy. That factor is numerically 1, so the reconstruction is bit-identical to the deployed path, while the gradient of the rate-distortion loss reaches the logits that an argmax would have blocked. A.7 describes the gate as it appears in the decode; the launcher for the pinned run does not pass the flag that switches it on.

## G.7. The operations underneath

Below the five procedures sit perhaps a dozen operations that have to be exactly right and are silent when they are not. Table 94 writes each one out with the file and the function it lives in. Shapes are given for the pinned geometry: T tiles, C = 384 channels, a feature tile of 32×32 and an RGB tile of 256×256.

| The operation                                                                                                                                                                                                                                                   | File and function                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>tiles and their order</b>                                                                                                                                                                                                                                    |                                                              |
| patchify: <code>x.view(B, C, nh, P, nw, P).permute(0, 2, 4, 1, 3, 5).reshape(B*nh*nw, C, P, P)</code> , with P the feature tile. No arithmetic, and it asserts divisibility rather than padding, so a ragged last row stops the run                             | decoder.py, patchify                                         |
| unpatchify: <code>x.view(B, nh, nw, C, P, P).permute(0, 3, 1, 4, 2, 5).reshape(B, C, nh*P, nw*P)</code> , the exact inverse                                                                                                                                     | decoder.py, unpatchify                                       |
| with a halo, in patchify_with_halo: replicate-pad by h, then unfold(2, P+2h, P).unfold(3, P+2h, P), permute(0, 2, 3, 1, 4, 5), reshape, contiguous                                                                                                              | decoder.py                                                   |
| per-tile error: <code>e = ((xhat - x)<sup>2</sup>).mean(1)</code> , then the same permutation at the RGB tile size, then mean over the tile's pixels                                                                                                            | eval.py, per_tile_mse                                        |
| per-tile router features: the same permutation at the feature tile size, then mean and standard deviation over the tile                                                                                                                                         | head2.py, _pool                                              |
| per-tile bits: the same permutation at the latent tile size, then a sum over the tile                                                                                                                                                                           | hybrid_curve.py, main                                        |
| <b>the exit map</b>                                                                                                                                                                                                                                             |                                                              |
| the clamp: <code>exit_map.to(device).long().clamp(min = j, max = K-1)</code> , before the per-tile loop and after the shape assertion. A.6 lists the four places that have to repeat it                                                                         | decoder.py, forward                                          |
| sorted execution: <code>order = argsort(m, descending, stable)</code> ; <code>bounds = (m_sorted[None, :] &gt; arange(j, K))[, None]</code> . <code>sum(1, tolist())</code> ; each group is then a slice, and the canvas is restored with <code>out[inv]</code> | decoder.py, _suffix_sorted                                   |
| <b>the modules the ladder adds</b>                                                                                                                                                                                                                              |                                                              |
| adapter choice: the FFN when the exit skips $(K-1-k) \cdot b \geq 4$ blocks, the 1×1 otherwise, and none at the deepest exit, whose feature goes to the head raw                                                                                                | decoder.py, build_adapter                                    |
| adapter form: $f + Wf$ for the 1×1; $f + \text{PW}(\text{act}(\text{PW}(f)))$ for the FFN, $C \rightarrow 4C$ then $C \rightarrow C$ because the gated activation folds 4:1                                                                                     | decoder.py, the two adapter classes                          |
| adapter initialisation: the LAST convolution's weight and bias are zeroed, and zeroed again by <code>_zero_init_adapters</code> after the released model's own initialisation pass has run over the whole network                                               | decoder.py, model.py                                         |
| seam repair: $x + G[r \bmod P, c \bmod P] \cdot \text{PW}(\text{WSiLU}(\text{DW}3 \times 3(x)))$ , with G a 32×32 gate initialised to $\exp(-d/\tau)$ , d the distance in feature pixels to the nearest tile edge, and PW zeroed                                | decoder.py, GridSeamRepair                                   |
| the gate laid on the canvas: <code>gate.repeat(1, 1, ceil(H/P), ceil(W/P))[:, :, :H, :W]</code>                                                                                                                                                                 | decoder.py, the gate in forward                              |
| tile padding: the 3×3 depthwise convolutions of groups j and deeper have their padding mode reassigned, or are wrapped when the scheme is not one of PyTorch's, and are restored afterwards                                                                     | decoder.py, _set_tile_padding; padding.py, wrap_tile_padding |
| <b>the two ways of counting</b>                                                                                                                                                                                                                                 |                                                              |

| The operation                                                                                                                                      | File and function         |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| the model: <code>c_k</code> assembled from the measured shares with <code>k_run = max(k, j)</code> applied to both the block count and the adapter | cost.py, exit_costs       |
| the meter: a forward hook on every Conv2d and Linear, MACs from the output shape, output positions taken as numel divided by channels              | measure.py, MacMeter      |
| the ratio: routed MACs on our decoder over released MACs on the reference, both on the same latent, so no calibration constant enters              | measure.py, measured_cost |

**Table 94.** The operations, with the file and the function for each, under the directories Table 88 gives. Take from it the six rows in the first two groups: the same row-major tile ordering is written out four times, at four different grid resolutions, and the exit map is indexed by all four. Only the first pair has an inverse and a test asserting round-trip exactness; the other two agree with it by construction and by inspection.

Shapes and module forms are read from the files named in the second column. The adapter selection and the two adapter costs are confirmed against `results/adapter_cost.json`, which measures the modules of the pinned checkpoint with hooks: the 1×1 costs 147,456 MAC per feature pixel and the FFN 737,280, against a trunk block's 1,035,648. B.3 prices them as shares of a decode.

## G.8. Where a natural implementation is wrong

Six operations, most of them in Table 94, are written the way they are because the obvious alternative fails, and each failure is silent in the sense that matters: it produces a decode that runs, returns a plausible picture and reports a plausible number.

**The four tile orderings.** A tile's exit, its measured error, its router features and its bit count are computed on four different grids, at the RGB, feature and latent resolutions, and every one of them is indexed by the same integer. A permutation that disagreed with the others would deliver a perfectly valid decode of the wrong allocation, and the symptom would be a router that predicts poorly or an oracle that underperforms its own bound, neither of which points at a reshape. The pair that has an inverse is pinned by a round-trip test with zero tolerance. The other two are pinned only by the permutation being written identically, which is why the table above writes it out three times rather than saying that they agree.

**The quantisation step and its batch.** The decoder multiplies the trunk output by a per-quality-index scale of shape [B, C, 1, 1] before the head. Handing a batch of those to a decode of one image broadcasts a [1, C, H, W] tensor back up to batch B, without an error and without a warning, and the wrong-shaped answer surfaces somewhere unrelated. It happened twice, in two files, because nothing tied the call sites together, and forward now raises when the scale's batch is neither 1 nor the latent's. It is also why one of the two training forwards decodes image by image with the scale sliced per image, while the other can decode the whole batch at once with a single exit map of length B·T.

**Installing a padding scheme.** Two of the four schemes are not PyTorch padding modes, so they are installed by wrapping the convolution: pre-pad by hand, then convolve with no padding. Three things about that wrapping are deliberate. The convolutions to be replaced are collected before any of them is replaced, because assigning a wrapper during a walk of the module tree inserts it into the tree being walked and the walk descends into it without end. A second wrapping is refused rather than allowed, because a wrapper whose inner convolution is another wrapper fails inside the convolution call with a message about a missing weight, far from the line that caused it. And the trained per-channel coefficient is one tensor held on the decoder and attached to each wrapper without being registered, since registering it would hand the optimiser one copy of the same parameter per wrapped convolution. The swap is applied only to the groups that run per tile and undone afterwards, which is what keeps the full-frame path bit-exact against the released decoder and the reference honest.

**The phase of the seam gate.** The repair pass is told where the seams are instead of having to infer them, by a gate indexed by position within a tile. That indexing is a modulo, and a modulo is only correct if the tile lattice starts at the origin. It does, because the frame is padded on its

bottom and right only: 1920×1080 becomes 2048×1280 with 128 columns and 200 rows added at the far edges. Centring the padding, which is the natural thing to do to an image, would put every seam out of phase with the gate that was trained on it. The gate is also the one module in the ladder that is tied to the tile size: it has one scalar per position in the tile, so a checkpoint trained at one tile size cannot be evaluated at another without turning it off, and the evaluation script that allows a tile-size override says so at the point of the override.

**The meter's batch dimension.** A convolution's MAC count is its output positions times its output channels times its taps times its input channels over its groups, and output positions here are the output's element count divided by its channel count rather than its height times its width. The per-tile trunk runs  $T$  tiles as  $T$  batch elements, so height times width would divide that part of the decode by the tile count; the comment in `flexuf/measure.py` records the deepest exit measuring 0.43 of a released decode under that reading, against the 1.011 in `results/ceiling_measured.json`. The hooks are registered when the context is entered and removed when it is left, so nothing leaks into the next measurement, and a module an early exit skipped costs nothing because its hook never fires. There is no bookkeeping to get wrong, which is the property the arithmetic model does not have.

**The clamp inside the cost model.** The decoder clamps the map, so a tile assigned an exit shallower than the split still runs the group at the split and wears that exit's adapter, and the model has to bill both the same way. A.6 gives the failure when it did not. What is left after the clamp is a residual with a closed form, measured per exit on the pinned checkpoint: the model under-bills by 0.0080 of a released decode at exits 0 to 3, 0.0027 at exit 4 and 0.0014 at the deepest.

That residual is why the reported saving is the meter's, and why the model's part is confined to the inside of an argmin. B.6 takes it apart term by term.

`results/ceiling_measured.json`, pinned checkpoint, q32, 2048×1280, 40 tiles. Its modelled column is `flexuf/cost.py` and its measured column is `flexuf/measure.py`, on the same latent. The reading attributed to the meter's earlier form is the comment in that file and is in no results file, which G.10 repeats.

## G.9. The controls that hold it in place

Every property above that could fail silently has a test that fails loudly, under tests/. They are asserted at zero tolerance wherever the arithmetic permits it, on the principle that a control which cannot fail is not a control, and the one tolerance that appears is a property of a library rather than of the algorithm.

- **test\_reference\_is\_the\_release.py** decodes with Microsoft's own class and forward function and requires our deepest exit to match it with a largest absolute difference of zero, so the claim rests on the two models producing the same pixels rather than on the key remap having been audited correctly.
- **test\_equivalence.py** holds 10 properties at zero tolerance: the warm-started ladder at its deepest exit is the released decoder; every untrained exit is that decoder truncated at its own depth; patchify and unpatchify round-trip losslessly; with  $j = K$  the tiled path reproduces the full decode, which exercises the halo, the loop, the canvas, the crop and the head at once; every seam-repair variant is the identity at initialisation and the grid gate is concentrated on the border; a mixed-depth map costs less than an all-deepest one; and one trunk pass tapped at every exit gives the same pixels as  $K$  separate passes, which is what makes the joint objective of Eq. (6)-(7) affordable and is the property the from-scratch run depends on.
- **test\_cost\_matches\_reality.py** runs the decode, counts its MACs with hooks and requires the arithmetic model to agree, which is the control that was missing when the model and the decoder drifted apart three times.
- **test\_sorted\_tiles.py** requires sorted execution to give the same picture as the masked loop, on a mixed exit map and on random content, because a uniform map makes the permutation the identity and a constant image makes any permutation exact.

- **test\_hybrid\_map.py** requires the override endpoints to be exactly configurations B and A at the level of the returned map, the overridden set to be the tiles of largest regret, the objective to be monotone in the signalled fraction, and the mask cost to be the binary entropy.
- **test\_router\_mask.py** requires the masked exits to be minus infinity, never the largest logit, and never selected by the tilted argmax at any tilt.
- **test\_router\_inputs.py** requires a zeroed input group to be unable to move a logit at all, and every ablation variant to have one architecture and one parameter count, so that the variants differ in information and in nothing else.

What they are chosen to catch is never a quality regression. Each pins a property whose violation would leave every downstream number internally consistent and wrong, which is the class of defect this project has actually shipped. Two of them are careful about their fixtures for the same reason: the equivalence tests initialise the adapters randomly where a zero would make the property trivial, and the sorted-execution test decodes content rather than zeros, where every tile is identical and any permutation is exact.

The sorted-execution comparison is exact on CUDA and carries a tolerance of  $1 \times 10^{-6}$  on CPU, where oneDNN selects a different blocking at some batch sizes and a plain DepthConvBlock is already order-dependent at 13 tiles while being exact at 4, 9, 16 and 40. That tolerance is the library's and is recorded in `tests/test_sorted_tiles.py`; no results file carries it.

## G.10. What this section does not settle

- **Nothing here is a quality measurement.** The numbers in this section price procedures and describe shapes. No quality figure is established here: the savings and the decibels are measured under the protocol A.4 fixes and reported in D and E.
- **The search timings are one sequence.** The cost of building the per-tile table is timed on Bosphorus at q63, 15 iterations on an NVIDIA RTX A6000, and it is a ratio to a decode on the same card rather than a distribution over the test set.
- **The step's time breakdown is not measured.** The four analysis passes are counted from the source. No file in results/ records what each term of the objective costs in wall clock, and A.10 records only the total for a step.
- **Two implemented paths are not the pinned ones.** The jointly trained router and canvas coupling are both implemented and both measured elsewhere in this document; the pinned checkpoint's configuration block records coupling off, and the launcher does not pass the joint-router flag. Where a number in this paper comes from either, it says so.
- **No test asserts that the four tile orderings agree.** Only the pair with an inverse is checked. The other two are checked by reading them.
- **The map cost is an entropy, not a coder.** No arithmetic coder is run over the exit map; the reported bits are the entropy of the per-frame histogram plus a flat byte a symbol for the histogram itself, which bounds what a real coder would spend from below on the payload and from above on the side information.
- **Three numbers here are typed from source comments.** The meter's earlier reading of the deepest exit, the CPU tolerance of the sorted-execution test, and the bits that merging the three equal-cost indices would save. Each is attributed where it is used, and no file in results/ carries any of them.

## H. Failure cases, limitations and open problems

The method removes decoder compute by letting easy tiles leave the trunk early, and there are frames where no tile is easy, frames with too few tiles to allocate over, and content where the quality a budget buys is spent in a place a viewer would look. This section measures those cases rather than describing them, then lists the mechanisms that were built for this work and dropped, each with the measurement that ended it, and

closes on the problems that are open at submission. Section D names five failure cases read off the operating-point files and is not repeated; what follows is the content axis, the abandoned work, and the state of the checkpoints every table in this document rests on.

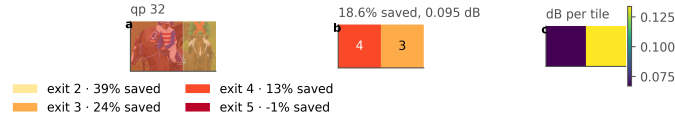
### H.1. The frames it does worst on

A frame is cut into tiles of 256 px, so the number of decisions the allocation has to make is fixed by the resolution. At 1920×1080 a padded frame carries 40 tiles and at 416×240 it carries two. Table 95 takes one frame from each of the five test classes at the same rate and the same budget and prints what the allocation did with it. Reading down the tile column, the exit map goes from a mixture over three exits to a constant, and the saving falls with it.

| frame                      | tiles | exits taken         | saved % | delivered dB | worst tile dB |
|----------------------------|-------|---------------------|---------|--------------|---------------|
| Bosphorus, 1920×1080       | 40    | e2:29 e3:8 e4:3     | 34.2    | 0.099        | 0.374         |
| BasketballDrive, 1920×1080 | 40    | e2:22 e3:9 e4:9     | 29.9    | 0.099        | 0.178         |
| FourPeople, 1280×720       | 15    | e2:6 e3:1 e4:3 e5:5 | 19.6    | 0.100        | 0.271         |
| PartyScene, 832×480        | 8     | e3:4 e4:3 e5:1      | 16.9    | 0.097        | 0.162         |
| RaceHorses, 416×240        | 2     | e3:1 e4:1           | 18.6    | 0.095        | 0.134         |

**Table 95. One frame from each class at q32 and a 0.1 dB budget.** The exits taken column is the histogram over the ladder, so e4:2 is two tiles at exit 4. Take from it that the allocation stops being an allocation at the bottom of the table: at 416×240 both tiles take the same exit, and the frame receives whatever that one rung happens to cost. The worst tile column is the largest per-tile loss in the frame and is between two and three times the frame's own delivered decibel on the three largest frames, which is the tail section D measures over the whole set.

The five results/supp\_exitmap\_\*\_q32\_b01.json files, all on runs/RECIPES12/ckpt\_PAPER.pth.tar, one frame each. These probes carry the modelled saving and no hook count, so each is optimistic by up to 0.8 points, the constant recorded in results/ceiling\_measured.json; they are printed for the allocation and the per-tile loss rather than for the headline, which section D reports hook-counted over the whole set.



**Figure 61. The smallest frame in the test set, and the whole of its allocation.** a, the decoded frame with the single tile boundary drawn, tinted by the exit each tile took. b, the exit map, two cells. c, the loss each tile pays. Both tiles take exit 4 and the map has nothing to trade; the legend gives what each rung would have saved had a tile taken it.

paper/figures/exitmap\_PAPER\_racehorses\_416x240\_q32\_b01.png, drawn from results/supp\_exitmap\_racehorses\_416x240\_q32\_b01.json on runs/RECIPES12/ckpt\_PAPER.pth.tar. RaceHorses at 416×240, q32, 0.1 dB.

The two frames at either end of that table are the same decoder at the same budget. Bosphorus at 1080p spreads 40 tiles over exits 2, 3, 4 and saves 34.2% of the modelled decode; RaceHorses at 416×240 puts both of its tiles on exit 4 and saves 18.6%. Nothing about the content separates them. What separates them is that a two-cell map has three admissible states above the split depth and a 40-cell map has more states than the sweep could enumerate, so the Lagrangian has somewhere to put the budget in one case and not in the other.

| sequence       | q  | tiles | PSNR lost dB | MS-SSIM lost dB | saved % | its rank |
|----------------|----|-------|--------------|-----------------|---------|----------|
| videoSRC26     | q0 | 40    | 0.137        | 0.118           | 39.1    | 49th     |
| KristenAndSara | q0 | 15    | 0.162        | 0.111           | 37.1    | 26th     |
| videoSRC03     | q0 | 40    | 0.147        | 0.107           | 37.4    | 28th     |

**Table 96. The three sequences that lose most on the perceptual metric.** All three are 1080p frames at the lowest rate and carry the full 40 tiles. The MS-SSIM column is  $-10 \log_{10}(1 - \text{MS-SSIM})$ , the unit section D puts the two metrics in, and on these sequences it is two to three times the set mean of 0.050 dB at this rate. The PSNR column is the same file's own difference in PSNR and runs from 0.137 to 0.162 dB, over the nominal budget on all three, which is the per-sequence overshoot the next subsection measures. The last column is where each sits in the set by compute saved, from the 26th percentile to the 49th: these are not the sequences that save least.

The worst\_sequences\_by\_ms\_ssim block of results/supp\_opquality\_PAPER.json, on runs/RECIPES12/ckpt\_PAPER.pth.tar, 53 sequences at one frame each, at the 0.1 dB operating point of results/supp\_paper\_curve\_PAPER.json. That block is the file's own shortlist of the fifteen largest MS-SSIM losses over the whole sweep, reordered here by the decibel rather than by the raw difference. The saving column is the same file's per-sequence saving against our own deepest exit and the rank is its percentile among the 53 sequences of results/supp\_per\_sequence\_PAPER\_b010.json at the same rate.

The ordering there is the one to take away. A budget in PSNR does not price the second metric evenly across content, and the sequences where the two disagree most are ordinary 1080p clips from the middle and upper part of the set by saving rather than the hard cases further down this section. On a clip with 40 tiles the multiplier can move a third of the frame two rungs down the ladder and stay inside 0.1 dB of PSNR, and the tiles it moves are the flat, low-texture ones a structural metric weighs differently from a mean squared error. The remedy is a budget written in the metric a deployment cares about, and no allocation in this work was bisected on anything but PSNR.

### H.2. What a set-level budget hides

| q   | over budget | worst dB | on         | least saved % | on             |
|-----|-------------|----------|------------|---------------|----------------|
| q0  | 32 of 53    | 0.248    | Jockey     | 24.2          | BQSquare       |
| q16 | 33 of 53    | 0.243    | videoSRC02 | -1.0          | BQSquare       |
| q32 | 31 of 53    | 0.258    | videoSRC02 | -1.0          | BQSquare       |
| q48 | 31 of 53    | 0.253    | videoSRC02 | -1.0          | BlowingBubbles |
| q63 | 31 of 53    | 0.215    | videoSRC02 | -1.0          | BQSquare       |

**Table 97. The 0.1 dB budget, sequence by sequence.** The multiplier is bisected once for the whole set, so the budget binds on the set mean and on nothing else. Rather more than half the sequences receive a decode worse than the budget they were nominally given, and the worst of them receives between two and three and a half times it. Section D.4 gives the spread on the compute axis; this is the same allocation seen on the quality axis.

results/supp\_per\_sequence\_PAPER\_b010.json, the rows block for the delivered decibel and rows\_vs\_release arithmetic for the saving, on runs/RECIPES12/ckpt\_PAPER.pth.tar. Decibels are pooled, the convention results/supp\_paper\_curve\_PAPER.json bisects in.

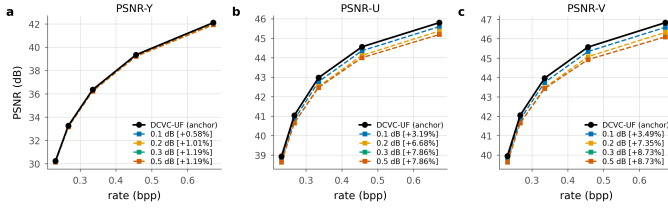
**The budget is spent unevenly across the colour components.** Every quality number in this work is DCVC-UF's own 6:1:1 weighted YUV420 PSNR, in which chroma carries an eighth of the weight, and the budget is set on it. Split into its parts at the 0.1 dB budget, luma falls by 0.066 dB and the two chroma planes by 0.205 and 0.223 – chroma loses 3.3 times what luma loses. The weighted mean lands on the budget exactly as it is supposed to, and says nothing about the split underneath it.

| budget | PSNR-Y | PSNR-U | PSNR-V | 6:1:1  |
|--------|--------|--------|--------|--------|
| 0.1 dB | -0.066 | -0.205 | -0.223 | -0.103 |
| 0.2 dB | -0.108 | -0.399 | -0.439 | -0.186 |
| 0.3 dB | -0.126 | -0.468 | -0.525 | -0.218 |
| 0.5 dB | -0.126 | -0.468 | -0.525 | -0.218 |

**Table 98. Quality given up per colour component,** dB below the released decoder, averaged over the five rates on the 53 test frames. The last column is the metric the budget is set on; the first three are what it is made of.

results/rd\_yuv\_PAPER.json. The operating points are not re-derived: each row of the signalled sweep records the multiplier its bisection settled on, and the exit map is a deterministic argmin given that multiplier, so the decode measured here is the decode the sweep priced. The split is checked against the weighted figure on every frame.

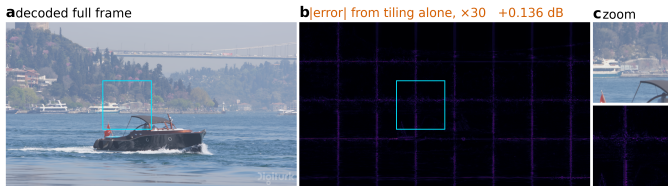
The mechanism is the objective, not the ladder. The allocation minimises absolute error, and chroma starts with far less of it – at the lowest rate the released decoder reaches 38.9 dB on U against 30.2 on Y, so chroma's mean squared error is about seven times smaller. The same absolute increase is therefore a much larger number of decibels on chroma, and an allocator that does not know the difference spends it there first. A budget set on the weighted metric cannot see this; one set per component could, and that is the change we would make.



**Figure 62. Rate-distortion per colour component**, ours against the released decoder, one panel per component, with each budget's BD-rate against the anchor in brackets. The anchor is the released decoder on the same bitstream, so a bracket is the price of decoding one file with fewer operations rather than one codec against another.

The other thing a set mean hides is how much of the budget is gone before any tile exits early. With every tile at full depth the decode already differs from the released decoder by 0.022 dB at q0 and 0.048 dB at q63, which is two thirds of the headline budget at the high rate. Two things make it up, and they are measured separately rather than subtracted from one another: the deepest exit's own departure from the released decoder, 0.028 dB at q63 on the same checkpoint, and the seam left by cutting the frame into tiles. Both are charged to the method, because the saving is measured against the released decoder rather than against our own full-depth decode, and both grow with rate while the budget does not.

Floors from results/signalised\_RECIPES12\_ctc53.json and drift from results/supp\_anchor\_PAPER.json, both on runs/RECIPES12/ckpt\_PAPER.pth.tar over 53 sequences. The two files do not pool their decibels alike, which is why the seam is described rather than given as the difference. Section C derives why a floor above the budget makes the feasible set empty and section E tabulates the three ways the floor itself can be measured.



**Figure 63. The seam, at frame scale.** a, the decoded frame. b, the absolute error against a full-frame decode of the same latent, amplified 30 $\times$ , with every tile at the deepest exit so that nothing here is early exiting. c, a zoom on one junction. The tile grid is the error: 0.136 dB at q63, paid before the allocation begins.

paper/figures/seam\_PAPER\_bosphorus\_q63.png, drawn from results/supp\_seam\_problem\_bosphorus\_q63.json on runs/RECIPES12/ckpt\_PAPER.pth.tar. The panel is the worst case rather than the shipped one: the file records tile\_pad\_mode zeros with the deblocking filter removed, which is how the artefact is made visible. The floors quoted above are the shipped configuration, replicate padding with the filter on.

### H.3. Built, measured and abandoned

Seven mechanisms were built, measured and then left out of the reported system. Two of them replace the border a tile is convolved against, one repairs the seam afterwards, one removes the seam by letting neighbouring tiles exchange features, one freezes the trunk so that the deepest exit cannot move, one gives the small resolutions a smaller tile, and the last is an explanation of the tiling penalty rather than a mechanism. Table 99 gives what settled each, and the file it was settled in. The cost of building them is not otherwise recoverable from the paper, and two of them are the first thing a reader would propose.

| what was built                    | the measurement that ended it                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------|
| First-order border extrapolation  | worse than replication at all three rates: 0.198, 0.286 and 0.384 dB against 0.071, 0.123 and 0.218           |
| Fitted AR(1) border padding       | ahead of replication by 0.021 dB at q63, and bought with wall clock that no file in results/ records          |
| A learned deblocking filter       | 0.0008 dB even with a perfect gate, against 1.09% of a decode                                                 |
| Halo exchange between tiles       | exact at uniform depth, and the saving falls from 31.2% to 5.5% at q32 and from 28.0% to 1.0% at q63          |
| Freezing the trunk                | drift exactly 0.000 dB, and the oracle ceiling collapses with it to 11.5% at q0, 0.4% at q32 and -1.0% at q63 |
| A tile size chosen per resolution | worth -0.06 points on the set mean over three rates, with the training confound running in its favour         |
| The affected-area fraction        | wrong by 160 to 264%, against 12 to 22% for a power law in the per-tile block count                           |

**Table 99. Seven mechanisms and what settled each.** Five were built and dropped, one is a proposed extension measurement did not support, and the last is an explanation of the tiling penalty that the measurement does not carry. None of the seven is in the reported system, whose configuration is replicate padding at 256 px with the deblocking filter left on, as the ablation table of section A records.

Rows in order: results/ctc\_seam\_p256.json (40 sequences, three rates); the same file with results/shrink\_ablation.log; results/seam\_spatial\_published.json with the module's share of a decode from results/supp\_module\_shapes.json; results/coupling\_ablation.json; results/first\_ckpt\_headonly.log; results/tilesize\_adaptive.json; results/supp\_seam\_vs\_split.json, refitted in the build. The coupling and heads-only files are on runs/RECIPES12/ckpt\_eval.pth.tar and runs/heads\_only\_j2\_p256/ckpt\_epo0.pth.tar; the seam-repair spatial file is on a runs/BEST checkpoint whose epoch it does not record. The rest are on runs/RECIPES12/ckpt\_PAPER.pth.tar.

Three of those rows need more than a line. The halo exchange does two of the three things asked of it: it is exact at uniform depth, and it removes 100% of the floor at q0 and 43% at q63. The third, that closing the floor is worth several points of saving, has the wrong sign at every rate above the lowest. A tile whose neighbour ran six more blocks is reading an activation the trained weights have never seen, and the error that introduces is far larger than the seam it removed. Exactness holds when every tile is at the same depth, and routing is the deliberate violation of that condition, so the two are in tension by construction. The one caveat that keeps this from being final is that the model was trained with replicate padding, so switching the exchange on at inference is a distribution shift; a run trained with it could reverse the sign, and no such run exists.

The deblocking filter is the reverse case, where the mechanism does what it was built to do and the amount is too small to pay for. Error falls monotonically with distance from the nearest tile boundary, 4.7e-05 in the innermost 4 px against 4.1e-05 more than 64 px away, so the seam is real and it is local. The filter's gate leaks: it gains in the boundary band and loses slightly everywhere else. Giving it a perfect gate, with its boundary gain kept and its interior loss set to zero, bounds it at 0.0008 dB for 1.09% of a decode. Tightening the gate cannot rescue the module because there is not enough in the band for it to win. It is still switched on in the reported system, because it was trained jointly with the decoder and switching it off at inference is not the same experiment as training without it, and no run has been trained without it at 256 px.

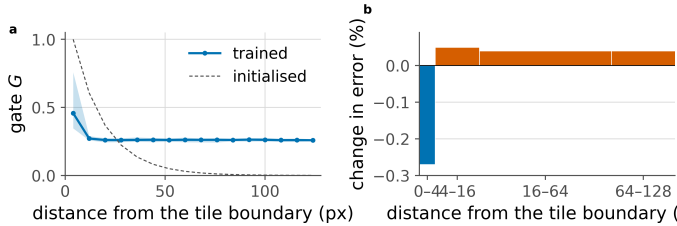
The affected-area fraction is an explanation rather than a mechanism, and it is the weaker of the two available. It is a correct statement about which pixels a tile border can reach and a poor predictor of what the border costs, because it saturates: at 12 blocks per tile it is already 0.94 and cannot rise further, while the penalty still climbs by a factor of 3.1 from 8 blocks to 12. Fitted with one free scale it is wrong by 160 to 264% on average. A power law in the block count is wrong by 12 to 22%, with exponents 2.38, 2.22 and 1.93 at q0, q32 and q63 and log-log correlations from 0.979 to 0.995. An exponent near two has a reading: the number of affected pixels grows with the border's reach and the damage in each grows with how many convolutions reached it, and the two are the same quantity. The drift from 2.38 to 1.93 with rate is not explained here. The control is the row that licenses reading any of it as seam: at zero blocks per tile the penalty is 0.0000 dB at every rate.

The tile-size row is the one a reader is most likely to propose, since the exit maps of Table 95 say the method is governed by tile count. Splicing 128 px tiles into the two smallest classes and leaving 256 px elsewhere is worth -0.06 points on the set mean, averaged over three rates. The direction differs by rate rather than being uniformly flat: at q0 the smaller tile is ahead in five of six classes and the splice gains +0.33 points, while at q63 the smaller tile is behind on every class and the splice loses -0.33. Halving the tile side doubles the seam at fixed depth, and above the lowest rate the extra seam costs more than the extra granularity returns. The comparison is not matched for training time: the 128 px column carries 24,000 more steps than the pinned checkpoint, an advantage that runs in the smaller tile's favour, which is why the conclusion is that the extension is not supported rather than that it is harmful.

results/tilesize\_adaptive.json, which splices results/per\_class\_RECIPES12.json on runs/RECIPES12/ckpt\_PAPER.pth.tar against results/per\_class\_BEST128.json on runs/BEST128/ckpt\_step.pth.tar. The file brackets the training confound with a third measurement on runs/RECIPES12/ckpt\_PIN\_e1.pth.tar and records the verdict per class and per rate.

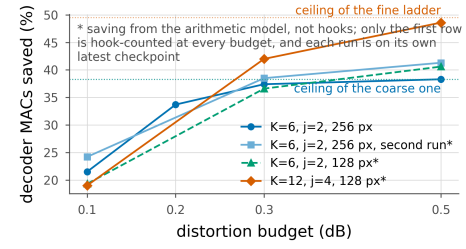
The AR(1) padding row is the one this section cannot close from results/. It is the best border estimator measured, 0.197 dB at q63 against 0.218 for replication, and it was rejected on wall clock, which no file in results/ holds. What results/ does hold is why nothing cheaper works. Its coefficient is fitted per channel and per tile, and replacing the fit with a fixed shrinkage recovers none of the gain: at a coefficient of 0.95 the seam is 0.2138 dB against replication's 0.2160 and the fit's 0.1797, and every coefficient below that is worse than replication at every rate. The gain is in the adaptivity, and the adaptivity is what costs.

results/shrink\_ablation.log, 10 CTC frames at  $j = 2$  and 128 px, typed into this module because it is a log rather than JSON; the seam figures it is compared against are results/ctc\_seam\_p256.json, 40 sequences at 256 px, so the two sets of absolute values are not on one protocol and only the ordering within each is read. The wall-clock measurement that decided the estimator is recorded in the project's decision log and in no file in results/.

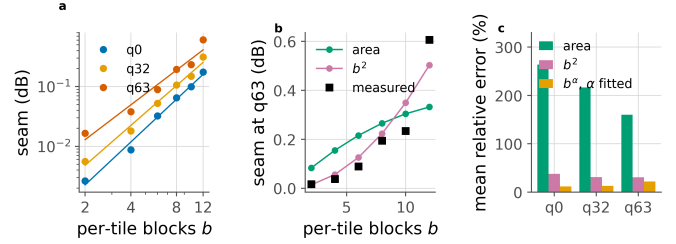


**Figure 64. The learned deblocking filter, and where it wins.** It is gated by position within a tile and applied once to the stitched frame, for 1.09% of the decode. **a**, the trained gate against distance from the boundary, beside its initialisation. **b**, the change in error by that distance; bar width is the share of pixels, so bar area is the contribution to the frame. It wins on the pixels nearest a boundary and loses on the rest, which is why a perfect gate still cannot pay for it.

The exactness claim for the halo exchange is worth writing out, because it is the condition and not the number that carries the argument. At uniform depth the largest absolute difference between a tiled decode with the exchange and a full-frame decode of the same latent is exactly zero once the deblocking pass is switched off; with the pass on it is  $4.27e-02$ , because that pass runs on the stitched frame and has no full-frame counterpart, and with replicate padding instead of the exchange it is larger again. The condition in that sentence is *uniform depth*, and routing is the deliberate violation of it, which is the whole of the tension.



**Figure 65. The right ladder depends on the budget.** Saving against budget for four ladders, with each one's architectural ceiling as a dotted line. A finer ladder has a higher ceiling and a higher floor, so the curves cross: the coarse ladder wins where the budget is tight and cannot be spent, and the fine one wins where it is loose enough to reach exits the coarse ladder does not have. The crossing is the design choice this section is about.



**Figure 66. The seam against per-tile depth.** **a**, the measurement with the fitted power law. **b**, both models against q63, one free scale each. **c**, mean relative error. The area fraction saturates once the border reaches every pixel and the penalty does not, which is where the area law fails as a predictor.

#### H.4. The checkpoints, and what a later one measures

Every table in this supplement is measured on runs/RECIPES12/ckpt\_PAPER.pth.tar, taken at the end of epoch 4. The per-checkpoint watcher measured the earlier epochs of the same run on the same 53 sequences at the same budget through the same code path, and Table 100 is what they say.

| checkpoint                 | q0   | q16  | q32  | q48  | q63  | mean |
|----------------------------|------|------|------|------|------|------|
| RECIPES12, epoch 0         | 29.2 | 25.0 | 20.3 | 17.8 | 15.2 | 21.5 |
| RECIPES12, epoch 1         | 30.1 | 26.4 | 21.4 | 19.4 | 16.8 | 22.8 |
| RECIPES12, epoch 2         | 32.8 | 28.1 | 22.5 | 19.9 | 16.9 | 24.0 |
| RECIPES12, epoch 3         | 33.2 | 29.5 | 24.2 | 22.4 | 19.9 | 25.8 |
| RECIPES12, epoch 4, pinned | 33.7 | 30.1 | 26.1 | 24.8 | 23.5 | 27.6 |
| BEST, epoch 1              | 34.0 | 27.7 | 22.3 | 19.8 | 17.2 | 24.2 |
| BEST, epoch 2              | 29.4 | 23.2 | 18.0 | 14.5 | 11.5 | 19.3 |
| BEST, epoch 3              | 30.8 | 24.7 | 19.2 | 15.6 | 12.2 | 20.5 |

**Table 100. Compute saved at the 0.1 dB budget as training continues.** RECIPES12 and BEST are two runs of one configuration, same architecture and same recipe, differing only in the draw. The pinned row is what the paper reports, and it is the last epoch the run had completed when we stopped. Every RECIPES12 row is hook-counted on the same 53 sequences, so the differences between them carry no convention offset. The series rises at every rate and at every step – 21.5% to 27.6% on the set mean – and the gain is larger at the high rates than the low ones, which is where the method had the least to give: q63 moves by 8.3 points against 4.5 at q0. Nothing here has flattened.

The RECIPES12 rows are every results/signalled\_RECIPES12\_\*.json over the full 53 set that carries a hook count, one row per epoch, selected by savings.epoch\_series – the same rule the main paper's macros and the plateau figure use. The BEST rows are results/signalled\_BEST\_ctc53.json, results/signalled\_BEST\_0819\_1050.json and results/signalled\_BEST\_0820\_0841.json on the ckpt\_eval and ckpt\_step files of that run, which the watchers overwrite; each records its own epoch. None of the three carries a hook count, so they are on one basis with each other and about 0.8 points optimistic against every RECIPES12 row. Comparisons within either block carry none of that offset; comparisons across the two do.

One condition governs any comparison between measurements made at different times. The test set grew from 40 sequences to 53 when MCL-JCV and the two smallest HEVC classes were added, and eight tiles at  $832 \times 480$  or two at  $416 \times 240$  save nothing like a 1080p frame does, so a saving measured on the smaller set is several points above one measured on the larger at the same checkpoint. Every row of this section

is on 53 sequences.

## H.5. How far two runs of one recipe are apart

RECIPE512 and BEST are the same K, the same split depth, the same tile size, the same adapters and the same 45,445,398 parameters. At epoch 2 they are 4.7 points apart on the set mean. The ladder table of the main paper reports a 256 px run at 24.2 against a 128 px run at 19.3 and a twelve-exit run at 19.0, the last over the four rates it reaches rather than five, which are differences of about five points. The gap between two runs of one configuration is the same size as the architectural differences the table is ranking.

results/signalled\_BEST\_ctc53.json, results/signalled\_BEST128\_ctc53.json and results/signalled\_FINE12\_ctc53.json, the three files behind the main paper's ladder table, each on the ckpt\_eval or ckpt\_step file of its own run; none of the three carries a hook count, so the three are on one basis with each other and about 0.8 points optimistic against the pinned row of Table 100. The epoch-2 comparison above is between two files that both carry one.

The spread is not only between runs. The watcher writes a measurement every time it finds a new checkpoint, so one epoch of one run yields several, and Table 101 prints six of them from a single epoch. They span 1.8 points on the four-rate mean and up to 3.0 at a single rate, on one test set, one budget and one code path, with every row hook-counted.

| checkpoint, in order | q0   | q16  | q32  | q48  | mean  |
|----------------------|------|------|------|------|-------|
| 0819_0923            | 30.3 | 24.8 | 16.9 | 10.8 | 20.70 |
| 0819_1239            | 28.9 | 22.9 | 18.2 | 12.4 | 20.60 |
| 0819_1555            | 30.8 | 25.3 | 18.3 | 12.6 | 21.74 |
| 0819_1911            | 30.1 | 23.5 | 18.0 | 12.2 | 20.94 |
| 0819_2157            | 31.5 | 25.6 | 18.8 | 13.7 | 22.38 |
| 0819_2227            | 31.0 | 25.9 | 18.4 | 12.4 | 21.94 |

**Table 101. Six checkpoints from one epoch of one run.** FINE12 is the twelve-exit ladder, whose highest rate has no admissible allocation at this budget, so the mean is over the four rates that do. Take from it that a comparison at a matched epoch is not a comparison at a matched checkpoint, and that the difference between two rows here is as large as several of the differences this work reads as results.

The six results/signalled\_FINE12\_0819\_\*.json files listed in this module, all at epoch 2 of runs/FINE12 on 53 sequences at a 0.1 dB budget, all carrying a hook count. The checkpoints themselves are the run's own per-epoch files, which it rewrites, so these measurements are what preserve them. A seventh epoch-2 measurement is excluded because it carries no hook count.

What follows from the two tables is a constraint on how this work may be read. No configuration was trained twice at two seeds, and no seed is set in the decoder's trainer, so there is no interval on any architectural comparison in this document. Section A says so and gives the sources of variation; what these files add is a magnitude. Until a configuration is trained twice, a difference of a few points between two rows of the ladder table is not separable from the difference between two runs of either of them.

## H.6. Anchor drift

| run                      | q0     | q32    | q63    | frames |
|--------------------------|--------|--------|--------|--------|
| RECIPE512, pinned        | 0.0004 | 0.0114 | 0.0283 | 53     |
| RECIPE512, epoch 0       | 0.0051 | 0.0145 | 0.0362 | 53     |
| BEST                     | 0.0015 | 0.0088 | 0.0154 | 53     |
| BEST128                  | 0.0036 | 0.0104 | 0.0216 | 53     |
| FINE12                   | 0.0056 | 0.0254 | 0.0600 | 53     |
| VERBATIM, no anchor term | 0.1101 | 0.1241 | 0.1594 | 40     |

**Table 102. How far the deepest exit has moved from the released decoder,** in decibels, one checkpoint per run. The deepest exit is supposed to be the released decoder and the training objective carries a term that holds it there. The term does real work: the run trained without it sits 0.110 to 0.159 dB away, more than the whole headline budget at every rate before a single tile has exited early. The term does not hold it exactly, and what it leaves behind grows with rate: 28% of a 0.1 dB budget at q63 on the pinned checkpoint and 60% on the twelve-exit run, whose floor exceeds that budget outright at the highest rate.

results/supp\_anchor\_PAPER.json on runs/RECIPE512/ckpt\_PAPER.pth.tar; results/anchor\_RECIPE512\_e0.json is the same measurement on the same frames and

rates when the pin held epoch 0 of that run; results/anchor\_BEST.json, results/anchor\_BEST128.json, results/anchor\_FINE12.json and results/anchor\_VERBATIM.json on the ckpt\_eval or ckpt\_step file of their runs, at the epoch each happened to be at, which none of them records. Frame counts differ by file and are printed, because two of these are 40 frames and are not comparable with the 53-frame rows at the fourth decimal.

Whether it grows with training is now settled, on this run at least. The two RECIPE512 rows are four epochs apart on the same 53 frames and the same rates, and the drift falls at 5 of the 5: 0.0051 to 0.0004 dB at q0 and 0.0362 to 0.0283 at q63. The anchor term tightens its hold as the run goes on rather than losing it, which is the direction the term is there to produce and not one we had evidence for before. The direction with rate survives the move and steepens: the deepest exit sits 76 times further from the released decoder at q63 than at q0 on the reported weights, against 7 times four epochs earlier, because q0 improved by more than the rest. The open problem is the size at the top of the ladder, not the trend: at q63 28% of a 0.1 dB budget is spent before the allocation begins, and a move to any later checkpoint has to remeasure this and report it as its own row rather than assume it carries over.

There is a structural fix and it was measured and rejected, which is row five of Table 99. Freezing the trunk makes the deepest exit the released decoder by construction, with no weight left that could move, and the drift measures exactly zero. The shallow exits then cannot improve either, because the exit heads alone cannot recover what six skipped blocks contained, and the oracle ceiling collapses to nothing above the lowest rate. Training the trunk is what makes the ladder worth having, and the drift is what training the trunk costs.

## H.7. How much of this can be repeated exactly

Every measurement file records the checkpoint it was taken on, and there are three kinds of answer. 64 of the files these two documents read name a checkpoint that does not move – a pin or a warm start – so the measurement can be repeated exactly. Another 17 name one the watcher overwrites but record the epoch or step it was at, so the weights are identified even though the path is not. The remaining 12 record neither, and their paths now point at whatever those runs have trained to since.

That last group is not hidden – several of the files in it are named as such where they are used, and the epoch tables of this section exist because of it – but it is worth stating once as a number rather than as remarks. Nothing in the main paper's headline is in it: the pinned checkpoint carries every table the paper reports. What is in it is the adapter and coupling ablations, the hybrid sweeps of configuration C, the wall-clock files and the anchor drift of the runs other than the pinned one. Repeating those exactly would need the checkpoints back, which is why scripts/check\_provenance.py refuses to let the count grow.

Counted in the build over results/\*.json against the file names scripts/build\_pdf.py, scripts/make\_paper\_tables.py and scripts/supp/\*.py mention. The same count is a check, with the present figure of 12 as its baseline.

## H.8. What is open

Every open problem in this supplement is listed here, so that a reader meets each of them once. Section A gives the provenance of each table and section D the spread behind each mean; what follows is what those two cannot close.

- **The reported checkpoint is early.** Table 100 says a later checkpoint of the same run saves more at every rate. Nothing in this document is measured on one, because 88 checked claims, every table and every figure rest on the pinned checkpoint, and moving it means remeasuring all of them. What a later checkpoint would change is the size of the headline and the anchor drift beneath it, in opposite directions.
- **No interval exists on any architectural comparison.** Two runs of one configuration are 4.7 points apart at a matched epoch, and six checkpoints inside one epoch of one run span 1.8. Both are the size of the differences the ladder table ranks. The remedy is repeated runs, which is a training cost this work has not spent.

- **The anchor is held by a loss term and not by construction.** The deepest exit is 0.028 dB from the released decoder at q63 on the pinned checkpoint, about a third of the headline budget, and the one construction that removes the drift removes the ladder's value with it.
- **Everything here is one decoder's intra path.** The ladder is built inside the intra decoder of one released codec and measured on intra frames of video sequences at 1080p and below. No inter frame, no second codec rearranged into a ladder, and no resolution above 1080p, the last because the evaluation card has about 5 GB free and a 4K decode does not fit. The construction needs a residual trunk with a shared head and nothing else, which is an argument rather than a measurement.
- **The window is measured with the losses known.** The allocations in this section are the exact per-tile search, which bounds any decoder-side predictor. Section F measures how much of it a trained head reaches and how much a rule with no learned parameters reaches, and neither closes the gap at high rate.
- **Part of the recipe is pinned outside results/.** Nothing that writes to results/ runs inside the trainer, so the hyperparameter table of A.11, the dataset statistics of A.8 and the training wall clock are read from the launcher, the trainer and the run's own log rather than from a measurement file. The warm-start report on disk is the K = 12 rebuild rather than the K = 6 one the pinned run descends from; its two zero-difference controls are asserted at every build instead, so a failure would stop the run.
- **One accelerator class.** Every millisecond and every joule in this work is an NVIDIA RTX A6000, all eight cards in this machine being that model, and the CPU rows of section B are the only second device class. Power comes from the driver counter, with no external meter to calibrate it against.

No internal check is outstanding. They all hold, including the one this list would otherwise have carried: the partially signalled configuration reduces to the fully signalled one to four decimals when every tile is overridden, with both sides read in the same saving convention, so the cost model's 0.8-point offset cannot present itself as a failure of the interpolation. As recorded in results/check\_paper.json, 82 of 88 checked claims pass.